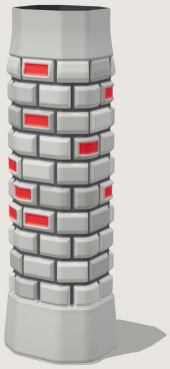
Pickleball Paddle Grips

Ten designs. One paddle slot.

Full specifications, true-size drawings and dimension data for every grip.



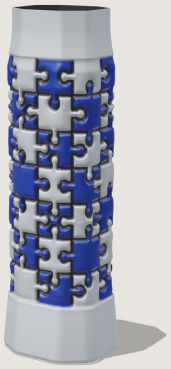
VORTEX GRIP
01 · 87.9 g PLA



CELLULAR MOD
02 · 89.2 g PLA



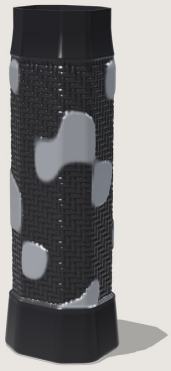
TESSEL-BLOCK
03 · 87.9 g PLA



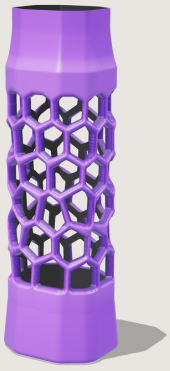
LOGIC-GRIP
04 · 89.5 g PLA



NEURO-TREAD
05 · 88.0 g PLA



CARBON MATRIX
06 · 89.5 g PLA



VORONOI-CORE
07 · 97.8 g PLA



TOPO-FLOW
08 · 89.2 g PLA



ERGO-CONTOUR
09 · 88.4 g PLA



HEXA-MOD
10 · 88.5 g PLA

132.82 mm
overall height

30.2 × 16.2 mm
paddle slot, identical on all ten

88 – 98 g
solid, 100 % infill, in PLA

Ø 3.5 mm
screw hole through the butt

Slot sized to the measured paddle · measured from the STL files · colours are preview only

10 specification sheets · 40 pages

Contents

Click any entry to jump to the page.

FRONT MATTER

All ten at a glance 3
Weight, size and print-readiness compared

The paddle slot and the reference 4
What is identical on every grip, and what changed from the reference sleeve

SPECIFICATION SHEETS · 4 PAGES EACH

01 **Vortex Grip** 5–8
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02 **Cellular Mod** 9–12
Overview 9 · Drawing 10 · Sections 11 · Surface 12

03 **Tessel-Block** 13–16
Overview 13 · Drawing 14 · Sections 15 · Surface 16

04 **Logic-Grip** 17–20
Overview 17 · Drawing 18 · Sections 19 · Surface 20

05 **Neuro-Tread** 21–24
Overview 21 · Drawing 22 · Sections 23 · Surface 24

06 **Carbon Matrix** 25–28
Overview 25 · Drawing 26 · Sections 27 · Surface 28

07 **Voronoi-Core** 29–32
Overview 29 · Drawing 30 · Sections 31 · Surface 32

08 **Topo-Flow** 33–36
Overview 33 · Drawing 34 · Sections 35 · Surface 36

09 **Ergo-Contour** 37–40
Overview 37 · Drawing 38 · Sections 39 · Surface 40

10 **Hexa-Mod** 41–44
Overview 41 · Drawing 42 · Sections 43 · Surface 44

BACK MATTER

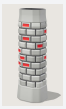
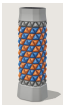
Notes, method and file manifest 45

HOW TO READ THESE SHEETS

Every dimension is measured from the STL, not copied from the design intent. Drawings on the second page of each sheet are true size when printed at 100 %. Renders and colours are a preview; the STLs are single-material. Weight = volume × density at 100 % infill.

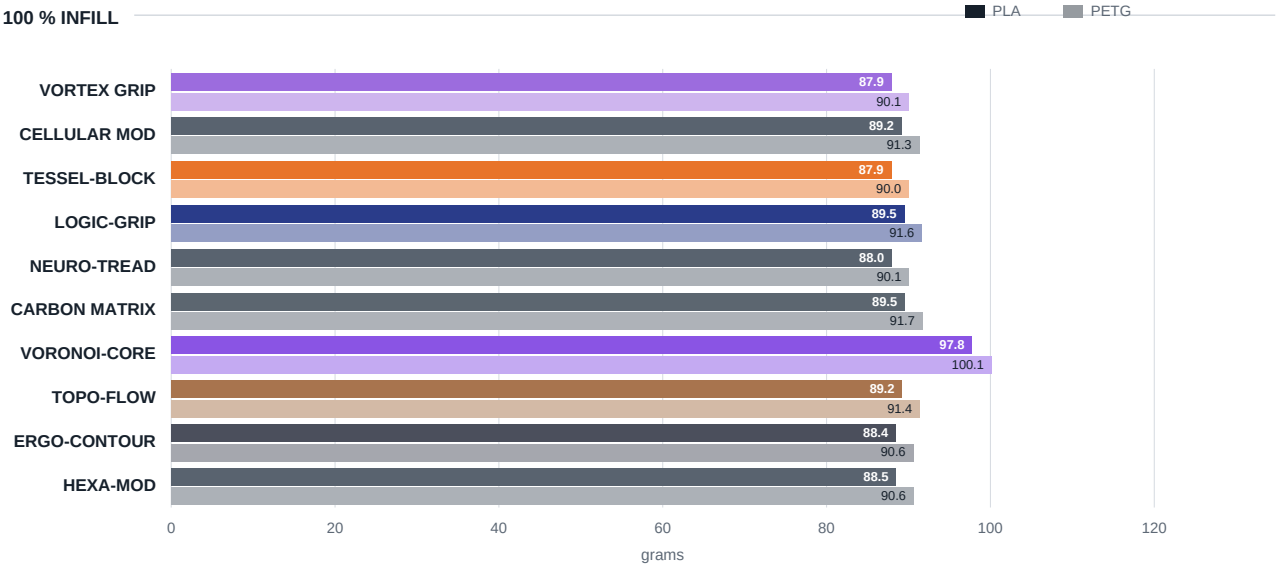
All ten at a glance

Same paddle slot, same foot, same envelope; ten different surfaces, all solid.

	DESIGN	VOLUME	PLA g	PETG g	TPU g	ABS g	PEAK	BAND W×D	BRIDGE	COLOURS
	**VORTEX GRIP** PG-01	70.9 cm³	**87.9**	90.1	85.8	73.7	2.85	37.8 × 31.7	none	
	**CELLULAR MOD** PG-02	71.9 cm³	**89.2**	91.3	87.0	74.8	3.04	38.1 × 31.9	none	
	**TESSEL-BLOCK** PG-03	70.9 cm³	**87.9**	90.0	85.8	73.7	3.44	38.9 × 32.9	none	
	**LOGIC-GRIP** PG-04	72.2 cm³	**89.5**	91.6	87.3	75.0	2.81	37.7 × 31.7	none	
	**NEURO-TREAD** PG-05	70.9 cm³	**88.0**	90.1	85.8	73.8	3.50	39.1 × 33.0	none	
	**CARBON MATRIX** PG-06	72.2 cm³	**89.5**	91.7	87.4	75.1	2.73	37.5 × 31.4	none	
	**VORONOI-CORE** PG-07	78.8 cm³	**97.8**	100.1	95.4	82.0	3.50	39.1 × 33.0	none	
	**TOPO-FLOW** PG-08	72.0 cm³	**89.2**	91.4	87.1	74.8	3.13	38.3 × 32.1	none	
	**ERGO-CONTOUR** PG-09	71.3 cm³	**88.4**	90.6	86.3	74.2	2.77	36.9 × 30.9	none	
	**HEXA-MOD** PG-10	71.4 cm³	**88.5**	90.6	86.3	74.2	3.50	38.7 × 33.0	none	

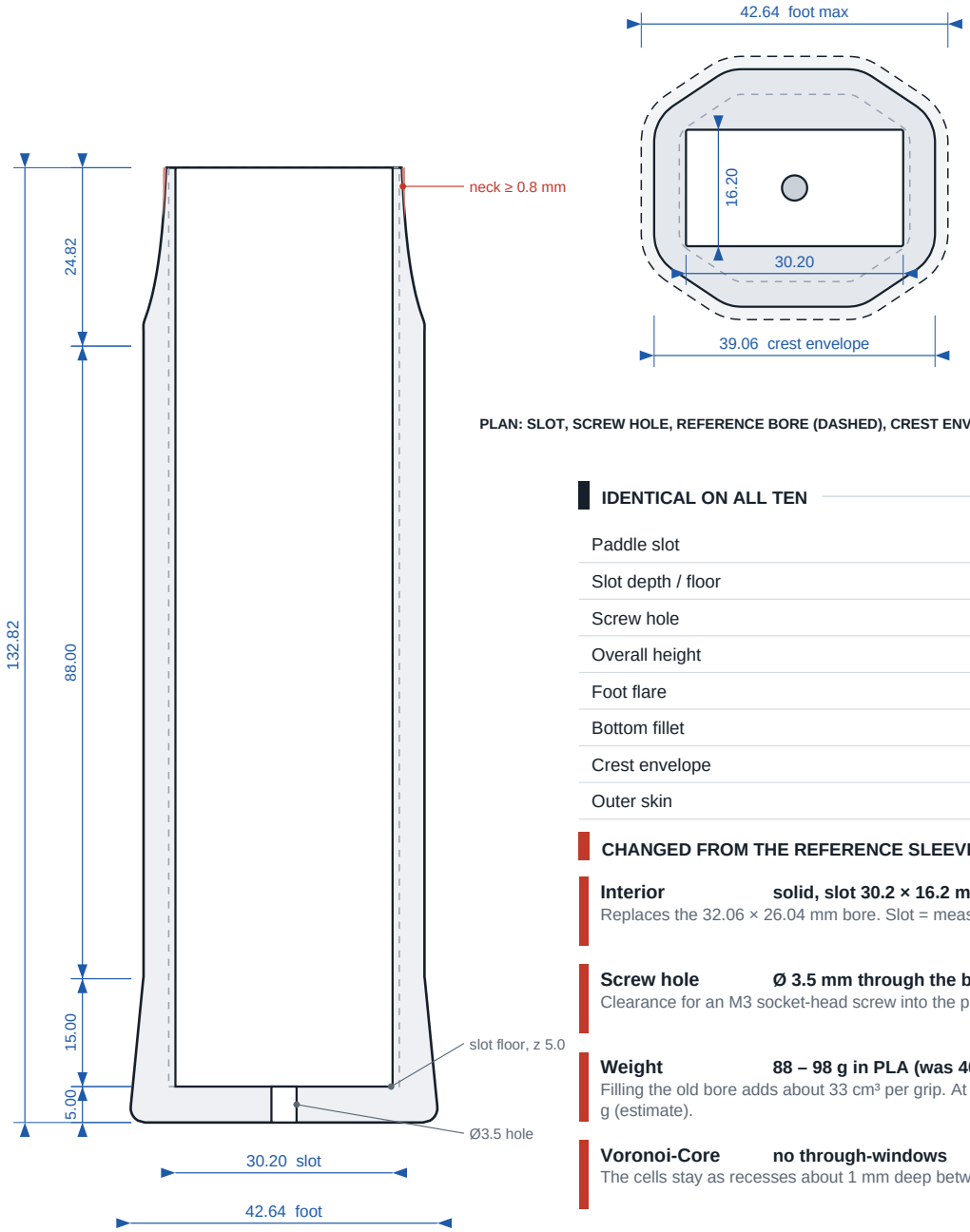
Volume and weights: 100 % infill, volume × density (g/cm³: PLA 1.24, PETG 1.27, TPU 1.21, ABS 1.04). Peak = tallest texture above the base line, mm. Bridge = longest unsupported run, mm.

WEIGHT AT 100 % INFILL



The paddle slot and the reference

Outer envelope measured from PickleballGrip_1.3mf; the interior is now a solid body with a slot for the paddle.



PLAN: SLOT, SCREW HOLE, REFERENCE BORE (DASHED), CREST ENVELOPE, FOOT (1 : 1)

FRONT ELEVATION OF THE ENVELOPE (1 : 1)

IDENTICAL ON ALL TEN

Paddle slot	**30.2 × 16.2 mm, corner radius 0.3**
Slot depth / floor	**127.82 mm / z 5.0**
Screw hole	**Ø 3.5 mm, z 0 – 5, on the slot axis**
Overall height	**132.823 mm**
Foot flare	**z 0 – 20, max 42.63 × 36.61**
Bottom fillet	**about 2 mm**
Crest envelope	**3.5 mm from the reference bore line**
Outer skin	**never exceeds the reference**

CHANGED FROM THE REFERENCE SLEEVE

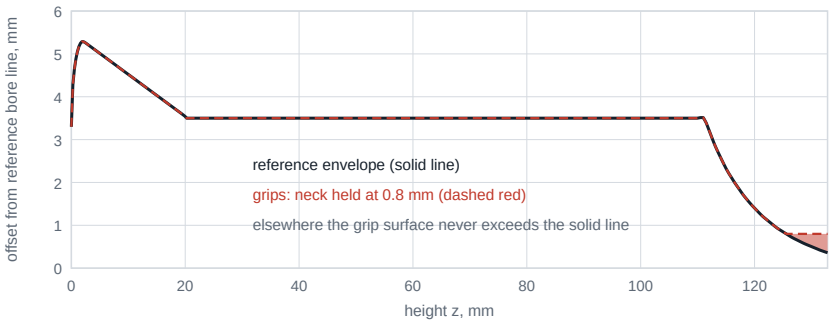
Interior — **solid, slot 30.2 × 16.2 mm**
Replaces the 32.06 × 26.04 mm bore. Slot = measured paddle tongue 30 × 16 mm + 0.2 mm.

Screw hole — **Ø 3.5 mm through the butt**
Clearance for an M3 socket-head screw into the paddle. No counterbore.

Weight — **88 – 98 g in PLA (was 40 – 49 g)**
Filling the old bore adds about 33 cm³ per grip. At 15 % infill with 4 walls expect roughly 57 – 61 g (estimate).

Voronoi-Core — **no through-windows**
The cells stay as recesses about 1 mm deep between the ribs; the body is solid like the others.

ENVELOPE VS HEIGHT: REFERENCE AND GRIPS



HOW IT WAS VERIFIED

- Every STL is cross-sectioned at 40+ heights.
- Slot: exactly one cavity from z 5 to the top, equal to the slot outline; no material inside it (0.000 mm²).
- Screw hole: one round Ø 3.5 mm hole on the axis through the butt, floor solid around it.
- Envelope: outer skin vs the reference, worst case +0.003 mm.
- Print check: every 0.20 mm layer sliced; one closed body, no enclosed void, no material under 0.4 mm wide.

01 VORTEX GRIP

Braided over/under helical straps for dynamic multi-point contact.

87.9 g

PLA · 100 % infill · solid

✓ SOLID · SLOT + SCREW HOLE



3/4 view · rendered from the STL, colours are preview only

AT A GLANCE

Overall height	**132.82 mm**
Max footprint (foot)	**42.63 × 36.61 mm**
Paddle slot	**30.2 × 16.2 mm, 127.8 deep**
Textured band	**z 21.5 – 116.0 mm (94.5 tall)**
Outer size in band (max)	**37.8 × 31.7 mm**
Peak texture height	**2.85 mm above base**
Screw hole (butt)	**Ø 3.5 mm, through z 0 – 5**
Volume / surface area	**70.91 cm³ / 307 cm²**

WEIGHT · 100 % INFILL

PLA	1.24 g/cm³	**87.9 g**
PETG	1.27 g/cm³	**90.1 g**
TPU	1.21 g/cm³	**85.8 g**
ABS	1.04 g/cm³	**73.7 g**

0 — 120 g

COLOURWAY

#3A225C #9C6CDE

3MF print palette

PRINT PROFILE

Infill / layer height	**100 % / 0.20 mm**
Layers	**665**
Orientation	**upright, butt end down**
Supports	**none required**
Filament, 1.75 mm	**≈ 29.5 m (100 % flow)**

DESIGN INTENT

Sixteen straps wound in opposite directions cross in an over/under braid, leaving dark valleys between them. Every crossing gives the hand an edge to bite on, so the grip works from any hand position on the handle.

Straps rise no faster than 45 degrees, so no supports are needed.

KEY FEATURES

- 8 right-hand + 8 left-hand straps on a 36 degree helix
- Over/under weave: the strap on top alternates at every crossing
- Valleys drop to about the 1.2 mm base wall, which keeps the mass low

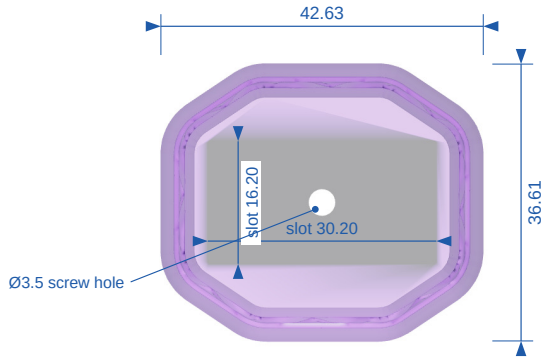
VERIFIED ON THE STL

Paddle slot	**30.2 × 16.2 mm (30 × 16 + 0.2)**
Screw hole	**Ø 3.5 mm through the butt**
Outer envelope vs reference	**+0.000 mm at most**
Bodies / enclosed voids	**1 / none**
Material under 0.4 mm wide	**0.000 % of volume**
Overhangs / bridges	**none**

FILES

stl/01_vortex_grip.stl	14.8 MB · 295,950 triangles
3mf/01_vortex_grip.3mf	3.2 MB · colour

✓ SLOT & HOLE VERIFIED ✓ SOLID · ONE CLOSED BODY ✓ PRINT-READY · 0.20 MM LAYERS ✓ 3MF VALIDATED

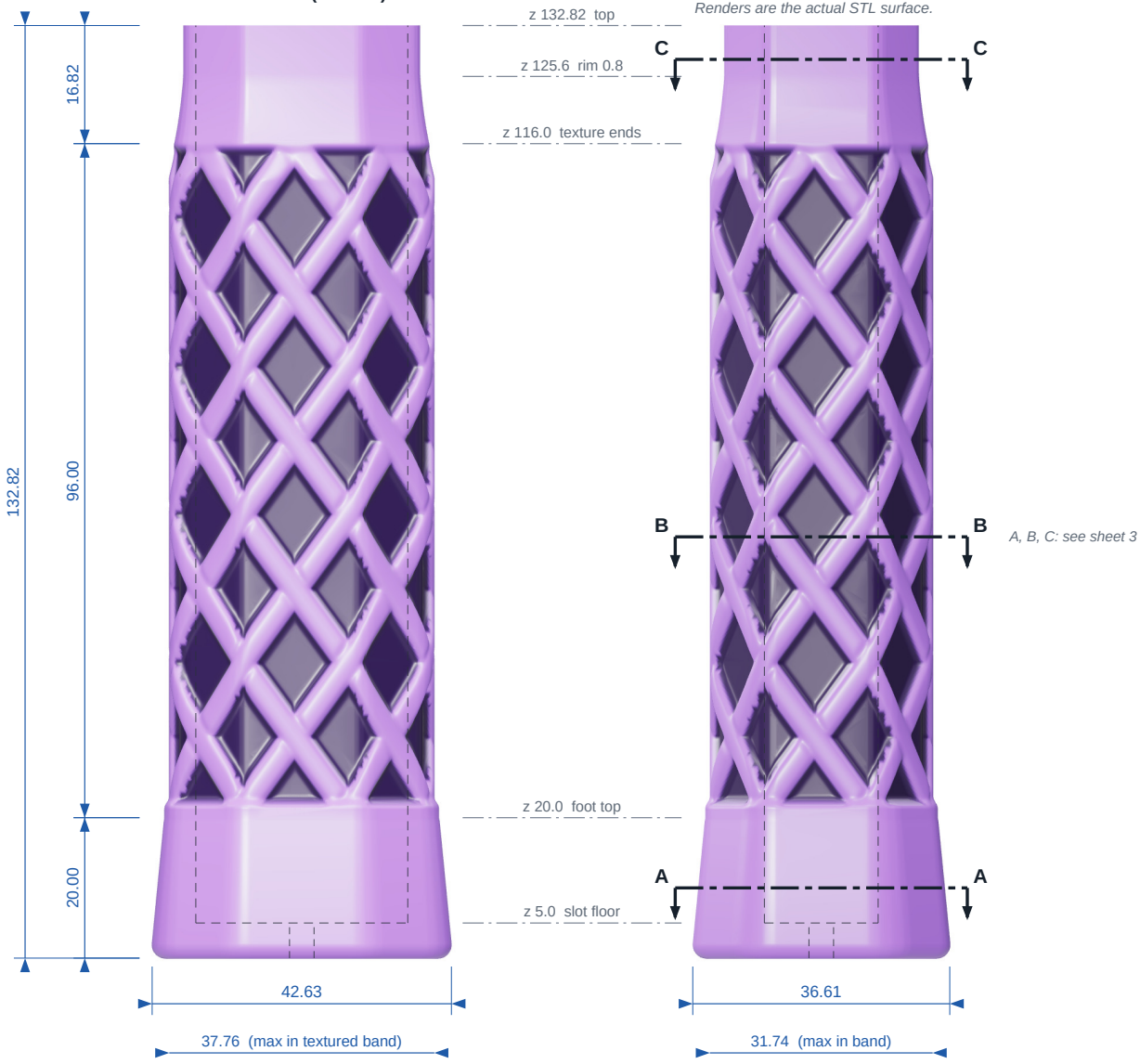


TOP VIEW (from +Z)

KEY DIMENSIONS

Overall height	**132.82**
Overall width × depth (foot)	**42.63 × 36.61**
Paddle slot width × depth	**30.20 × 16.20**
Slot depth (floor at z 5.0)	**127.82**
Outer W × D, textured band (max)	**37.76 × 31.74**
Foot top (z 20)	**39.16 × 33.13**
Butt face (z 0)	**40.56 × 34.54**
Peak texture height above base	**2.85**
Screw hole through butt (z 0 – 5)	**Ø 3.50**

All values in mm. Dashed lines: paddle slot and screw hole (hidden edges). Renders are the actual STL surface.



FRONT VIEW (from −Y)

RIGHT VIEW (from +X)

SCALE 1 : 1



PART

VORTEX GRIP

Pickleball paddle grip sleeve, closed butt, open throat

SCALE	UNITS	PROJECTION
1 : 1	millimetres	
MATERIAL	**INFILL**	**LAYER HEIGHT**
PLA / PETG / TPU	100 %	0.20 mm
MASS (PLA)	**VOLUME**	**SURFACE AREA**
87.9 g	70.91 cm³	307 cm²
REVISION	**DATE**	**SHEET**
3	2026-10-17	2 of 4

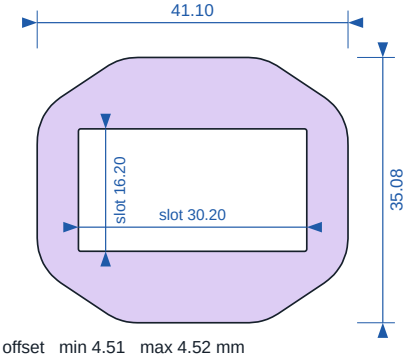
GENERAL NOTES

1. Views are true size: print this sheet at 100 %, do not fit to page.
2. Dimensions are measured from the STL, not nominal.
3. Solid body: only the 30.2 × 16.2 mm paddle slot and the Ø3.5 mm screw hole are open.
4. Outer envelope matches the reference PickleballGrip_1.3mf (rim held ≥ 0.8 mm).

Sections & profiles

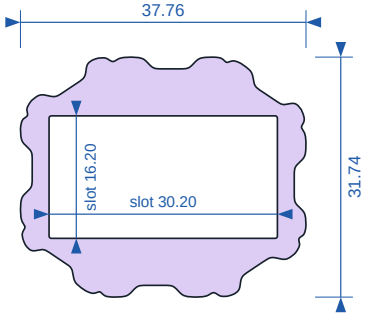
True-size cross-sections, how the outer surface and material change with height, and the dimension tables.

SECTION A–A z = 10 mm · foot



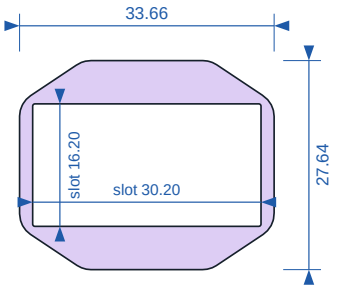
Skin offset min 4.51 max 4.52 mm
Area 780 mm² · outer perimeter 128.9 mm

SECTION B–B z = 60 mm · textured band



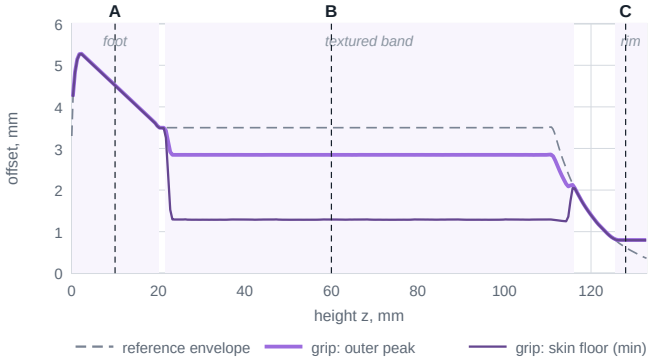
Skin offset min 1.29 max 2.85 mm
Area 491 mm² · outer perimeter 118.0 mm

SECTION C–C z = 128 mm · rim

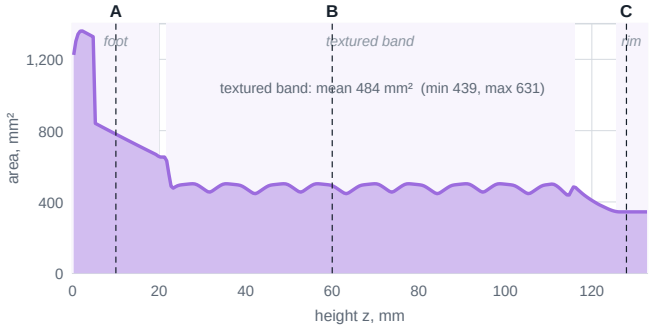


Skin offset min 0.79 max 0.80 mm
Area 344 mm² · outer perimeter 105.5 mm

OUTER OFFSET FROM THE REFERENCE OUTLINE VS HEIGHT



MATERIAL CROSS-SECTION AREA VS HEIGHT



OVERALL & SLOT

Overall length (Z)	**132.82 mm**
Overall width (X)	**42.63 mm**
Overall depth (Y)	**36.61 mm**
Slot width × depth	**30.20 × 16.20 mm**
Slot area / perimeter	**489.2 mm² / 92.3 mm**
Slot depth (floor at z 5.0)	**127.82 mm**
Screw hole (butt, z 0 – 5)	**Ø 3.50 mm**
Butt face (z 0)	**40.56 × 34.54 mm**
Outer W × D in band (max)	**37.76 × 31.74 mm**
Volume / surface area	**70.91 cm³ / 307.4 cm²**

ZONES BY HEIGHT

ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	**5.0**
Flared foot	0.0	20.0	**20.0**
Smooth collar	20.0	21.5	**1.5**
Textured band	21.5	116.0	**94.5**
Smooth neck (taper)	116.0	125.6	**9.6**
Rim (flat top, slot opening)	125.6	132.82	**7.3**

Relief floor above reference line (design / min)	**1.20 / 1.29 mm**
Min skin at the neck (design / measured)	**0.80 / 0.80 mm**
Peak texture height above base	**2.85 mm**

OUTER DIMENSIONS BY HEIGHT

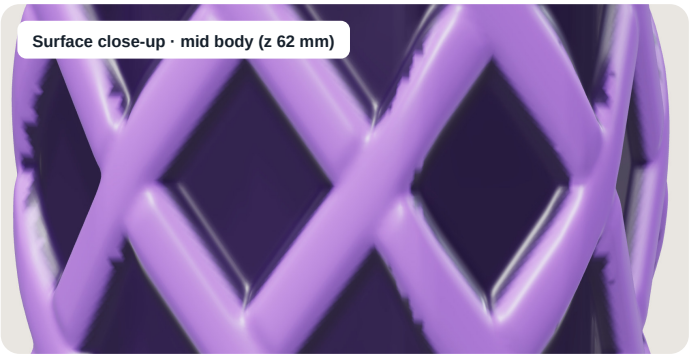
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM²	SKIN MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	1,084	5.00	5.01
10.0	41.10	35.08	128.9	780	4.51	4.52
20.0	39.11	33.09	122.6	655	3.52	3.52
30.0	37.76	31.74	114.5	477	1.29	2.85
45.0	37.76	31.74	118.0	487	1.29	2.85
60.0	37.76	31.74	118.0	491	1.29	2.85
75.0	37.76	31.74	114.1	471	1.29	2.85
90.0	37.76	31.74	117.1	500	1.29	2.85
105.0	37.75	31.74	117.1	449	1.29	2.85
116.0	36.24	30.22	113.4	483	2.04	2.09
125.0	33.77	27.75	105.8	350	0.85	0.85
132.0	33.66	27.64	105.5	344	0.79	0.80

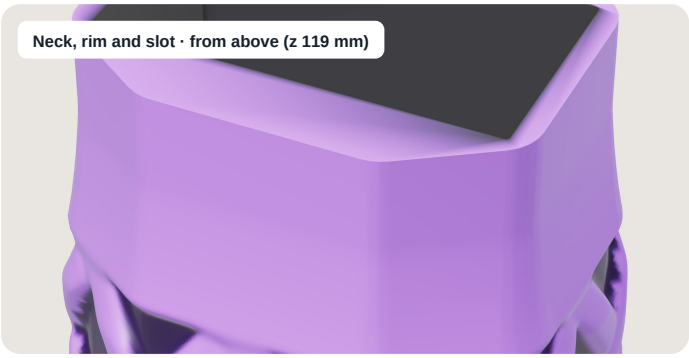
Views & surface

Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.

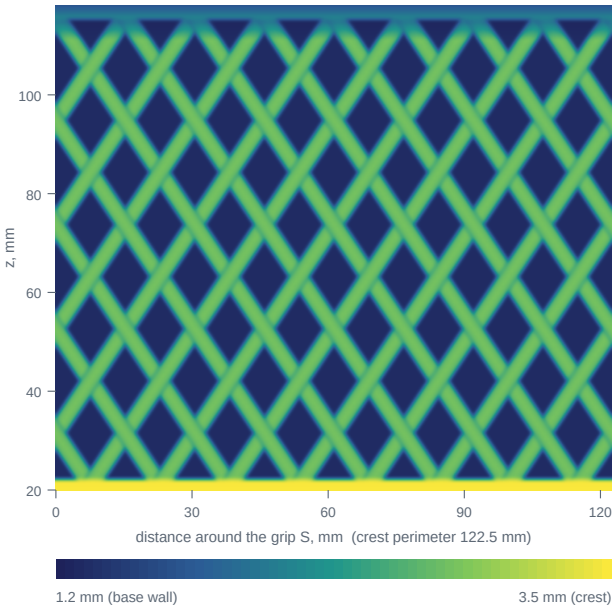


Four views, 90° apart · rendered from the STL, colours are preview only





UNROLLED SURFACE · HEIGHT ABOVE BASE



DESIGN PARAMETERS

Straps	**8 + 8, counter-wound**
Helix angle	**36 deg from vertical**
Strap pitch (around)	**15.31 mm**
Strap width	**6.1 mm around / 5.0 mm across**
Strap top height	**2.35 – 2.86 mm above base**
Valley floor	**1.29 mm above base**
Over/under step	**0.37 mm**

PRINT & VERIFICATION

Result: slot & hole / print-ready	**PASS / PASS**
Bodies · genus	**1 · 1 (screw hole)**
Paddle slot	**30.2 × 16.2 mm**
Envelope over reference	**+0.000 mm**
Material under 0.4 mm wide	**0.000 %**
Overhangs / bridges	**none**

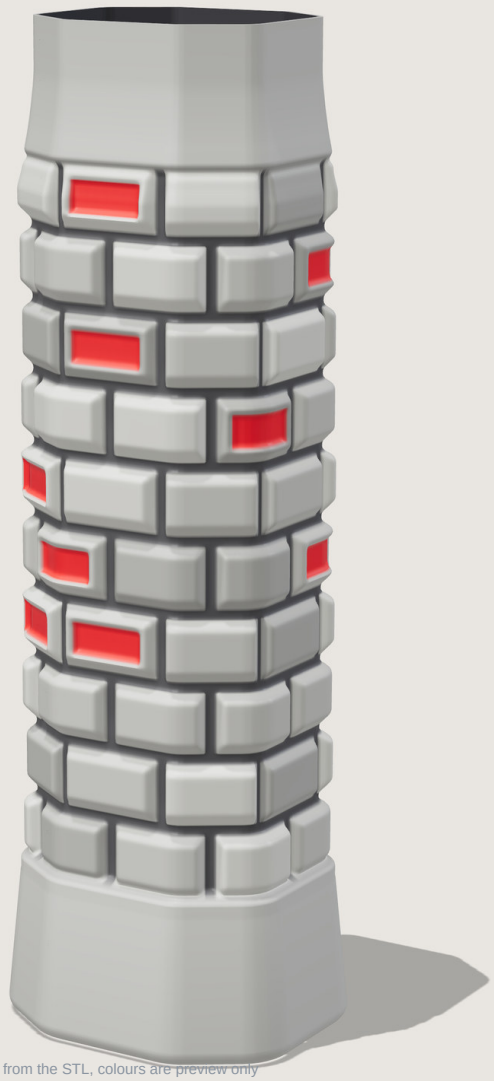
02 CELLULAR MOD

Staggered stacked cells; red cells are recessed pockets for swappable inserts.

89.2 g

PLA · 100 % infill · solid

✓ SOLID · SLOT + SCREW HOLE



3/4 view · rendered from the STL, colours are preview only

AT A GLANCE

Overall height	**132.82 mm**
Max footprint (foot)	**42.63 × 36.61 mm**
Paddle slot	**30.2 × 16.2 mm, 127.8 deep**
Textured band	**z 21.5 – 116.0 mm (94.5 tall)**
Outer size in band (max)	**38.1 × 31.9 mm**
Peak texture height	**3.04 mm above base**
Screw hole (butt)	**Ø 3.5 mm, through z 0 – 5**
Volume / surface area	**71.91 cm³ / 312 cm²**

WEIGHT · 100 % INFILL

PLA	1.24 g/cm³		**89.2 g**
PETG	1.27 g/cm³		**91.3 g**
TPU	1.21 g/cm³		**87.0 g**
ABS	1.04 g/cm³		**74.8 g**

0 — 120 g

COLOURWAY

#464648	#969692	#C81E22

3MF print palette

PRINT PROFILE

Infill / layer height	**100 % / 0.20 mm**
Layers	**665**
Orientation	**upright, butt end down**
Supports	**none required**
Filament, 1.75 mm	**≈ 29.9 m (100 % flow)**

DESIGN INTENT

A wall of staggered, individually pillowed cells. Roughly one cell in five is recessed as an insert pocket (red in the colourway), sized to take a contrasting insert. No insert parts are included.

Cell edges are chamfered; nothing overhangs beyond 45 degrees.

KEY FEATURES

- 10 rows × 8 staggered cells with 1.4 mm gaps
- Three cell heights give a subtly stepped surface
- About 22% of cells are recessed 0.7 – 1.2 mm as insert pockets

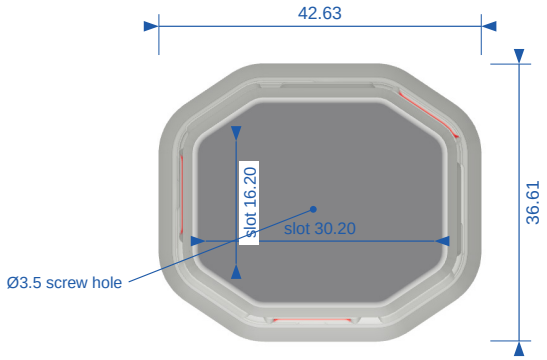
VERIFIED ON THE STL

Paddle slot	**30.2 × 16.2 mm (30 × 16 + 0.2)**
Screw hole	**Ø 3.5 mm through the butt**
Outer envelope vs reference	**+0.000 mm at most**
Bodies / enclosed voids	**1 / none**
Material under 0.4 mm wide	**0.000 % of volume**
Overhangs / bridges	**none**

FILES

stl/02_cellular_mod.stl	6.0 MB · 119,164 triangles
3mf/02_cellular_mod.3mf	1.4 MB · colour

✓ SLOT & HOLE VERIFIED ✓ SOLID · ONE CLOSED BODY ✓ PRINT-READY · 0.20 MM LAYERS ✓ 3MF VALIDATED

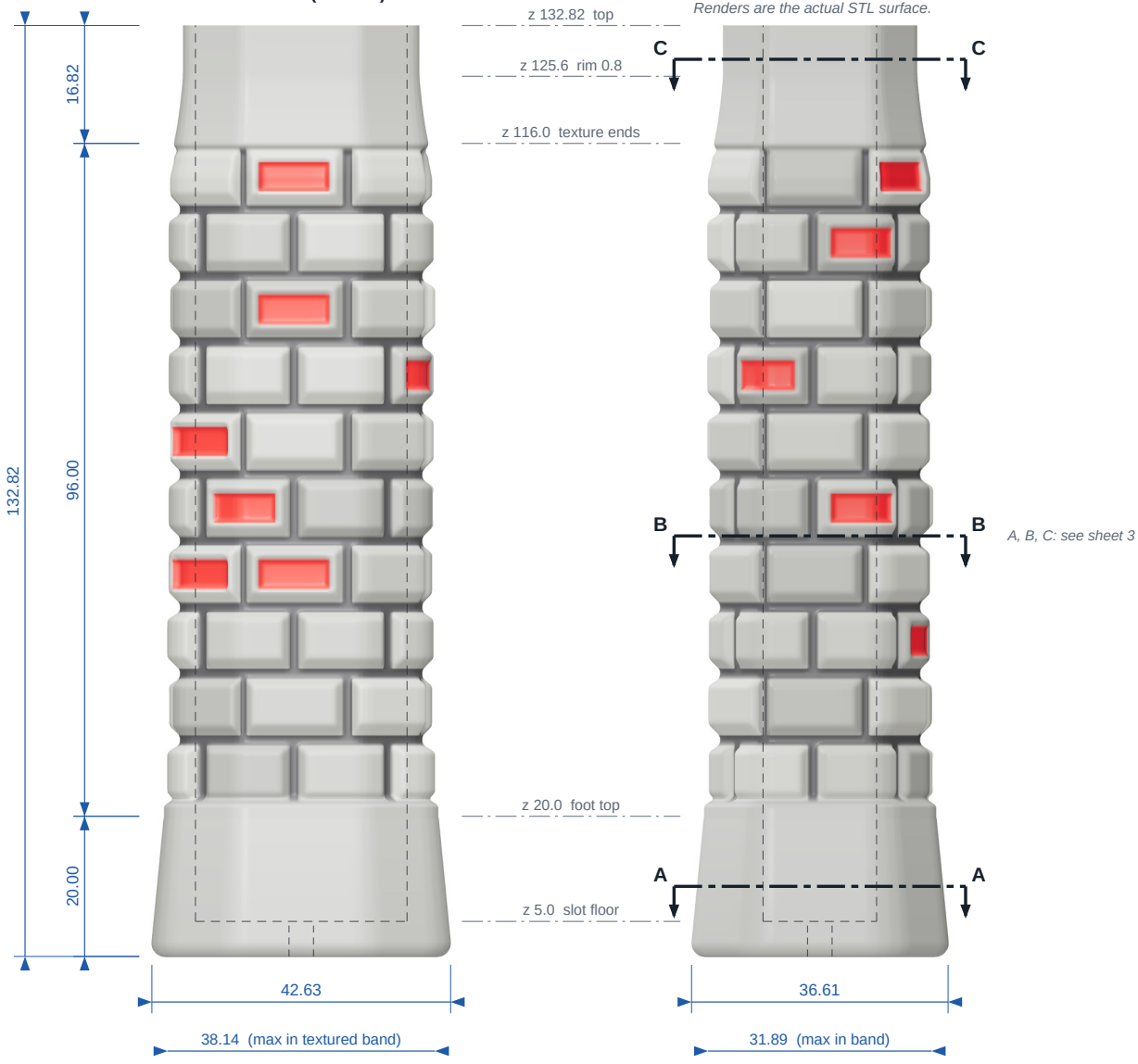


TOP VIEW (from +Z)

KEY DIMENSIONS

Overall height	**132.82**
Overall width × depth (foot)	**42.63 × 36.61**
Paddle slot width × depth	**30.20 × 16.20**
Slot depth (floor at z 5.0)	**127.82**
Outer W × D, textured band (max)	**38.14 × 31.89**
Foot top (z 20)	**39.15 × 33.13**
Butt face (z 0)	**40.56 × 34.54**
Peak texture height above base	**3.04**
Screw hole through butt (z 0 – 5)	**Ø 3.50**

All values in mm. Dashed lines: paddle slot and screw hole (hidden edges). Renders are the actual STL surface.



FRONT VIEW (from −Y)

RIGHT VIEW (from +X)

SCALE 1 : 1



PART

CELLULAR MOD

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

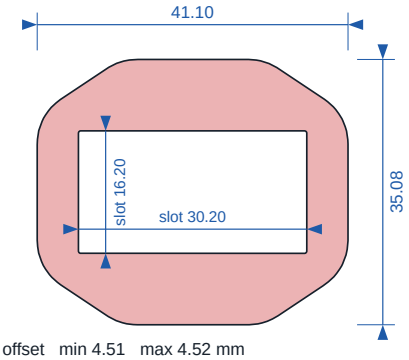
1. Views are true size: print this sheet at 100 %, do not fit to page.
2. Dimensions are measured from the STL, not nominal.
3. Solid body: only the 30.2 × 16.2 mm paddle slot and the Ø3.5 mm screw hole are open.
4. Outer envelope matches the reference PickleballGrip_1.3mf (rim held ≥ 0.8 mm).

SCALE **1 : 1**	UNITS **millimetres**	PROJECTION
MATERIAL **PLA / PETG / TPU**	INFILL **100 %**	LAYER HEIGHT **0.20 mm**
MASS (PLA) **89.2 g**	VOLUME **71.91 cm³**	SURFACE AREA **312 cm²**
REVISION **3**	DATE **2026-10-17**	SHEET **2 of 4**

Sections & profiles

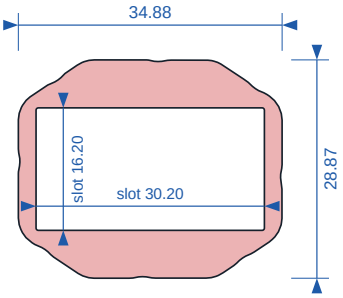
True-size cross-sections, how the outer surface and material change with height, and the dimension tables.

SECTION A–A z = 10 mm · foot



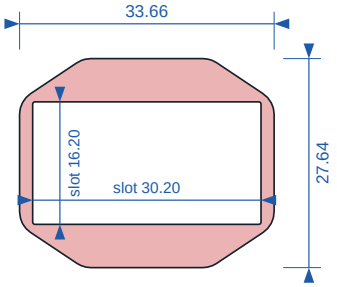
Skin offset min 4.51 max 4.52 mm
Area 780 mm² · outer perimeter 128.9 mm

SECTION B–B z = 60 mm · textured band



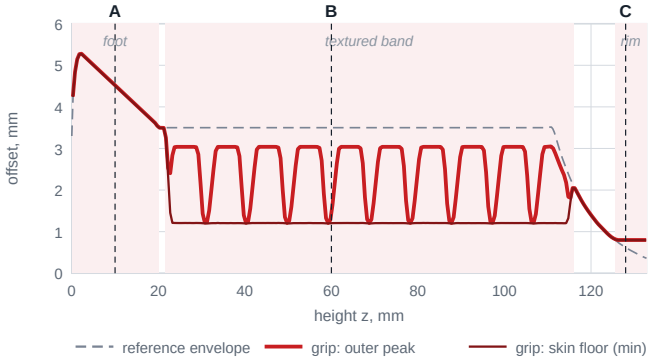
Skin offset min 1.20 max 1.46 mm
Area 406 mm² · outer perimeter 109.3 mm

SECTION C–C z = 128 mm · rim

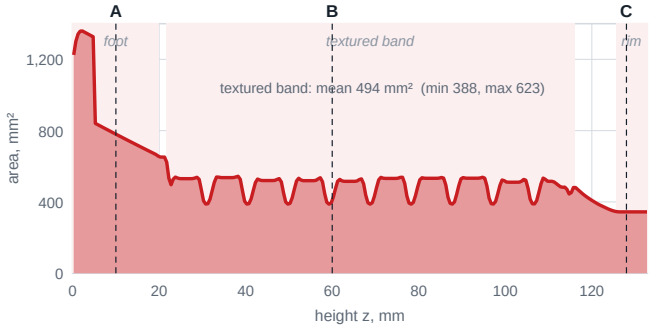


Skin offset min 0.79 max 0.80 mm
Area 344 mm² · outer perimeter 105.5 mm

OUTER OFFSET FROM THE REFERENCE OUTLINE VS HEIGHT



MATERIAL CROSS-SECTION AREA VS HEIGHT



OVERALL & SLOT

Overall length (Z)	**132.82 mm**
Overall width (X)	**42.63 mm**
Overall depth (Y)	**36.61 mm**
Slot width × depth	**30.20 × 16.20 mm**
Slot area / perimeter	**489.2 mm² / 92.3 mm**
Slot depth (floor at z 5.0)	**127.82 mm**
Screw hole (butt, z 0 – 5)	**Ø 3.50 mm**
Butt face (z 0)	**40.56 × 34.54 mm**
Outer W × D in band (max)	**38.14 × 31.89 mm**
Volume / surface area	**71.91 cm³ / 312.4 cm²**

ZONES BY HEIGHT

ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	**5.0**
Flared foot	0.0	20.0	**20.0**
Smooth collar	20.0	21.5	**1.5**
Textured band	21.5	116.0	**94.5**
Smooth neck (taper)	116.0	125.6	**9.6**
Rim (flat top, slot opening)	125.6	132.82	**7.3**

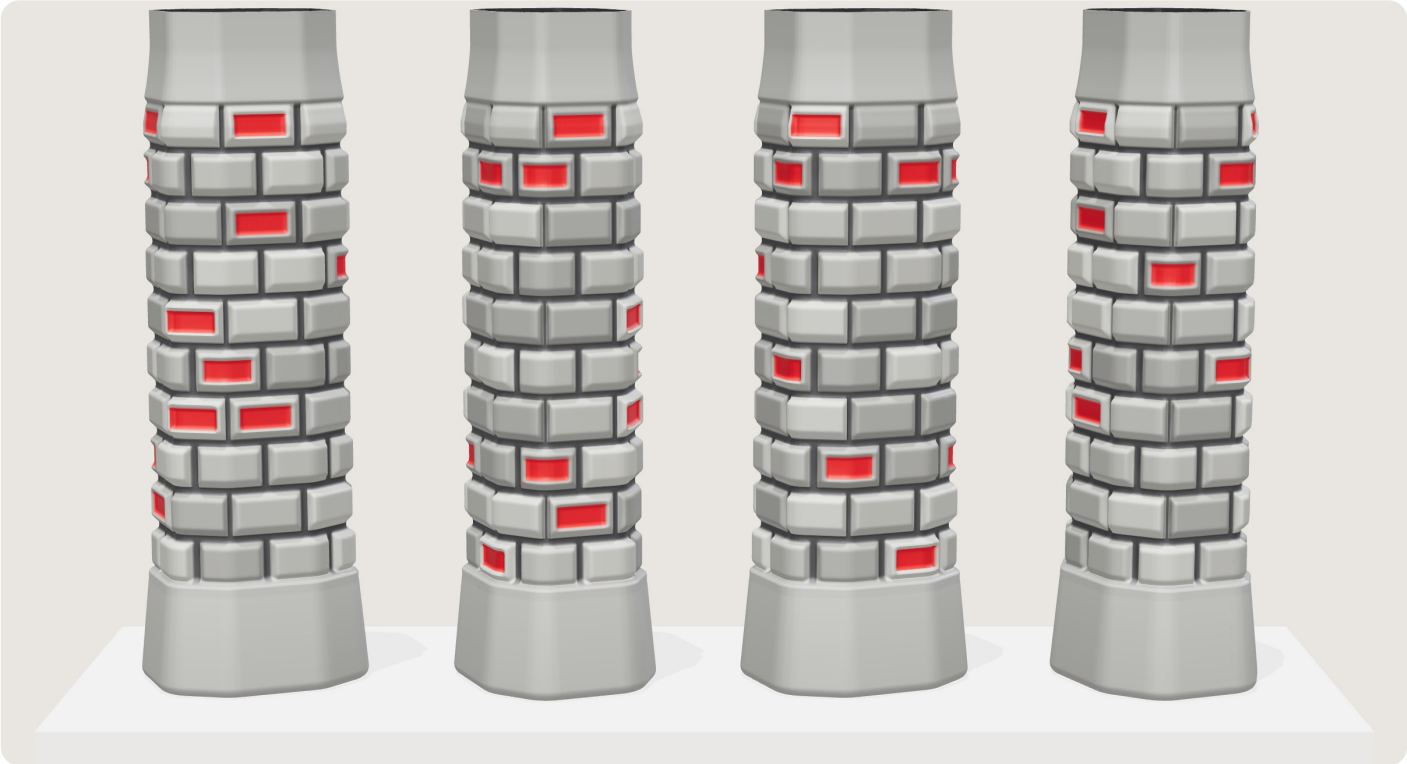
Relief floor above reference line (design / min)	**1.20 / 1.20 mm**
Min skin at the neck (design / measured)	**0.80 / 0.80 mm**
Peak texture height above base	**3.04 mm**

OUTER DIMENSIONS BY HEIGHT

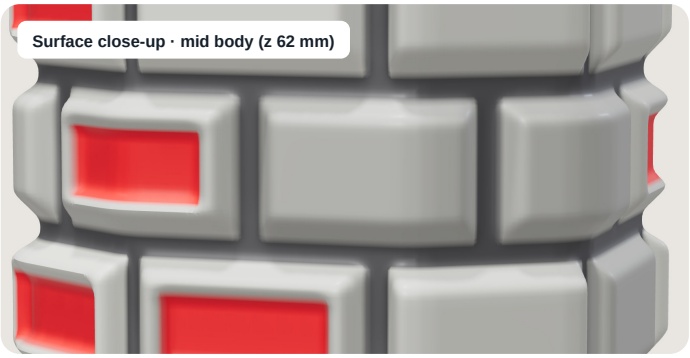
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM²	SKIN MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	1,084	5.00	5.01
10.0	41.10	35.08	128.9	780	4.51	4.52
20.0	39.11	33.09	122.6	655	3.52	3.52
30.0	35.66	29.57	111.5	438	1.20	1.80
45.0	37.91	31.43	117.0	520	1.21	3.04
60.0	34.96	28.95	109.6	410	1.20	1.46
75.0	37.46	31.66	117.7	533	1.21	3.04
90.0	37.54	31.46	117.3	533	1.20	2.91
105.0	36.98	30.96	115.5	499	1.21	2.64
116.0	36.16	30.14	113.4	481	2.04	2.05
125.0	33.77	27.75	105.8	350	0.85	0.85
132.0	33.66	27.64	105.5	344	0.79	0.80

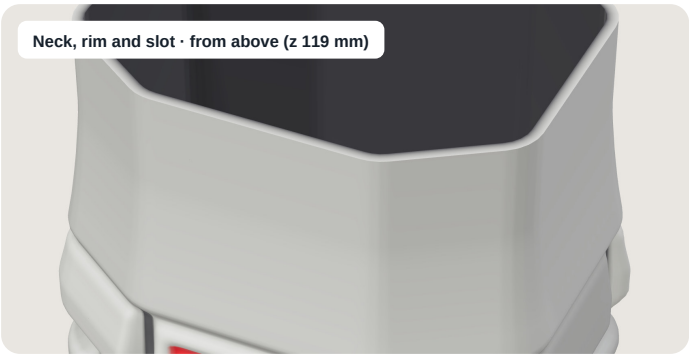
Views & surface

Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.

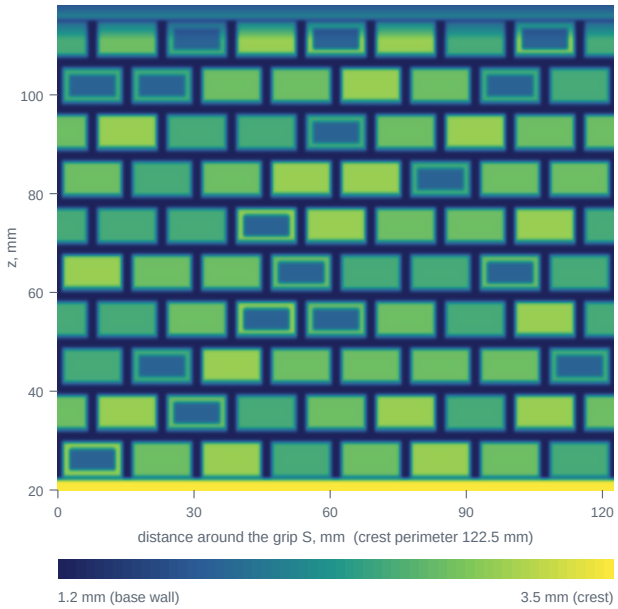


Four views, 90° apart · rendered from the STL, colours are preview only





UNROLLED SURFACE · HEIGHT ABOVE BASE



DESIGN PARAMETERS

Rows × cells	**10 × 8 (staggered)**
Row pitch	**9.45 mm**
Cell size	**15.31 × 8.05 mm**
Gap between cells	**1.4 mm**
Cell heights	**2.58 / 2.81 / 3.04 mm**
Pocket floor	**1.84 mm above base**
Groove floor	**1.20 mm (base wall)**

PRINT & VERIFICATION

Result: slot & hole / print-ready	**PASS / PASS**
Bodies · genus	**1 · 1 (screw hole)**
Paddle slot	**30.2 × 16.2 mm**
Envelope over reference	**+0.000 mm**
Material under 0.4 mm wide	**0.000 %**
Overhangs / bridges	**none**

03 TESSEL-BLOCK

Faceted triangular pyramids in staggered rows for enhanced traction.

87.9 g

PLA · 100 % infill · solid

✓ SOLID · SLOT + SCREW HOLE



3/4 view · rendered from the STL, colours are preview only

AT A GLANCE

Overall height	**132.82 mm**
Max footprint (foot)	**42.63 × 36.61 mm**
Paddle slot	**30.2 × 16.2 mm, 127.8 deep**
Textured band	**z 21.5 – 116.0 mm (94.5 tall)**
Outer size in band (max)	**38.9 × 32.9 mm**
Peak texture height	**3.44 mm above base**
Screw hole (butt)	**Ø 3.5 mm, through z 0 – 5**
Volume / surface area	**70.88 cm³ / 310 cm²**

WEIGHT · 100 % INFILL

PLA	1.24 g/cm³		**87.9 g**
PETG	1.27 g/cm³		**90.0 g**
TPU	1.21 g/cm³		**85.8 g**
ABS	1.04 g/cm³		**73.7 g**

0 — 120 g

COLOURWAY

#848482	#E8742A	#387AC4

3MF print palette

PRINT PROFILE

Infill / layer height	**100 % / 0.20 mm**
Layers	**665**
Orientation	**upright, butt end down**
Supports	**none required**
Filament, 1.75 mm	**≈ 29.5 m (100 % flow)**

DESIGN INTENT

A tessellation of triangular pyramids. Traction comes from many small angled edges rather than one deep texture, and up-pointing pyramids stand taller than down-pointing ones, giving a two-level, gem-like surface.

Facet slopes stay below 45 degrees; rim rows are flat grey bands in the colourway.

KEY FEATURES

- 11 rows × 24 triangles (12 up + 12 down per row)
- Sharp V-grooves between facets
- Up-pyramids 3.44 mm, down-pyramids 3.04 mm above the base

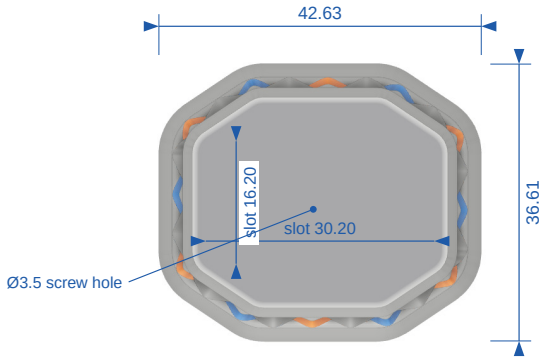
VERIFIED ON THE STL

Paddle slot	**30.2 × 16.2 mm (30 × 16 + 0.2)**
Screw hole	**Ø 3.5 mm through the butt**
Outer envelope vs reference	**+0.000 mm at most**
Bodies / enclosed voids	**1 / none**
Material under 0.4 mm wide	**0.000 % of volume**
Overhangs / bridges	**none**

FILES

stl/03_tessel_block.stl	15.0 MB · 300,964 triangles
3mf/03_tessel_block.3mf	3.3 MB · colour

✓ SLOT & HOLE VERIFIED ✓ SOLID · ONE CLOSED BODY ✓ PRINT-READY · 0.20 MM LAYERS ✓ 3MF VALIDATED

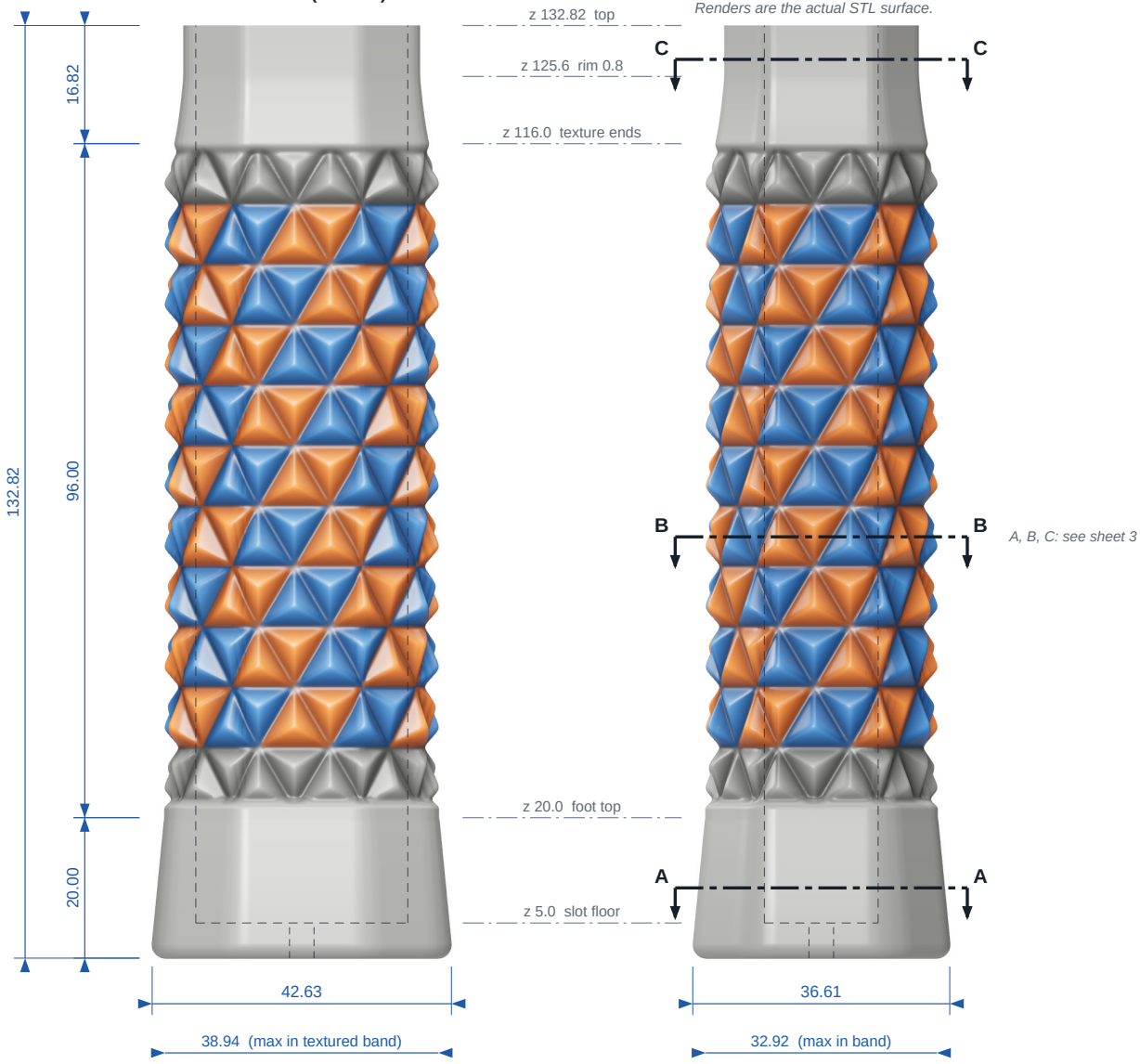


TOP VIEW (from +Z)

KEY DIMENSIONS

Overall height	**132.82**
Overall width × depth (foot)	**42.63 × 36.61**
Paddle slot width × depth	**30.20 × 16.20**
Slot depth (floor at z 5.0)	**127.82**
Outer W × D, textured band (max)	**38.94 × 32.92**
Foot top (z 20)	**39.16 × 33.13**
Butt face (z 0)	**40.56 × 34.54**
Peak texture height above base	**3.44**
Screw hole through butt (z 0 – 5)	**Ø 3.50**

All values in mm. Dashed lines: paddle slot and screw hole (hidden edges). Renders are the actual STL surface.



FRONT VIEW (from −Y)

RIGHT VIEW (from +X)

SCALE 1 : 1



PART

TESSEL-BLOCK

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

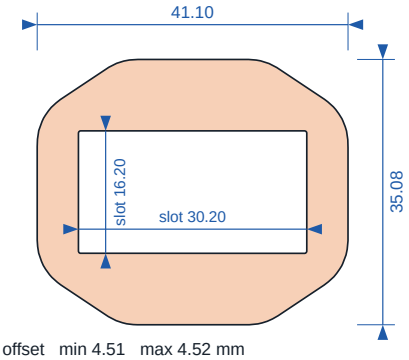
1. Views are true size: print this sheet at 100 %, do not fit to page.
2. Dimensions are measured from the STL, not nominal.
3. Solid body: only the 30.2 × 16.2 mm paddle slot and the Ø3.5 mm screw hole are open.
4. Outer envelope matches the reference PickleballGrip_1.3mf (rim held ≥ 0.8 mm).

SCALE **1 : 1**	UNITS **millimetres**	PROJECTION
MATERIAL **PLA / PETG / TPU**	INFILL **100 %**	LAYER HEIGHT **0.20 mm**
MASS (PLA) **87.9 g**	VOLUME **70.88 cm³**	SURFACE AREA **310 cm²**
REVISION **3**	DATE **2026-10-17**	SHEET **2 of 4**

Sections & profiles

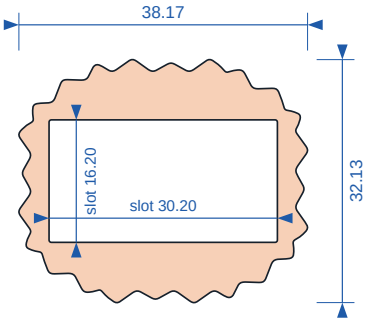
True-size cross-sections, how the outer surface and material change with height, and the dimension tables.

SECTION A–A z = 10 mm · foot



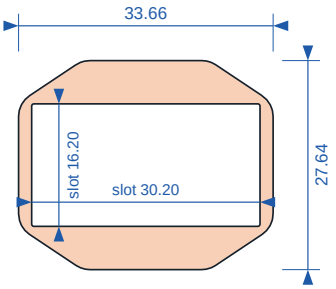
Skin offset min 4.51 max 4.52 mm
Area 780 mm² · outer perimeter 128.9 mm

SECTION B–B z = 60 mm · textured band



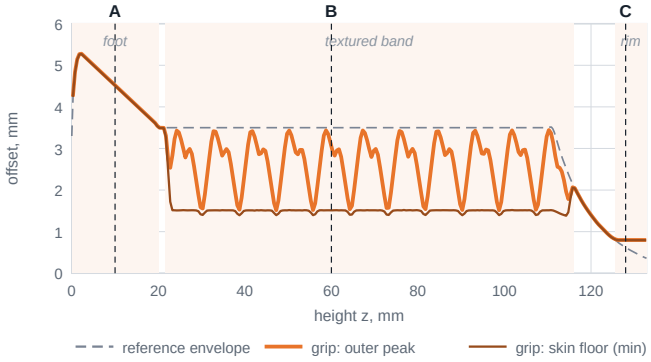
Skin offset min 1.51 max 3.05 mm
Area 495 mm² · outer perimeter 117.2 mm

SECTION C–C z = 128 mm · rim

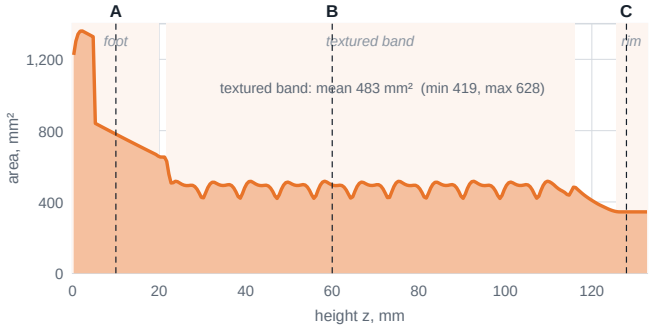


Skin offset min 0.79 max 0.80 mm
Area 344 mm² · outer perimeter 105.5 mm

OUTER OFFSET FROM THE REFERENCE OUTLINE VS HEIGHT



MATERIAL CROSS-SECTION AREA VS HEIGHT



OVERALL & SLOT

Overall length (Z)	**132.82 mm**
Overall width (X)	**42.63 mm**
Overall depth (Y)	**36.61 mm**
Slot width × depth	**30.20 × 16.20 mm**
Slot area / perimeter	**489.2 mm² / 92.3 mm**
Slot depth (floor at z 5.0)	**127.82 mm**
Screw hole (butt, z 0 – 5)	**Ø 3.50 mm**
Butt face (z 0)	**40.56 × 34.54 mm**
Outer W × D in band (max)	**38.94 × 32.92 mm**
Volume / surface area	**70.88 cm³ / 310.3 cm²**

ZONES BY HEIGHT

ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	**5.0**
Flared foot	0.0	20.0	**20.0**
Smooth collar	20.0	21.5	**1.5**
Textured band	21.5	116.0	**94.5**
Smooth neck (taper)	116.0	125.6	**9.6**
Rim (flat top, slot opening)	125.6	132.82	**7.3**

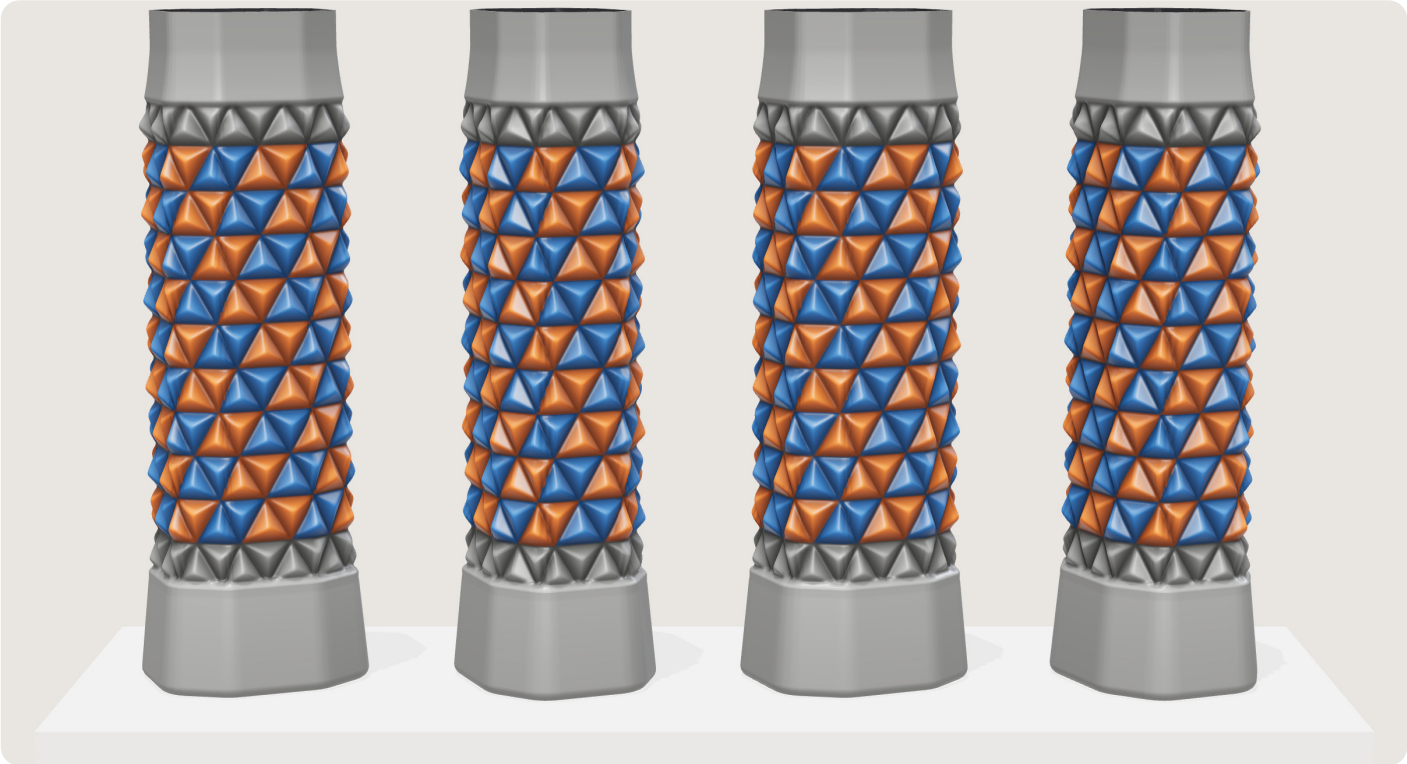
Relief floor above reference line (design / min)	**1.20 / 1.39 mm**
Min skin at the neck (design / measured)	**0.80 / 0.80 mm**
Peak texture height above base	**3.44 mm**

OUTER DIMENSIONS BY HEIGHT

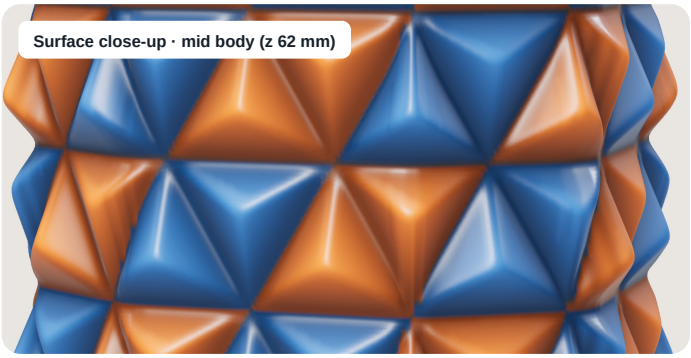
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM²	SKIN MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	1,084	5.00	5.01
10.0	41.10	35.08	128.9	780	4.51	4.52
20.0	39.11	33.09	122.6	655	3.52	3.52
30.0	35.21	29.18	110.3	424	1.41	1.57
45.0	37.70	31.68	117.3	492	1.51	2.82
60.0	38.17	32.13	117.3	495	1.51	3.05
75.0	38.05	32.03	116.9	511	1.51	3.00
90.0	35.24	29.22	110.3	426	1.42	1.59
105.0	37.81	31.79	115.9	496	1.51	2.87
116.0	36.18	30.16	113.4	482	2.05	2.06
125.0	33.77	27.75	105.8	350	0.85	0.85
132.0	33.66	27.64	105.5	344	0.79	0.80

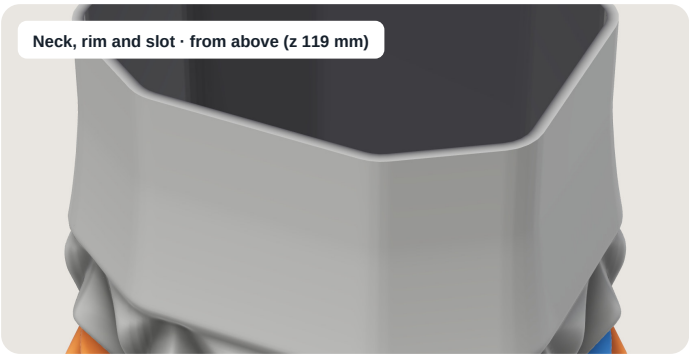
Views & surface

Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.

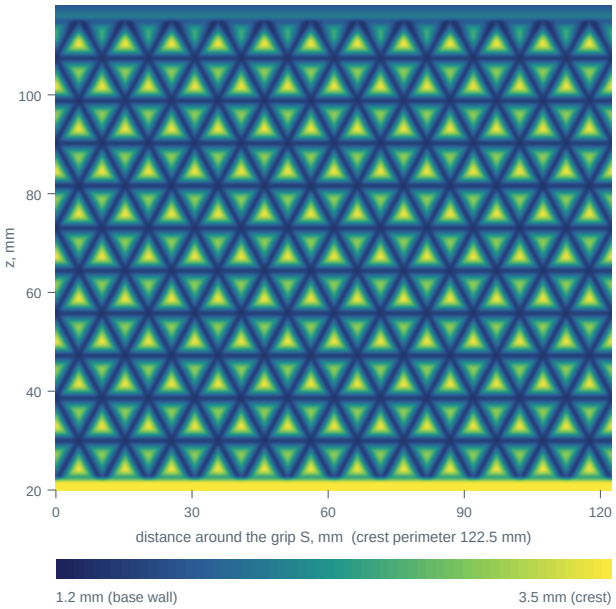


Four views, 90° apart · rendered from the STL, colours are preview only





UNROLLED SURFACE · HEIGHT ABOVE BASE



DESIGN PARAMETERS

Rows	**11**
Triangles per row	**24 (12 up + 12 down)**
Triangle base	**10.21 mm**
Row height	**8.59 mm**
Up-pyramid apex	**3.44 mm above base**
Down-pyramid apex	**3.04 mm above base**
Groove floor	**about 1.4 mm above base**

PRINT & VERIFICATION

Result: slot & hole / print-ready	**PASS / PASS**
Bodies · genus	**1 · 1 (screw hole)**
Paddle slot	**30.2 × 16.2 mm**
Envelope over reference	**+0.000 mm**
Material under 0.4 mm wide	**0.000 %**
Overhangs / bridges	**none**

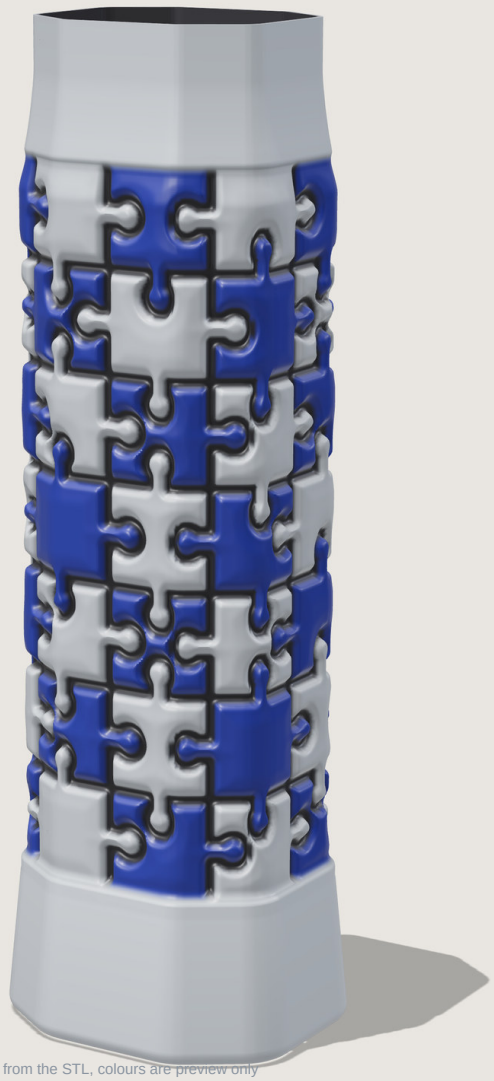
04 LOGIC-GRIP

Seamlessly interlocking ergonomic puzzle design.

89.5 g

PLA · 100 % infill · solid

✓ SOLID · SLOT + SCREW HOLE



3/4 view · rendered from the STL, colours are preview only

AT A GLANCE

Overall height	**132.82 mm**
Max footprint (foot)	**42.63 × 36.61 mm**
Paddle slot	**30.2 × 16.2 mm, 127.8 deep**
Textured band	**z 21.5 – 116.0 mm (94.5 tall)**
Outer size in band (max)	**37.7 × 31.7 mm**
Peak texture height	**2.81 mm above base**
Screw hole (butt)	**Ø 3.5 mm, through z 0 – 5**
Volume / surface area	**72.16 cm³ / 320 cm²**

WEIGHT · 100 % INFILL

PLA	1.24 g/cm³	**89.5 g**
PETG	1.27 g/cm³	**91.6 g**
TPU	1.21 g/cm³	**87.3 g**
ABS	1.04 g/cm³	**75.0 g**

0 — 120 g

COLOURWAY

#323236	#2A3C8A	#9EA4AA

3MF print palette

PRINT PROFILE

Infill / layer height	**100 % / 0.20 mm**
Layers	**665**
Orientation	**upright, butt end down**
Supports	**none required**
Filament, 1.75 mm	**≈ 30.0 m (100 % flow)**

DESIGN INTENT

Fifty-six jigsaw pieces tile the grip in alternating colours. Each piece is a shallow pillow separated by fine grooves and locked to its neighbours by mushroom knobs; knob directions are randomised so no two rows repeat.

Grooves are 0.84 mm wide, about two nozzle widths, and 1.6 mm deep.

KEY FEATURES

- 8 × 7 = 56 pieces in a checkerboard colourway
- Mushroom knobs: 2.9 mm neck, 4.2 mm head, 5.1 mm reach
- 0.84 mm grooves cut down to the base wall

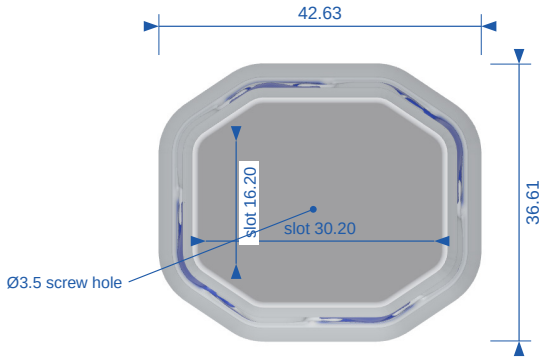
VERIFIED ON THE STL

Paddle slot	**30.2 × 16.2 mm (30 × 16 + 0.2)**
Screw hole	**Ø 3.5 mm through the butt**
Outer envelope vs reference	**+0.000 mm at most**
Bodies / enclosed voids	**1 / none**
Material under 0.4 mm wide	**0.000 % of volume**
Overhangs / bridges	**none**

FILES

stl/04_logic_grip.stl	14.4 MB · 287,530 triangles
3mf/04_logic_grip.3mf	3.1 MB · colour

✓ SLOT & HOLE VERIFIED ✓ SOLID · ONE CLOSED BODY ✓ PRINT-READY · 0.20 MM LAYERS ✓ 3MF VALIDATED

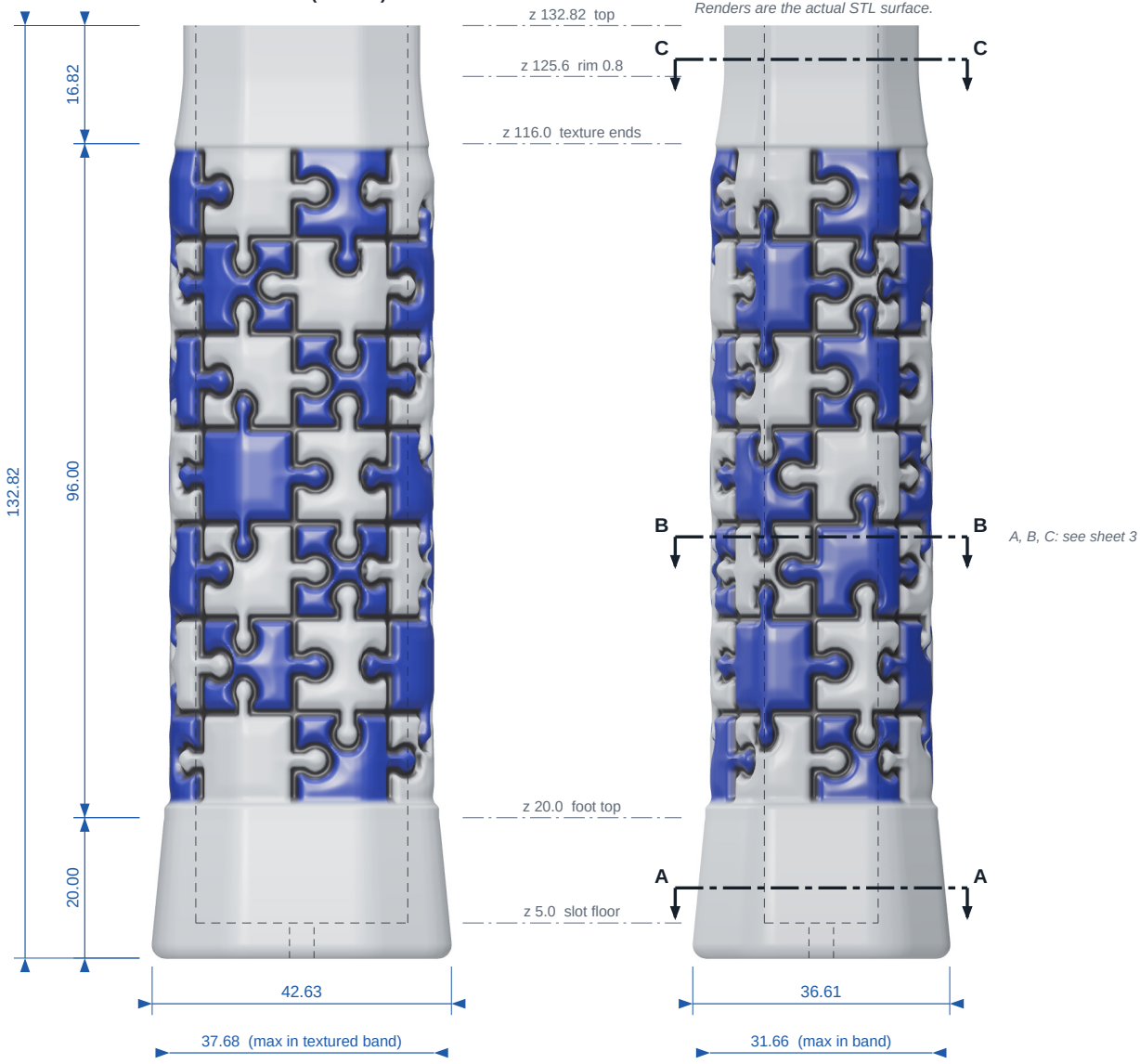


TOP VIEW (from +Z)

KEY DIMENSIONS

Overall height	**132.82**
Overall width × depth (foot)	**42.63 × 36.61**
Paddle slot width × depth	**30.20 × 16.20**
Slot depth (floor at z 5.0)	**127.82**
Outer W × D, textured band (max)	**37.68 × 31.66**
Foot top (z 20)	**39.16 × 33.13**
Butt face (z 0)	**40.56 × 34.54**
Peak texture height above base	**2.81**
Screw hole through butt (z 0 – 5)	**Ø 3.50**

All values in mm. Dashed lines: paddle slot and screw hole (hidden edges). Renders are the actual STL surface.



A, B, C: see sheet 3

FRONT VIEW (from −Y)

RIGHT VIEW (from +X)

SCALE 1 : 1

0 10 20 30 40 50 mm

PART

LOGIC-GRIP

Pickleball paddle grip sleeve, closed butt, open throat

SCALE	UNITS	PROJECTION
1 : 1	millimetres	
MATERIAL	**INFILL**	**LAYER HEIGHT**
PLA / PETG / TPU	100 %	0.20 mm
MASS (PLA)	**VOLUME**	**SURFACE AREA**
89.5 g	72.16 cm³	320 cm²
REVISION	**DATE**	**SHEET**
3	2026-10-17	2 of 4

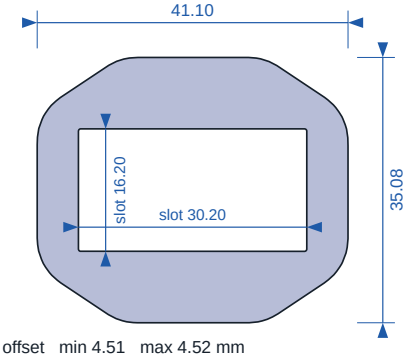
GENERAL NOTES

1. Views are true size: print this sheet at 100 %, do not fit to page.
2. Dimensions are measured from the STL, not nominal.
3. Solid body: only the 30.2 × 16.2 mm paddle slot and the Ø3.5 mm screw hole are open.
4. Outer envelope matches the reference PickleballGrip_1.3mf (rim held ≥ 0.8 mm).

Sections & profiles

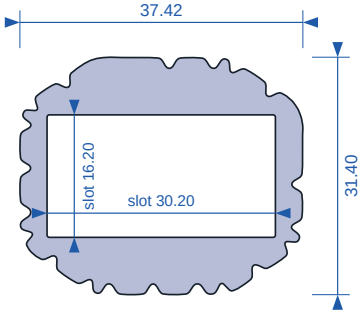
True-size cross-sections, how the outer surface and material change with height, and the dimension tables.

SECTION A–A z = 10 mm · foot



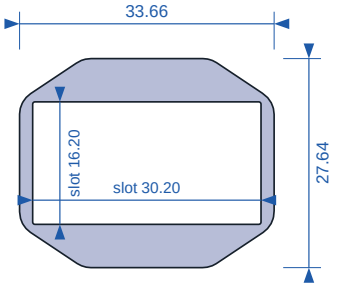
Skin offset min 4.51 max 4.52 mm
Area 780 mm² · outer perimeter 128.9 mm

SECTION B–B z = 60 mm · textured band



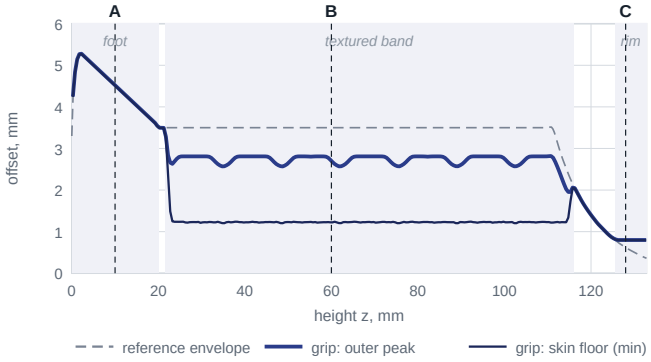
Skin offset min 1.22 max 2.70 mm
Area 508 mm² · outer perimeter 116.7 mm

SECTION C–C z = 128 mm · rim

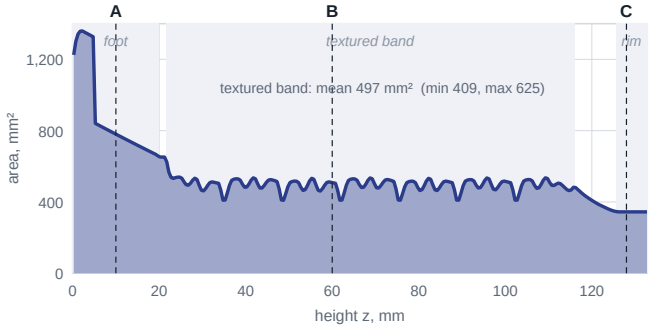


Skin offset min 0.79 max 0.80 mm
Area 344 mm² · outer perimeter 105.5 mm

OUTER OFFSET FROM THE REFERENCE OUTLINE VS HEIGHT



MATERIAL CROSS-SECTION AREA VS HEIGHT



OVERALL & SLOT

Overall length (Z)	**132.82 mm**
Overall width (X)	**42.63 mm**
Overall depth (Y)	**36.61 mm**
Slot width × depth	**30.20 × 16.20 mm**
Slot area / perimeter	**489.2 mm² / 92.3 mm**
Slot depth (floor at z 5.0)	**127.82 mm**
Screw hole (butt, z 0 – 5)	**Ø 3.50 mm**
Butt face (z 0)	**40.56 × 34.54 mm**
Outer W × D in band (max)	**37.68 × 31.66 mm**
Volume / surface area	**72.16 cm³ / 319.7 cm²**

ZONES BY HEIGHT

ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	**5.0**
Flared foot	0.0	20.0	**20.0**
Smooth collar	20.0	21.5	**1.5**
Textured band	21.5	116.0	**94.5**
Smooth neck (taper)	116.0	125.6	**9.6**
Rim (flat top, slot opening)	125.6	132.82	**7.3**

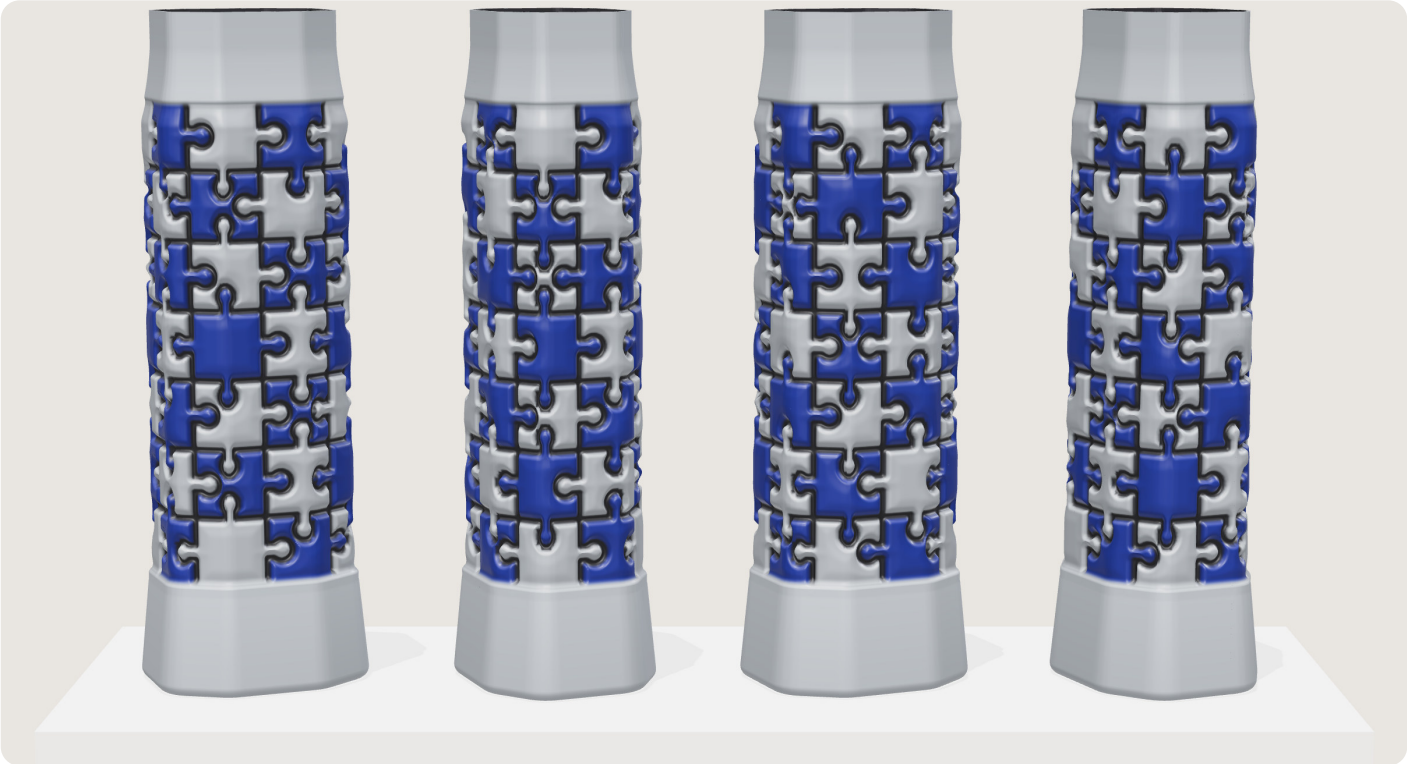
Relief floor above reference line (design / min)	**1.20 / 1.20 mm**
Min skin at the neck (design / measured)	**0.80 / 0.80 mm**
Peak texture height above base	**2.81 mm**

OUTER DIMENSIONS BY HEIGHT

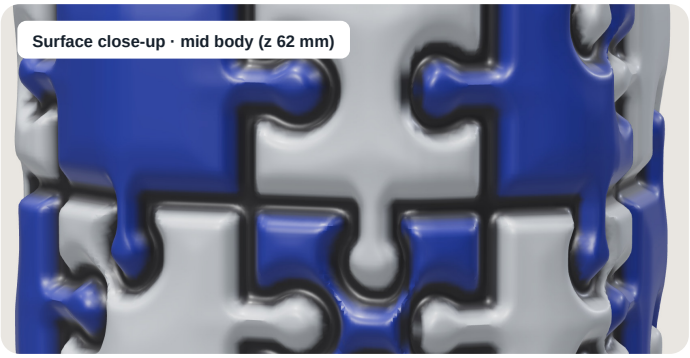
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM²	SKIN MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	1,084	5.00	5.01
10.0	41.10	35.08	128.9	780	4.51	4.52
20.0	39.11	33.09	122.6	655	3.52	3.52
30.0	37.57	31.60	115.1	464	1.22	2.81
45.0	37.59	31.65	117.1	516	1.23	2.81
60.0	37.42	31.40	116.7	509	1.22	2.70
75.0	37.23	31.19	114.9	441	1.21	2.58
90.0	37.24	31.27	115.3	470	1.23	2.61
105.0	37.56	31.50	117.2	524	1.23	2.75
116.0	36.17	30.15	113.4	482	2.04	2.05
125.0	33.77	27.75	105.8	350	0.85	0.85
132.0	33.66	27.64	105.5	344	0.79	0.80

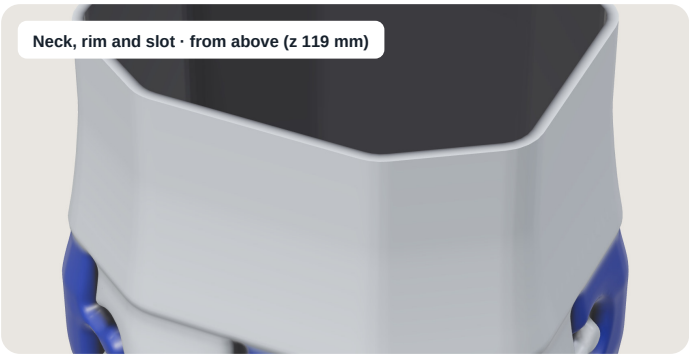
Views & surface

Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.

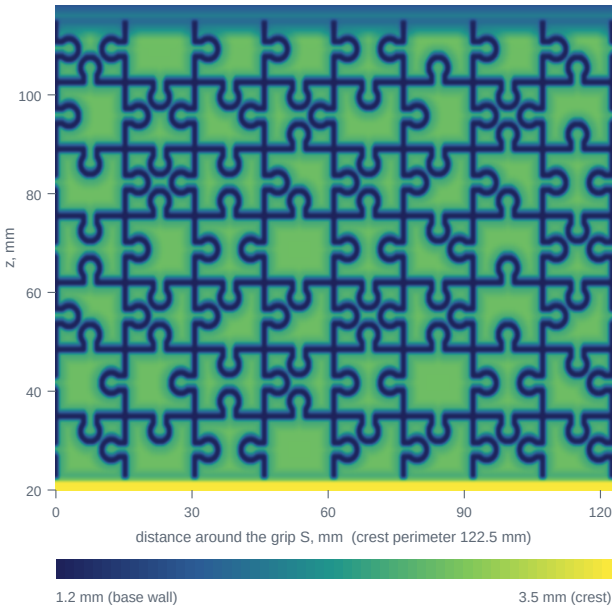


Four views, 90° apart · rendered from the STL, colours are preview only





UNROLLED SURFACE · HEIGHT ABOVE BASE



DESIGN PARAMETERS

Pieces	**8 × 7 = 56**
Piece size	**15.31 × 13.50 mm**
Knob neck / head	**2.9 mm / dia 4.2 mm**
Knob reach	**5.1 mm**
Groove width	**0.84 mm**
Groove floor	**1.20 mm (base wall)**
Piece top height	**up to 2.81 mm above base**

PRINT & VERIFICATION

Result: slot & hole / print-ready	**PASS / PASS**
Bodies · genus	**1 · 1 (screw hole)**
Paddle slot	**30.2 × 16.2 mm**
Envelope over reference	**+0.000 mm**
Material under 0.4 mm wide	**0.000 %**
Overhangs / bridges	**none**

05 NEURO-TREAD

Bio-inspired raised neural network over a dimpled adaptive base.

88.0 g

PLA · 100 % infill · solid

✓ SOLID · SLOT + SCREW HOLE



3/4 view · rendered from the STL, colours are preview only

AT A GLANCE

Overall height	**132.82 mm**
Max footprint (foot)	**42.63 × 36.61 mm**
Paddle slot	**30.2 × 16.2 mm, 127.8 deep**
Textured band	**z 21.5 – 116.0 mm (94.5 tall)**
Outer size in band (max)	**39.1 × 33.0 mm**
Peak texture height	**3.50 mm above base**
Screw hole (butt)	**Ø 3.5 mm, through z 0 – 5**
Volume / surface area	**70.93 cm³ / 320 cm²**

WEIGHT · 100 % INFILL

PLA	1.24 g/cm³	**88.0 g**
PETG	1.27 g/cm³	**90.1 g**
TPU	1.21 g/cm³	**85.8 g**
ABS	1.04 g/cm³	**73.8 g**

0 — 120 g

COLOURWAY

#18181A #CE1820

3MF print palette

PRINT PROFILE

Infill / layer height	**100 % / 0.20 mm**
Layers	**665**
Orientation	**upright, butt end down**
Supports	**none required**
Filament, 1.75 mm	**≈ 29.5 m (100 % flow)**

DESIGN INTENT

A network of 48 neurons joined by curved axon ridges rises to the full 3.5 mm crest, standing proud of a field of small bumps. The ridges guide the fingers while the bump field provides fine grip.

Ridges and bumps are gentle slopes; the network is printed in the same material as the base.

KEY FEATURES

- 48 somas (8 × 6) joined by curved axons about 1.5 mm wide
- Bump field on a 2.66 mm hex pitch
- Network and somas reach the full 3.5 mm crest

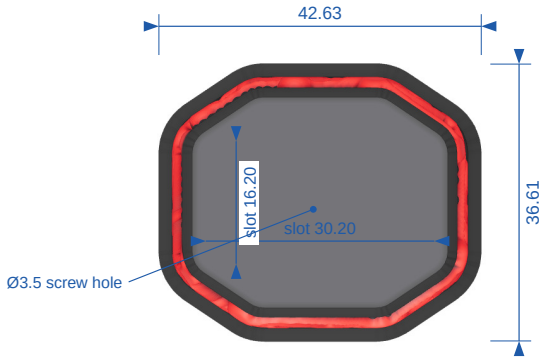
VERIFIED ON THE STL

Paddle slot	**30.2 × 16.2 mm (30 × 16 + 0.2)**
Screw hole	**Ø 3.5 mm through the butt**
Outer envelope vs reference	**+0.000 mm at most**
Bodies / enclosed voids	**1 / none**
Material under 0.4 mm wide	**0.000 % of volume**
Overhangs / bridges	**none**

FILES

stl/05_neuro_tread.stl	18.6 MB · 371,624 triangles
3mf/05_neuro_tread.3mf	4.1 MB · colour

✓ SLOT & HOLE VERIFIED ✓ SOLID · ONE CLOSED BODY ✓ PRINT-READY · 0.20 MM LAYERS ✓ 3MF VALIDATED

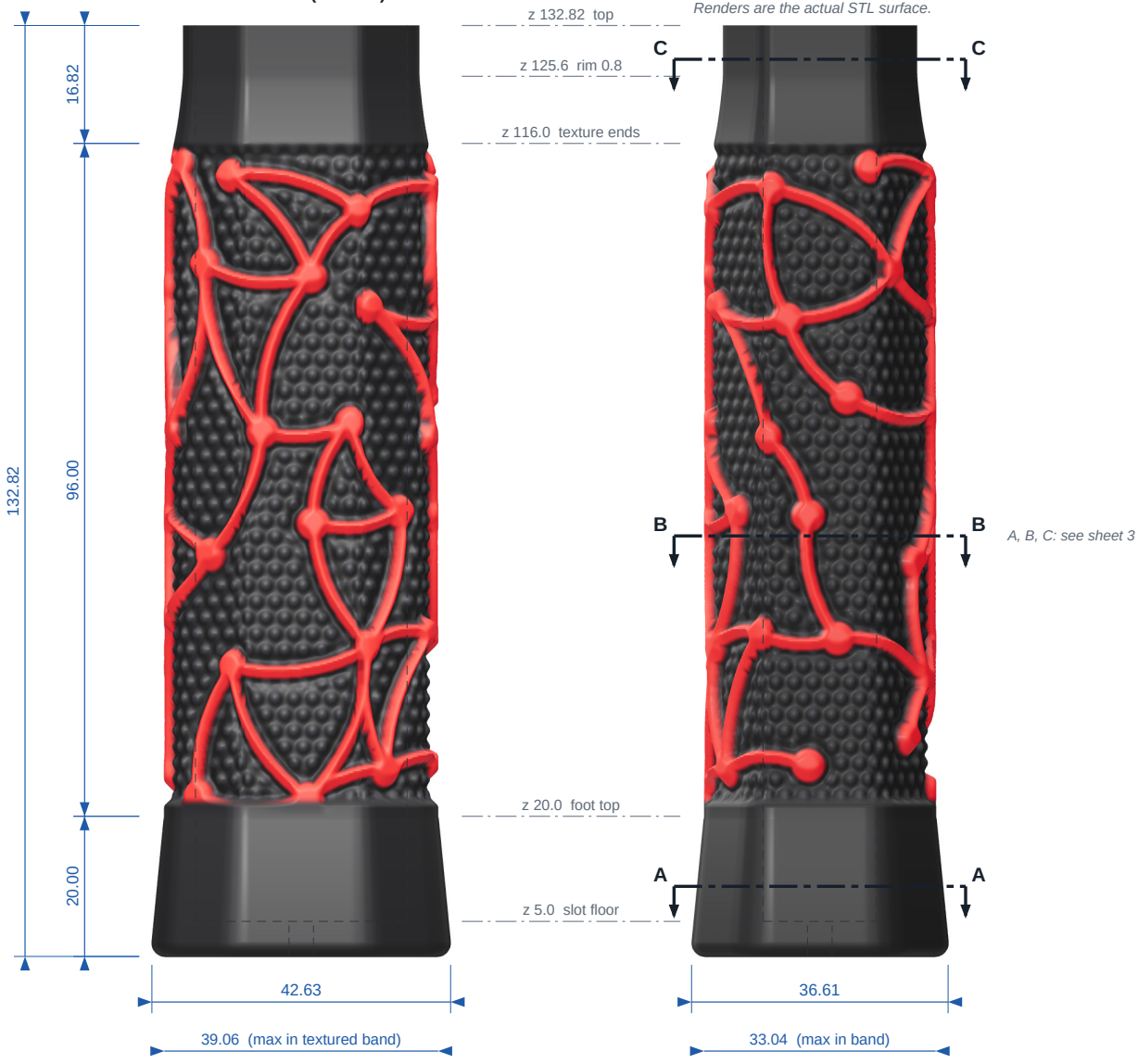


TOP VIEW (from +Z)

KEY DIMENSIONS

Overall height	**132.82**
Overall width × depth (foot)	**42.63 × 36.61**
Paddle slot width × depth	**30.20 × 16.20**
Slot depth (floor at z 5.0)	**127.82**
Outer W × D, textured band (max)	**39.06 × 33.04**
Foot top (z 20)	**39.16 × 33.13**
Butt face (z 0)	**40.56 × 34.54**
Peak texture height above base	**3.50**
Screw hole through butt (z 0 – 5)	**Ø 3.50**

All values in mm. Dashed lines: paddle slot and screw hole (hidden edges). Renders are the actual STL surface.



FRONT VIEW (from −Y)

RIGHT VIEW (from +X)

SCALE 1 : 1

0 10 20 30 40 50 mm

PART

NEURO-TREAD

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

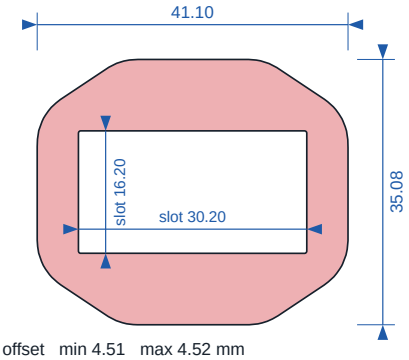
1. Views are true size: print this sheet at 100 %, do not fit to page.
2. Dimensions are measured from the STL, not nominal.
3. Solid body: only the 30.2 × 16.2 mm paddle slot and the Ø3.5 mm screw hole are open.
4. Outer envelope matches the reference PickleballGrip_1.3mf (rim held ≥ 0.8 mm).

SCALE **1 : 1**	UNITS **millimetres**	PROJECTION
MATERIAL **PLA / PETG / TPU**	INFILL **100 %**	LAYER HEIGHT **0.20 mm**
MASS (PLA) **88.0 g**	VOLUME **70.93 cm³**	SURFACE AREA **320 cm²**
REVISION **3**	DATE **2026-10-17**	SHEET **2 of 4**

Sections & profiles

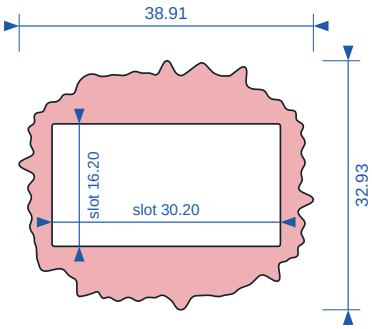
True-size cross-sections, how the outer surface and material change with height, and the dimension tables.

SECTION A–A — z = 10 mm · foot



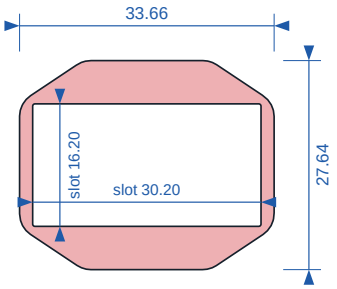
Skin offset min 4.51 max 4.52 mm
Area 780 mm² · outer perimeter 128.9 mm

SECTION B–B — z = 60 mm · textured band



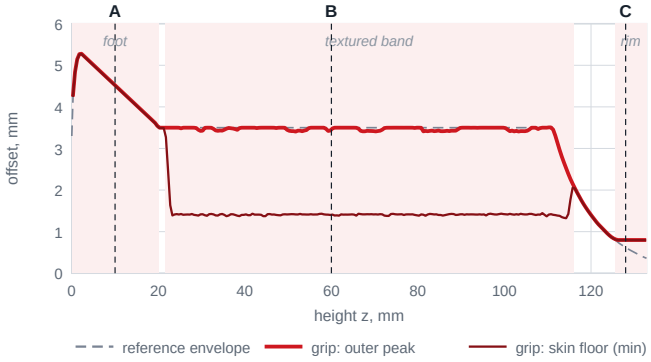
Skin offset min 1.41 max 3.46 mm
Area 480 mm² · outer perimeter 116.9 mm

SECTION C–C — z = 128 mm · rim

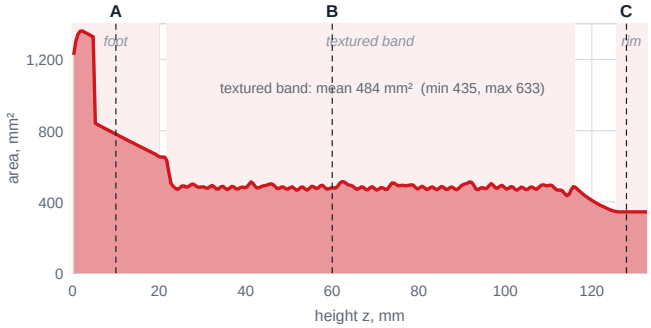


Skin offset min 0.79 max 0.80 mm
Area 344 mm² · outer perimeter 105.5 mm

OUTER OFFSET FROM THE REFERENCE OUTLINE VS HEIGHT



MATERIAL CROSS-SECTION AREA VS HEIGHT



OVERALL & SLOT

Overall length (Z)	**132.82 mm**
Overall width (X)	**42.63 mm**
Overall depth (Y)	**36.61 mm**
Slot width × depth	**30.20 × 16.20 mm**
Slot area / perimeter	**489.2 mm² / 92.3 mm**
Slot depth (floor at z 5.0)	**127.82 mm**
Screw hole (butt, z 0 – 5)	**Ø 3.50 mm**
Butt face (z 0)	**40.56 × 34.54 mm**
Outer W × D in band (max)	**39.06 × 33.04 mm**
Volume / surface area	**70.93 cm³ / 319.6 cm²**

ZONES BY HEIGHT

ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	**5.0**
Flared foot	0.0	20.0	**20.0**
Smooth collar	20.0	21.5	**1.5**
Textured band	21.5	116.0	**94.5**
Smooth neck (taper)	116.0	125.6	**9.6**
Rim (flat top, slot opening)	125.6	132.82	**7.3**

Relief floor above reference line (design / min)	**1.20 / 1.37 mm**
Min skin at the neck (design / measured)	**0.80 / 0.80 mm**
Peak texture height above base	**3.50 mm**

OUTER DIMENSIONS BY HEIGHT

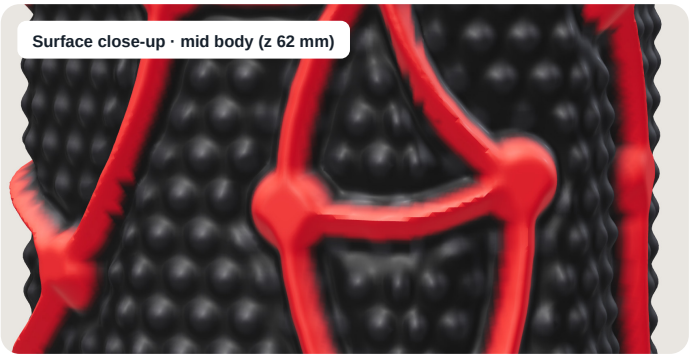
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM²	SKIN MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	1,084	5.00	5.01
10.0	41.10	35.08	128.9	780	4.51	4.52
20.0	39.11	33.09	122.6	655	3.52	3.52
30.0	37.77	31.75	116.8	484	1.42	3.43
45.0	38.99	32.88	117.1	496	1.40	3.50
60.0	38.91	32.93	116.9	479	1.41	3.46
75.0	38.94	32.96	119.3	494	1.42	3.50
90.0	38.99	32.86	117.2	493	1.42	3.50
105.0	38.74	32.79	116.7	470	1.41	3.42
116.0	36.24	30.19	113.5	483	2.05	2.10
125.0	33.77	27.75	105.8	350	0.85	0.85
132.0	33.66	27.64	105.5	344	0.79	0.80

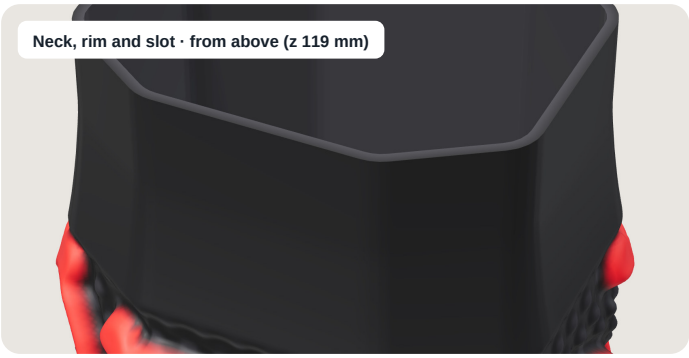
Views & surface

Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.

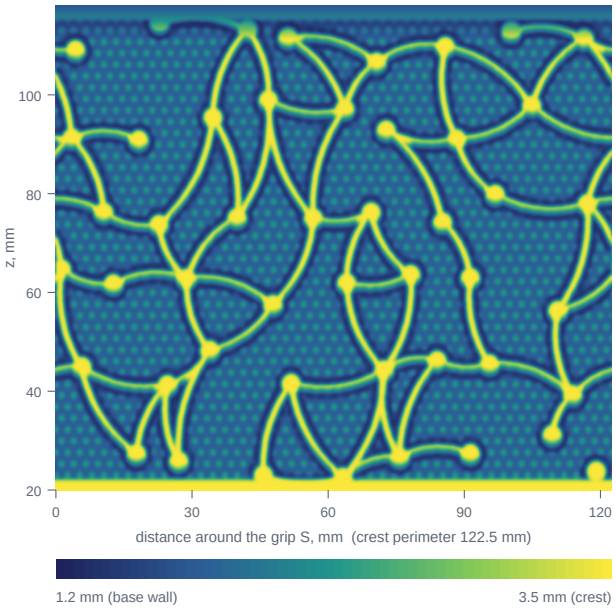


Four views, 90° apart · rendered from the STL, colours are preview only





UNROLLED SURFACE · HEIGHT ABOVE BASE



DESIGN PARAMETERS

Neurons (somas)	**48 (8 × 6)**
Axon ridge width	**about 1.5 mm**
Soma diameter	**about 4.2 mm**
Network height	**3.50 mm above base**
Bump pitch	**2.66 mm (hex)**
Bump field height	**1.4 – 2.4 mm above base**
Ridge / bump colours	**red / black**

PRINT & VERIFICATION

Result: slot & hole / print-ready	**PASS / PASS**
Bodies · genus	**1 · 1 (screw hole)**
Paddle slot	**30.2 × 16.2 mm**
Envelope over reference	**+0.000 mm**
Material under 0.4 mm wide	**0.000 %**
Overhangs / bridges	**none**

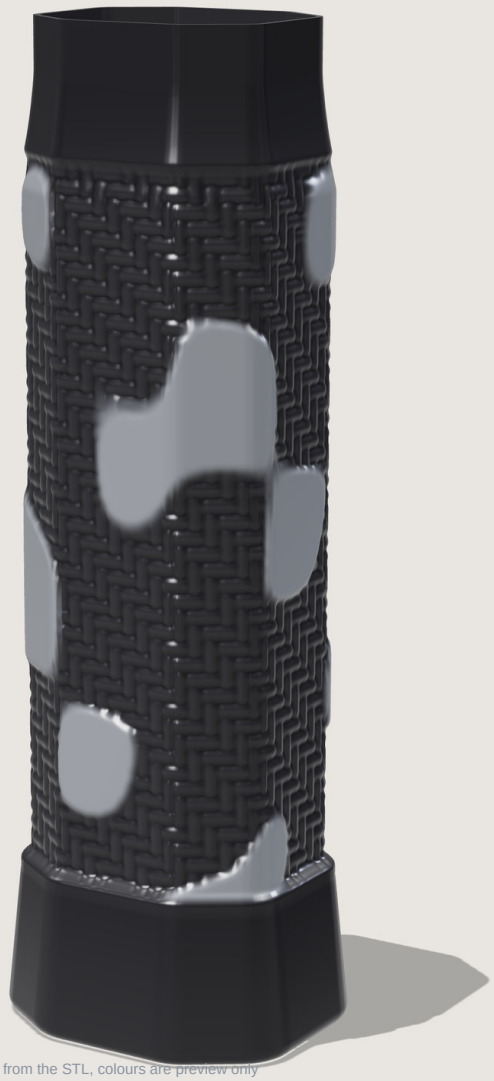
06 CARBON MATRIX

Continuous 2/2 twill weave with selective reinforcement patches.

89.5 g

PLA · 100 % infill · solid

✓ SOLID · SLOT + SCREW HOLE



3/4 view · rendered from the STL, colours are preview only

AT A GLANCE

Overall height	**132.82 mm**
Max footprint (foot)	**42.63 × 36.61 mm**
Paddle slot	**30.2 × 16.2 mm, 127.8 deep**
Textured band	**z 21.5 – 116.0 mm (94.5 tall)**
Outer size in band (max)	**37.5 × 31.4 mm**
Peak texture height	**2.73 mm above base**
Screw hole (butt)	**Ø 3.5 mm, through z 0 – 5**
Volume / surface area	**72.21 cm³ / 296 cm²**

WEIGHT · 100 % INFILL

PLA	1.24 g/cm³	**89.5 g**
PETG	1.27 g/cm³	**91.7 g**
TPU	1.21 g/cm³	**87.4 g**
ABS	1.04 g/cm³	**75.1 g**

0 — 120 g

COLOURWAY

#1C1C20 #80848A

3MF print palette

PRINT PROFILE

Infill / layer height	**100 % / 0.20 mm**
Layers	**665**
Orientation	**upright, butt end down**
Supports	**none required**
Filament, 1.75 mm	**≈ 30.0 m (100 % flow)**

DESIGN INTENT

A woven-carbon look built as real geometry: every tow rises over and dives under its neighbours in a 2/2 twill. Smooth, slightly raised patches suggest selective reinforcement where the palm loads the handle.

The weave is the finest texture in the set (0.4 mm relief on a 2 mm pitch).

KEY FEATURES

- 60 tows around, 2.04 mm pitch, 2/2 twill
- Weave relief about 0.4 mm; every tow is a rounded ridge
- Smooth reinforcement patches stand about 0.5 mm above the weave

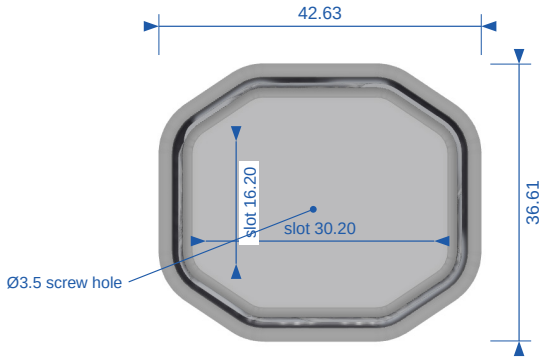
VERIFIED ON THE STL

Paddle slot	**30.2 × 16.2 mm (30 × 16 + 0.2)**
Screw hole	**Ø 3.5 mm through the butt**
Outer envelope vs reference	**+0.000 mm at most**
Bodies / enclosed voids	**1 / none**
Material under 0.4 mm wide	**0.000 % of volume**
Overhangs / bridges	**none**

FILES

stl/06_carbon_matrix.stl	15.4 MB · 308,248 triangles
3mf/06_carbon_matrix.3mf	3.3 MB · colour

✓ SLOT & HOLE VERIFIED ✓ SOLID · ONE CLOSED BODY ✓ PRINT-READY · 0.20 MM LAYERS ✓ 3MF VALIDATED

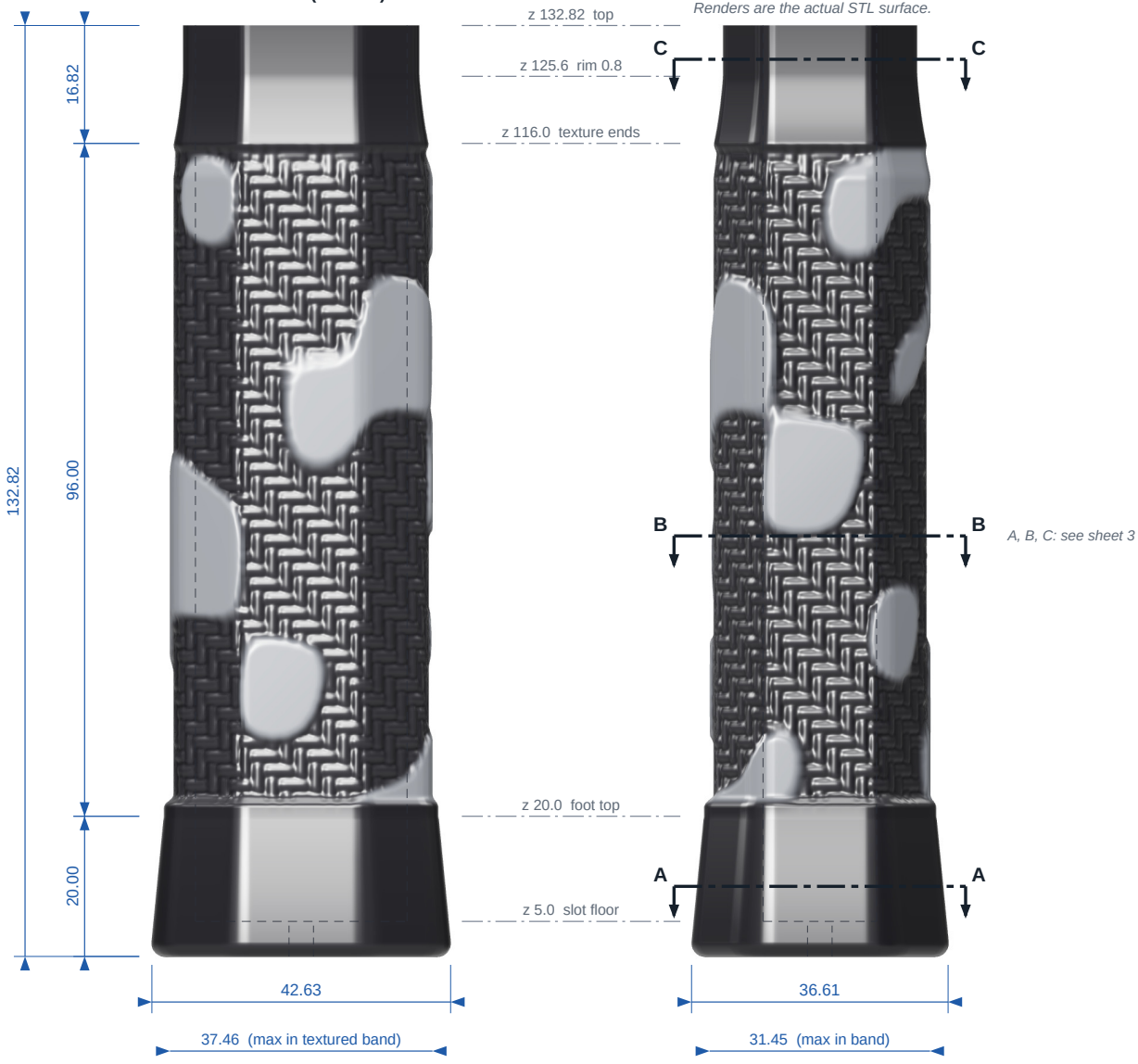


TOP VIEW (from +Z)

KEY DIMENSIONS

Overall height	**132.82**
Overall width × depth (foot)	**42.63 × 36.61**
Paddle slot width × depth	**30.20 × 16.20**
Slot depth (floor at z 5.0)	**127.82**
Outer W × D, textured band (max)	**37.46 × 31.45**
Foot top (z 20)	**39.16 × 33.13**
Butt face (z 0)	**40.56 × 34.54**
Peak texture height above base	**2.73**
Screw hole through butt (z 0 – 5)	**Ø 3.50**

All values in mm. Dashed lines: paddle slot and screw hole (hidden edges). Renders are the actual STL surface.



FRONT VIEW (from −Y)

RIGHT VIEW (from +X)

SCALE 1 : 1

0 10 20 30 40 50 mm

PART

CARBON MATRIX

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

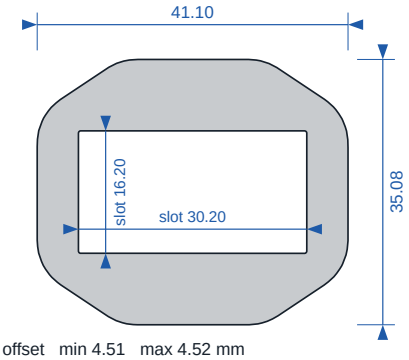
1. Views are true size: print this sheet at 100 %, do not fit to page.
2. Dimensions are measured from the STL, not nominal.
3. Solid body: only the 30.2 × 16.2 mm paddle slot and the Ø3.5 mm screw hole are open.
4. Outer envelope matches the reference PickleballGrip_1.3mf (rim held ≥ 0.8 mm).

SCALE **1 : 1**	UNITS **millimetres**	PROJECTION
MATERIAL **PLA / PETG / TPU**	INFILL **100 %**	LAYER HEIGHT **0.20 mm**
MASS (PLA) **89.5 g**	VOLUME **72.21 cm³**	SURFACE AREA **296 cm²**
REVISION **3**	DATE **2026-10-17**	SHEET **2 of 4**

Sections & profiles

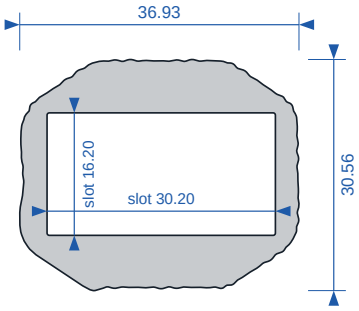
True-size cross-sections, how the outer surface and material change with height, and the dimension tables.

SECTION A–A z = 10 mm · foot



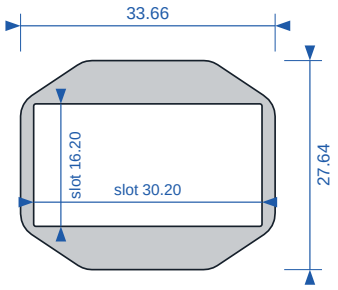
Skin offset min 4.51 max 4.52 mm
Area 780 mm² · outer perimeter 128.9 mm

SECTION B–B z = 60 mm · textured band



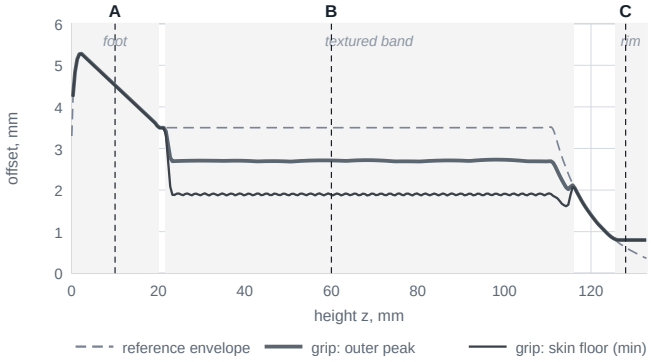
Skin offset min 1.88 max 2.71 mm
Area 491 mm² · outer perimeter 114.4 mm

SECTION C–C z = 128 mm · rim

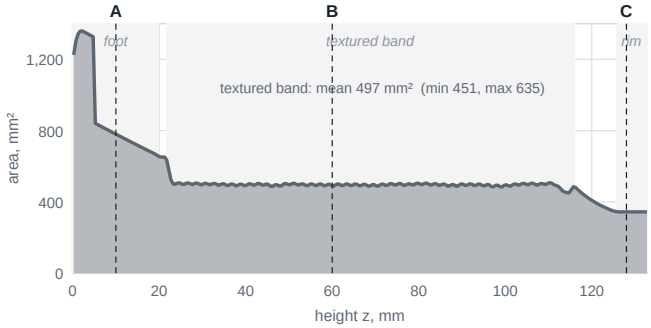


Skin offset min 0.79 max 0.80 mm
Area 344 mm² · outer perimeter 105.5 mm

OUTER OFFSET FROM THE REFERENCE OUTLINE VS HEIGHT



MATERIAL CROSS-SECTION AREA VS HEIGHT



OVERALL & SLOT

Overall length (Z)	**132.82 mm**
Overall width (X)	**42.63 mm**
Overall depth (Y)	**36.61 mm**
Slot width × depth	**30.20 × 16.20 mm**
Slot area / perimeter	**489.2 mm² / 92.3 mm**
Slot depth (floor at z 5.0)	**127.82 mm**
Screw hole (butt, z 0 – 5)	**Ø 3.50 mm**
Butt face (z 0)	**40.56 × 34.54 mm**
Outer W × D in band (max)	**37.46 × 31.45 mm**
Volume / surface area	**72.21 cm³ / 296.5 cm²**

ZONES BY HEIGHT

ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	**5.0**
Flared foot	0.0	20.0	**20.0**
Smooth collar	20.0	21.5	**1.5**
Textured band	21.5	116.0	**94.5**
Smooth neck (taper)	116.0	125.6	**9.6**
Rim (flat top, slot opening)	125.6	132.82	**7.3**

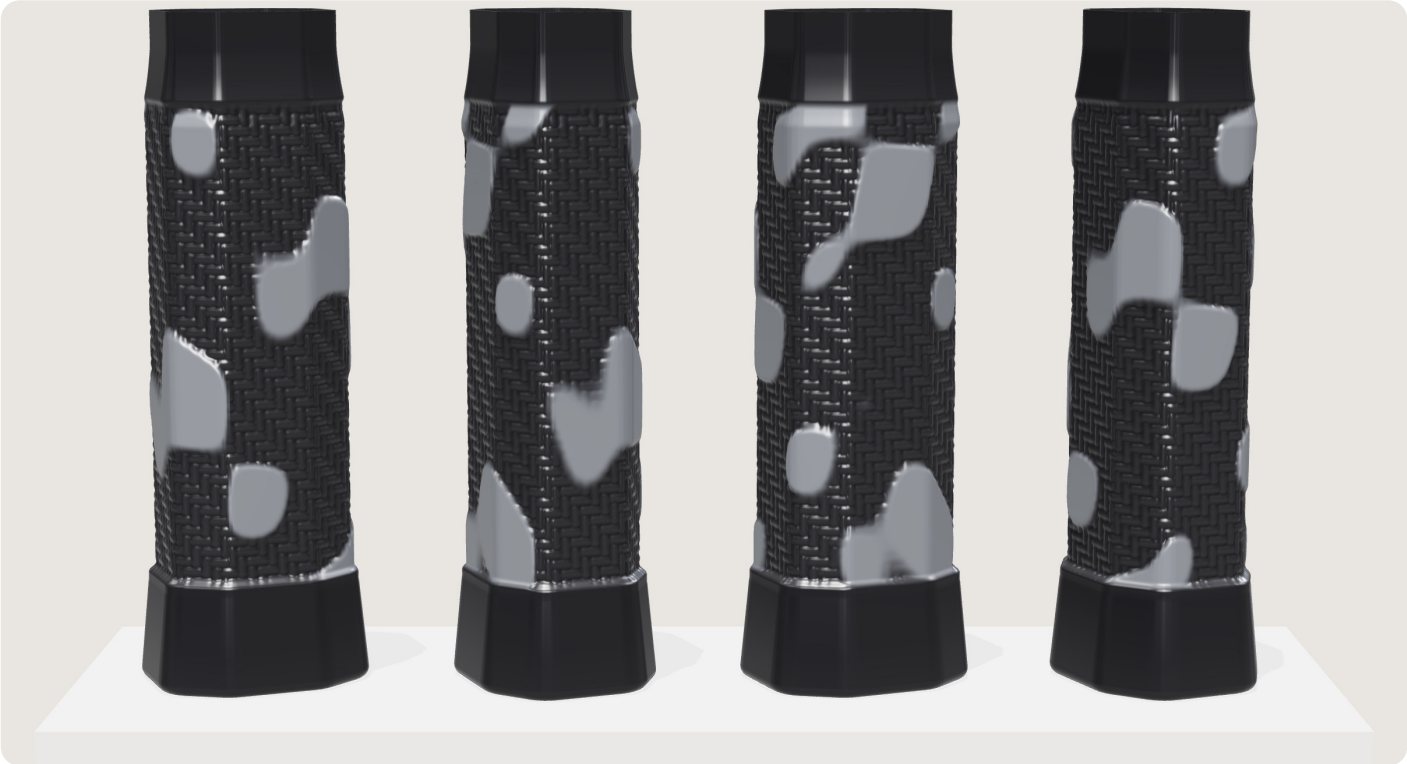
Relief floor above reference line (design / min)	**1.20 / 1.87 mm**
Min skin at the neck (design / measured)	**0.80 / 0.80 mm**
Peak texture height above base	**2.73 mm**

OUTER DIMENSIONS BY HEIGHT

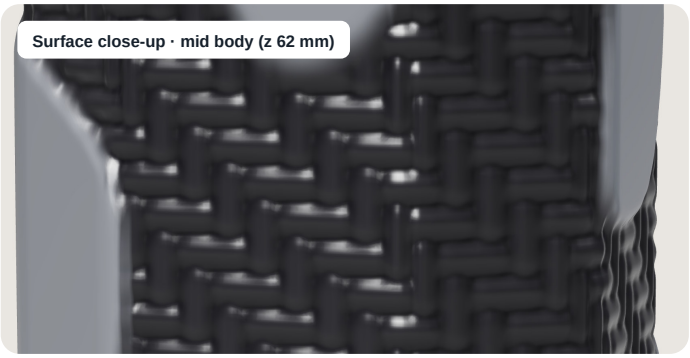
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM²	SKIN MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	1,084	5.00	5.01
10.0	41.10	35.08	128.9	780	4.51	4.52
20.0	39.11	33.09	122.6	655	3.52	3.52
30.0	36.88	30.88	114.8	499	1.89	2.71
45.0	37.26	31.01	114.5	499	1.91	2.69
60.0	36.96	30.56	114.5	492	1.88	2.71
75.0	36.88	30.87	114.5	499	1.89	2.69
90.0	36.78	31.08	114.9	500	1.91	2.71
105.0	36.89	30.87	114.7	498	1.88	2.69
116.0	36.21	30.20	113.5	483	2.06	2.08
125.0	33.77	27.75	105.8	350	0.85	0.85
132.0	33.66	27.64	105.5	344	0.79	0.80

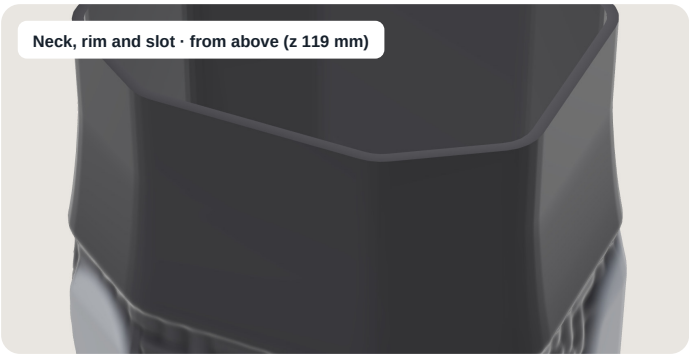
Views & surface

Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.

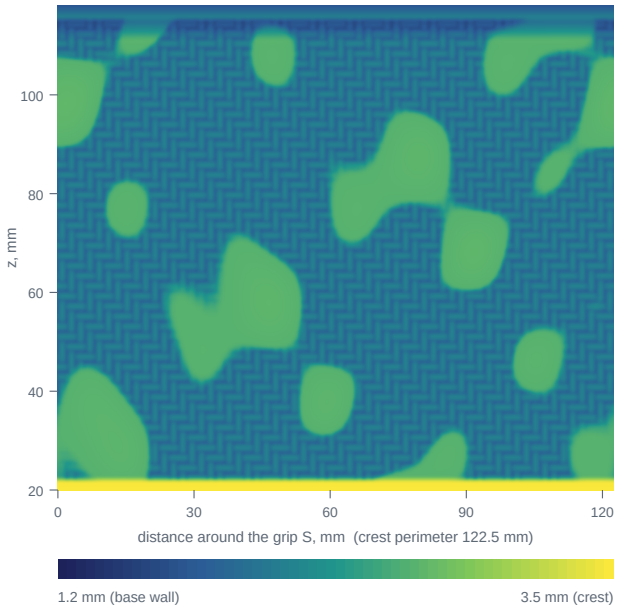


Four views, 90° apart · rendered from the STL, colours are preview only





UNROLLED SURFACE · HEIGHT ABOVE BASE



DESIGN PARAMETERS

Tows around	**60 (2/2 twill)**
Tow pitch	**2.04 mm**
Weave heights	**1.7 – 2.2 mm above base**
Patch heights	**2.3 – 2.7 mm above base**
Patch coverage	**about 23% of the band**
Weave / patch colours	**carbon / grey**
Textured band	**94.5 mm tall**

PRINT & VERIFICATION

Result: slot & hole / print-ready	**PASS / PASS**
Bodies · genus	**1 · 1 (screw hole)**
Paddle slot	**30.2 × 16.2 mm**
Envelope over reference	**+0.000 mm**
Material under 0.4 mm wide	**0.000 %**
Overhangs / bridges	**none**

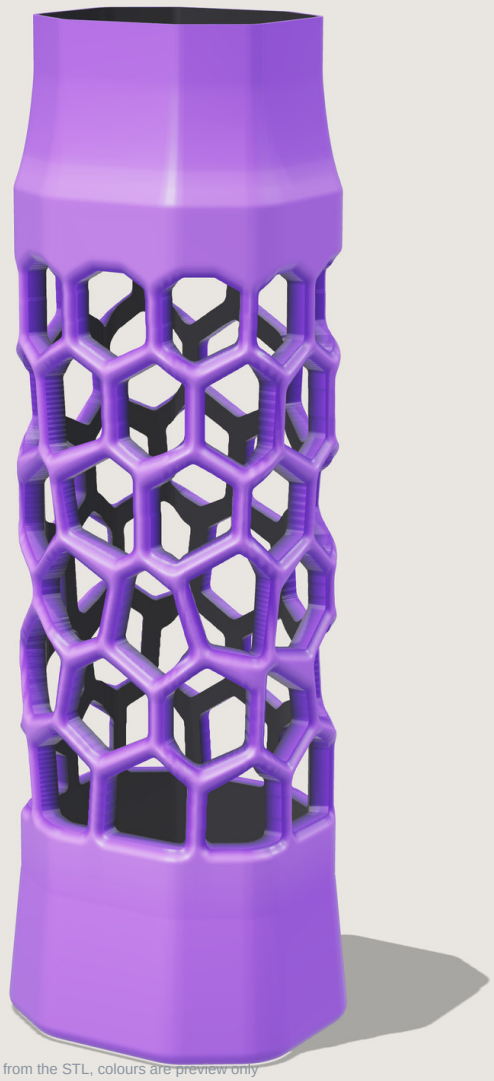
07 VORONOI-CORE

Voronoi rib lattice with recessed cells between rounded ribs.

97.8 g

PLA · 100 % infill · solid

✓ SOLID · SLOT + SCREW HOLE



3/4 view · rendered from the STL, colours are preview only

AT A GLANCE

Overall height	**132.82 mm**
Max footprint (foot)	**42.63 × 36.61 mm**
Paddle slot	**30.2 × 16.2 mm, 127.8 deep**
Textured band	**z 21.5 – 116.0 mm (94.5 tall)**
Outer size in band (max)	**39.1 × 33.0 mm**
Peak texture height	**3.50 mm above base**
Screw hole (butt)	**Ø 3.5 mm, through z 0 – 5**
Volume / surface area	**78.85 cm³ / 303 cm²**

WEIGHT · 100 % INFILL

PLA	1.24 g/cm³	**97.8 g**
PETG	1.27 g/cm³	**100.1 g**
TPU	1.21 g/cm³	**95.4 g**
ABS	1.04 g/cm³	**82.0 g**

0 — 120 g

COLOURWAY

#8A54E4

3MF print palette

PRINT PROFILE

Infill / layer height	**100 % / 0.20 mm**
Layers	**665**
Orientation	**upright, butt end down**
Supports	**none required**
Filament, 1.75 mm	**≈ 32.8 m (100 % flow)**

DESIGN INTENT

Irregular Voronoi cells become 2.6 mm ribs with rounded crests, and the cells between the ribs are recessed about 1 mm. The lattice is relaxed for even cells but keeps a natural irregularity. Like the other nine it is a solid body.

Rib flanks rise at no more than 45 degrees, so no supports are needed.

KEY FEATURES

- 54 recessed cells in the band z 27.5 – 103.5 mm
- 2.6 mm ribs standing at the full 3.5 mm crest
- Cell floors 2.46 mm above base, about 1 mm below the rib crests

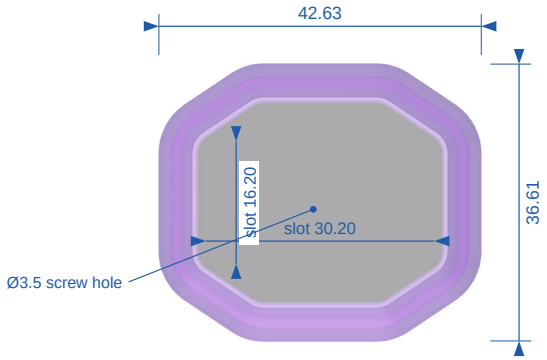
VERIFIED ON THE STL

Paddle slot	**30.2 × 16.2 mm (30 × 16 + 0.2)**
Screw hole	**Ø 3.5 mm through the butt**
Outer envelope vs reference	**+0.003 mm at most**
Bodies / enclosed voids	**1 / none**
Material under 0.4 mm wide	**0.000 % of volume**
Overhangs / bridges	**none**

FILES

stl/07_voronoi_core.stl	7.9 MB · 158,992 triangles
3mf/07_voronoi_core.3mf	1.8 MB · colour

✓ SLOT & HOLE VERIFIED ✓ SOLID · ONE CLOSED BODY ✓ PRINT-READY · 0.20 MM LAYERS ✓ 3MF VALIDATED

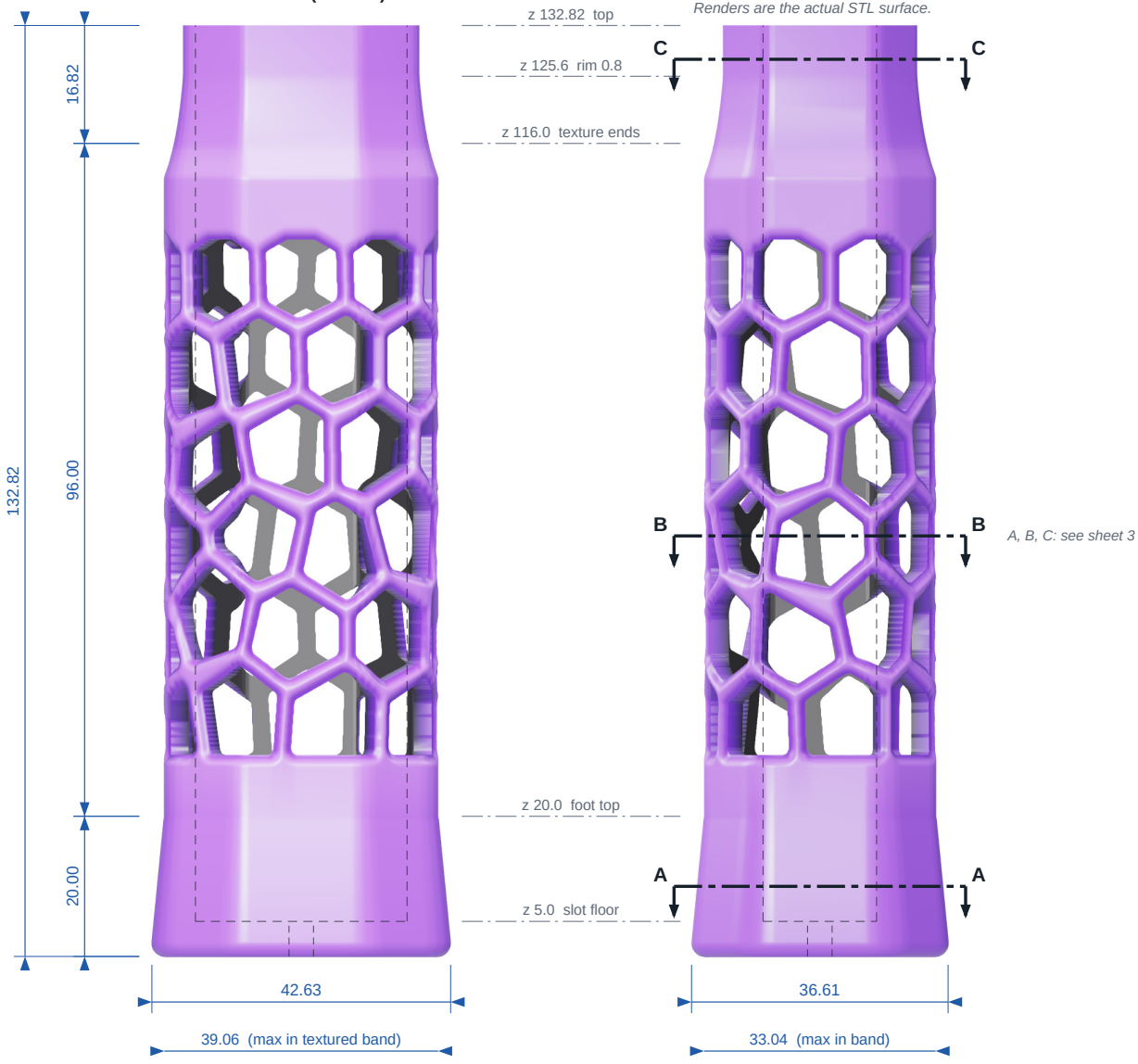


TOP VIEW (from +Z)

KEY DIMENSIONS

Overall height	**132.82**
Overall width × depth (foot)	**42.63 × 36.61**
Paddle slot width × depth	**30.20 × 16.20**
Slot depth (floor at z 5.0)	**127.82**
Outer W × D, textured band (max)	**39.06 × 33.04**
Foot top (z 20)	**39.16 × 33.13**
Butt face (z 0)	**40.56 × 34.54**
Peak texture height above base	**3.50**
Screw hole through butt (z 0 – 5)	**Ø 3.50**

All values in mm. Dashed lines: paddle slot and screw hole (hidden edges). Renders are the actual STL surface.



FRONT VIEW (from −Y)

RIGHT VIEW (from +X)

SCALE 1 : 1



PART

VORONOI-CORE

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

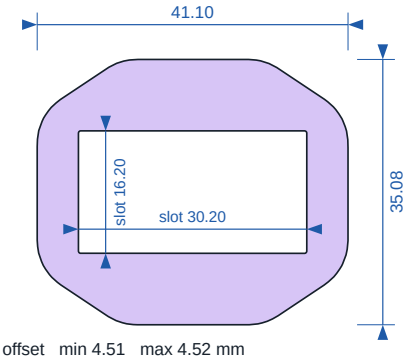
1. Views are true size: print this sheet at 100 %, do not fit to page.
2. Dimensions are measured from the STL, not nominal.
3. Solid body: only the 30.2 × 16.2 mm paddle slot and the Ø3.5 mm screw hole are open.
4. Outer envelope matches the reference PickleballGrip_1.3mf (rim held ≥ 0.8 mm).

SCALE **1 : 1**	UNITS **millimetres**	PROJECTION
MATERIAL **PLA / PETG / TPU**	INFILL **100 %**	LAYER HEIGHT **0.20 mm**
MASS (PLA) **97.8 g**	VOLUME **78.85 cm³**	SURFACE AREA **303 cm²**
REVISION **3**	DATE **2026-10-17**	SHEET **2 of 4**

Sections & profiles

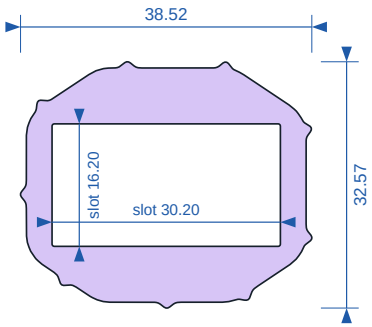
True-size cross-sections, how the outer surface and material change with height, and the dimension tables.

SECTION A–A z = 10 mm · foot



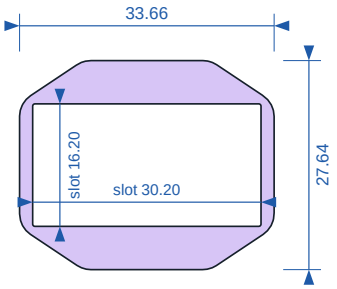
Skin offset min 4.51 max 4.52 mm
Area 780 mm² · outer perimeter 128.9 mm

SECTION B–B z = 60 mm · textured band



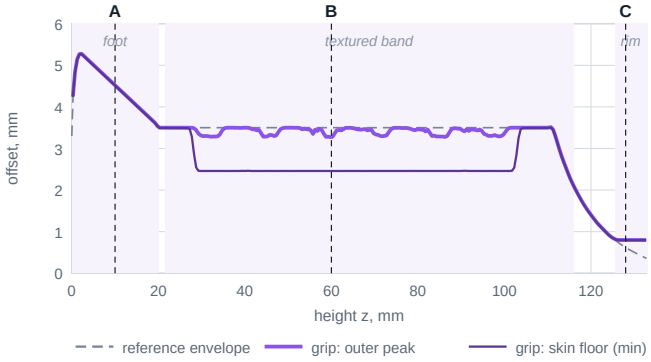
Skin offset min 2.46 max 3.28 mm
Area 539 mm² · outer perimeter 118.3 mm

SECTION C–C z = 128 mm · rim

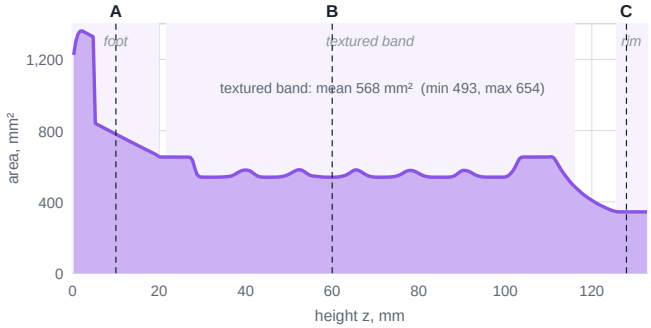


Skin offset min 0.79 max 0.80 mm
Area 344 mm² · outer perimeter 105.5 mm

OUTER OFFSET FROM THE REFERENCE OUTLINE VS HEIGHT



MATERIAL CROSS-SECTION AREA VS HEIGHT



OVERALL & SLOT

Overall length (Z)	**132.82 mm**
Overall width (X)	**42.63 mm**
Overall depth (Y)	**36.61 mm**
Slot width × depth	**30.20 × 16.20 mm**
Slot area / perimeter	**489.2 mm² / 92.3 mm**
Slot depth (floor at z 5.0)	**127.82 mm**
Screw hole (butt, z 0 – 5)	**Ø 3.50 mm**
Butt face (z 0)	**40.56 × 34.54 mm**
Outer W × D in band (max)	**39.06 × 33.04 mm**
Volume / surface area	**78.85 cm³ / 302.8 cm²**

ZONES BY HEIGHT

ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	**5.0**
Flared foot	0.0	20.0	**20.0**
Smooth collar	20.0	21.5	**1.5**
Textured band	21.5	116.0	**94.5**
Smooth neck (taper)	116.0	125.6	**9.6**
Rim (flat top, slot opening)	125.6	132.82	**7.3**

Relief floor above reference line (design / min)	**1.20 / 2.46 mm**
Min skin at the neck (design / measured)	**0.80 / 0.80 mm**
Peak texture height above base	**3.50 mm**

OUTER DIMENSIONS BY HEIGHT

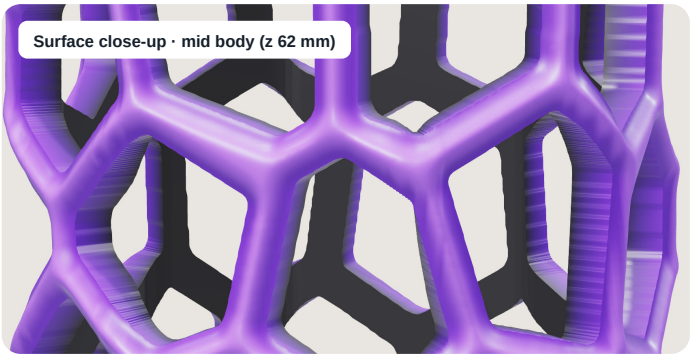
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM²	SKIN MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	1,084	5.00	5.01
10.0	41.10	35.08	128.9	780	4.51	4.52
20.0	39.11	33.09	122.6	655	3.52	3.52
30.0	38.63	32.65	118.1	539	2.46	3.31
45.0	38.57	32.62	118.3	539	2.46	3.29
60.0	38.52	32.57	118.3	539	2.46	3.28
75.0	38.81	32.61	118.5	546	2.46	3.50
90.0	38.91	32.86	120.5	575	2.46	3.46
105.0	39.06	33.04	122.5	652	3.49	3.50
116.0	36.27	30.24	113.7	487	2.10	2.10
125.0	33.77	27.75	105.8	350	0.85	0.85
132.0	33.66	27.64	105.5	344	0.79	0.80

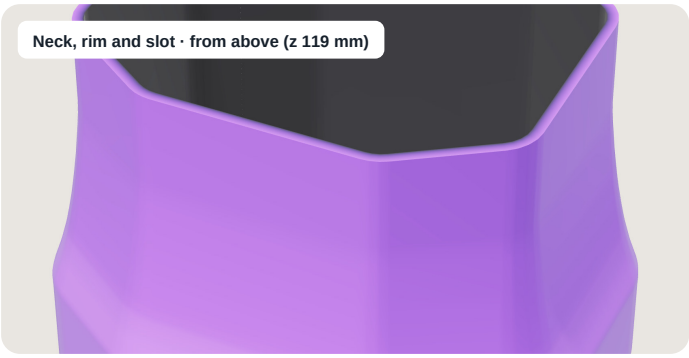
Views & surface

Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.

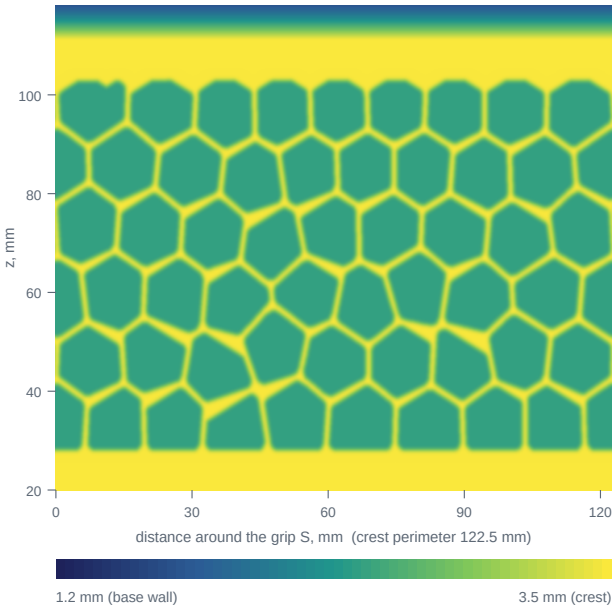


Four views, 90° apart · rendered from the STL, colours are preview only





UNROLLED SURFACE · HEIGHT ABOVE BASE



DESIGN PARAMETERS

Cells	**54, recessed about 1 mm**
Rib width	**2.6 mm**
Rib height	**3.50 mm above base**
Textured band	**z 27.5 – 103.5 mm (76 mm)**
Cell corner radius	**1.0 mm**
Cell floor	**2.46 mm above base**
Recessed area	**about 64% of the band**

PRINT & VERIFICATION

Result: slot & hole / print-ready	**PASS / PASS**
Bodies · genus	**1 · 1 (screw hole)**
Paddle slot	**30.2 × 16.2 mm**
Envelope over reference	**+0.003 mm**
Material under 0.4 mm wide	**0.000 %**
Overhangs / bridges	**none**

08 TOPO-FLOW

Continuous undulating contours mimicking wood grain and terrain.

89.2 g

PLA · 100 % infill · solid

✓ SOLID · SLOT + SCREW HOLE



3/4 view · rendered from the STL, colours are preview only

AT A GLANCE

Overall height	**132.82 mm**
Max footprint (foot)	**42.63 × 36.61 mm**
Paddle slot	**30.2 × 16.2 mm, 127.8 deep**
Textured band	**z 21.5 – 116.0 mm (94.5 tall)**
Outer size in band (max)	**38.3 × 32.1 mm**
Peak texture height	**3.13 mm above base**
Screw hole (butt)	**Ø 3.5 mm, through z 0 – 5**
Volume / surface area	**71.96 cm³ / 309 cm²**

WEIGHT · 100 % INFILL

PLA	1.24 g/cm³	**89.2 g**
PETG	1.27 g/cm³	**91.4 g**
TPU	1.21 g/cm³	**87.1 g**
ABS	1.04 g/cm³	**74.8 g**

0 — 120 g

COLOURWAY

#D6AC80 #A8744E #784828

3MF print palette

PRINT PROFILE

Infill / layer height	**100 % / 0.20 mm**
Layers	**665**
Orientation	**upright, butt end down**
Supports	**none required**
Filament, 1.75 mm	**≈ 29.9 m (100 % flow)**

DESIGN INTENT

The surface is a contour map of terrain with four wood-like knots. Ridges follow the contours so the lines flow around the handle like grain in a cut log, giving directional grip that is softer on the hand than sharp knurling.

Ridge flanks are rounded and below 45 degrees; the grain is geometry, not colour alone.

KEY FEATURES

- Four knots, contour ridges spaced about 2.5 – 7 mm
- Rounded ridges: crests about 3.1 mm, valleys 1.66 mm above the base
- Three-tone wood colourway (light / mid / dark)

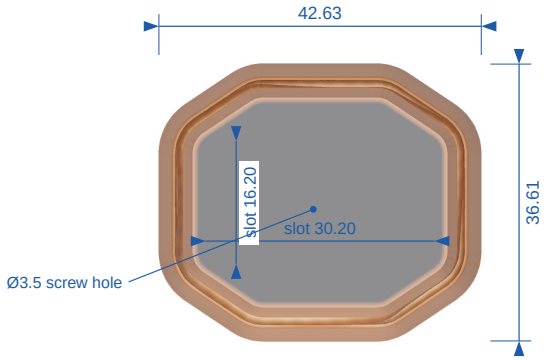
VERIFIED ON THE STL

Paddle slot	**30.2 × 16.2 mm (30 × 16 + 0.2)**
Screw hole	**Ø 3.5 mm through the butt**
Outer envelope vs reference	**+0.000 mm at most**
Bodies / enclosed voids	**1 / none**
Material under 0.4 mm wide	**0.000 % of volume**
Overhangs / bridges	**none**

FILES

stl/08_topo_flow.stl	17.0 MB · 339,714 triangles
3mf/08_topo_flow.3mf	3.8 MB · colour

✓ SLOT & HOLE VERIFIED ✓ SOLID · ONE CLOSED BODY ✓ PRINT-READY · 0.20 MM LAYERS ✓ 3MF VALIDATED

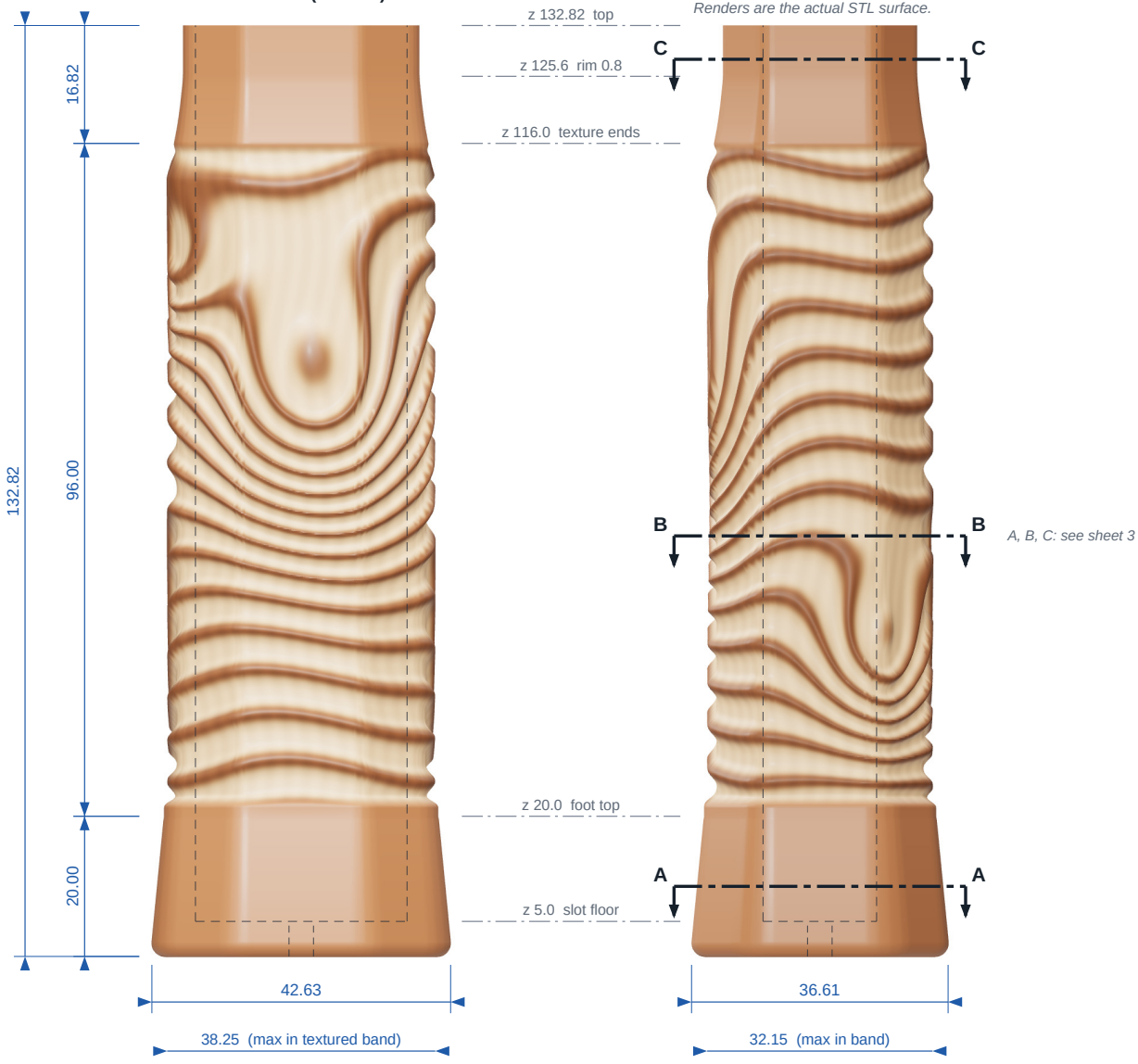


TOP VIEW (from +Z)

KEY DIMENSIONS

Overall height	**132.82**
Overall width × depth (foot)	**42.63 × 36.61**
Paddle slot width × depth	**30.20 × 16.20**
Slot depth (floor at z 5.0)	**127.82**
Outer W × D, textured band (max)	**38.25 × 32.15**
Foot top (z 20)	**39.16 × 33.13**
Butt face (z 0)	**40.56 × 34.54**
Peak texture height above base	**3.13**
Screw hole through butt (z 0 – 5)	**Ø 3.50**

All values in mm. Dashed lines: paddle slot and screw hole (hidden edges). Renders are the actual STL surface.



FRONT VIEW (from −Y)

RIGHT VIEW (from +X)

SCALE 1 : 1



PART

TOPO-FLOW

Pickleball paddle grip sleeve, closed butt, open throat

SCALE	UNITS	PROJECTION
1 : 1	millimetres	
MATERIAL	**INFILL**	**LAYER HEIGHT**
PLA / PETG / TPU	100 %	0.20 mm
MASS (PLA)	**VOLUME**	**SURFACE AREA**
89.2 g	71.96 cm³	309 cm²
REVISION	**DATE**	**SHEET**
3	2026-10-17	2 of 4

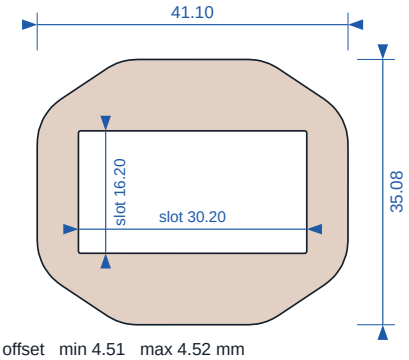
GENERAL NOTES

1. Views are true size: print this sheet at 100 %, do not fit to page.
2. Dimensions are measured from the STL, not nominal.
3. Solid body: only the 30.2 × 16.2 mm paddle slot and the Ø3.5 mm screw hole are open.
4. Outer envelope matches the reference PickleballGrip_1.3mf (rim held ≥ 0.8 mm).

Sections & profiles

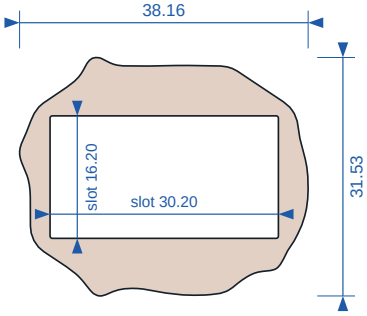
True-size cross-sections, how the outer surface and material change with height, and the dimension tables.

SECTION A–A z = 10 mm · foot



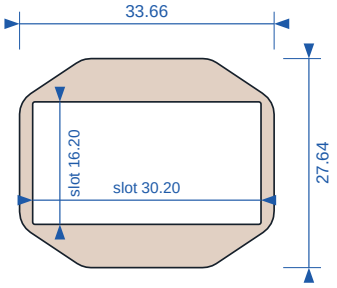
Skin offset min 4.51 max 4.52 mm
Area 780 mm² · outer perimeter 128.9 mm

SECTION B–B z = 60 mm · textured band



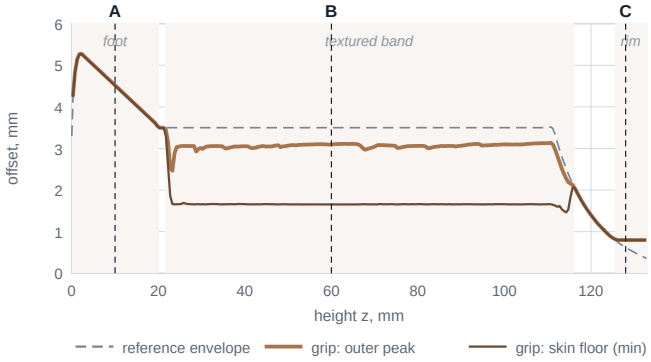
Skin offset min 1.65 max 3.10 mm
Area 479 mm² · outer perimeter 114.1 mm

SECTION C–C z = 128 mm · rim

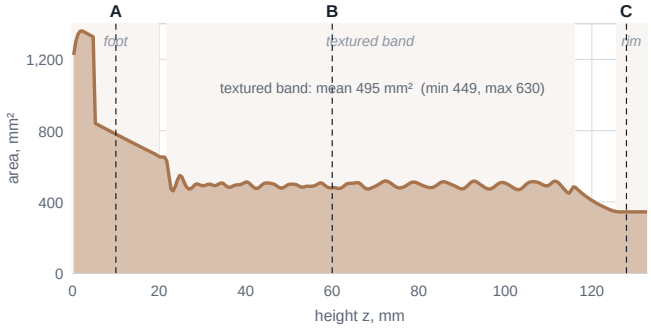


Skin offset min 0.79 max 0.80 mm
Area 344 mm² · outer perimeter 105.5 mm

OUTER OFFSET FROM THE REFERENCE OUTLINE VS HEIGHT



MATERIAL CROSS-SECTION AREA VS HEIGHT



OVERALL & SLOT

Overall length (Z)	**132.82 mm**
Overall width (X)	**42.63 mm**
Overall depth (Y)	**36.61 mm**
Slot width × depth	**30.20 × 16.20 mm**
Slot area / perimeter	**489.2 mm² / 92.3 mm**
Slot depth (floor at z 5.0)	**127.82 mm**
Screw hole (butt, z 0 – 5)	**Ø 3.50 mm**
Butt face (z 0)	**40.56 × 34.54 mm**
Outer W × D in band (max)	**38.25 × 32.15 mm**
Volume / surface area	**71.96 cm³ / 309.3 cm²**

ZONES BY HEIGHT

ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	**5.0**
Flared foot	0.0	20.0	**20.0**
Smooth collar	20.0	21.5	**1.5**
Textured band	21.5	116.0	**94.5**
Smooth neck (taper)	116.0	125.6	**9.6**
Rim (flat top, slot opening)	125.6	132.82	**7.3**

Relief floor above reference line (design / min)	**1.20 / 1.65 mm**
Min skin at the neck (design / measured)	**0.80 / 0.80 mm**
Peak texture height above base	**3.13 mm**

OUTER DIMENSIONS BY HEIGHT

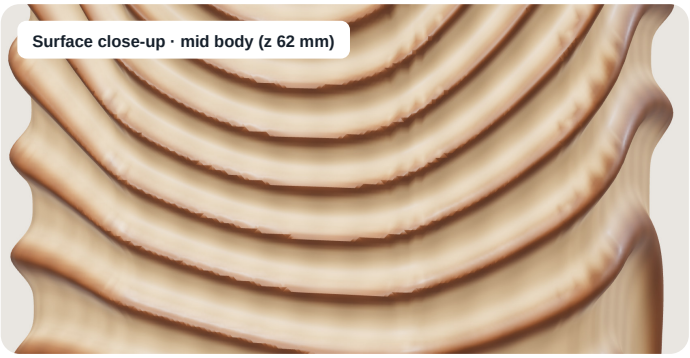
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM²	SKIN MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	1,084	5.00	5.01
10.0	41.10	35.08	128.9	780	4.51	4.52
20.0	39.11	33.09	122.6	655	3.52	3.52
30.0	37.98	31.02	113.6	491	1.66	3.00
45.0	38.00	32.05	117.4	506	1.66	3.05
60.0	38.13	31.60	114.1	479	1.65	3.10
75.0	37.02	31.88	115.4	482	1.66	3.04
90.0	37.70	31.13	115.4	474	1.66	3.07
105.0	38.18	29.98	116.0	502	1.66	3.11
116.0	36.19	30.19	113.4	483	2.05	2.09
125.0	33.77	27.75	105.8	350	0.85	0.85
132.0	33.66	27.64	105.5	344	0.79	0.80

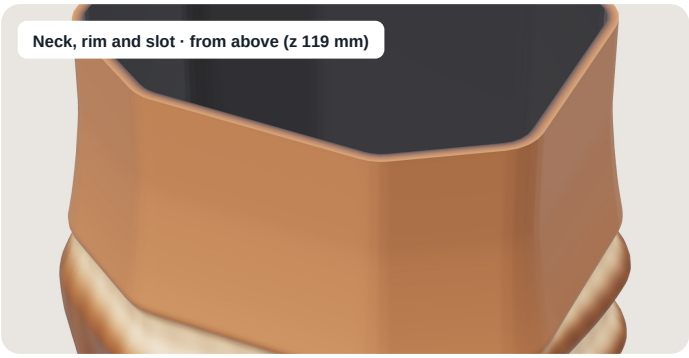
Views & surface

Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.

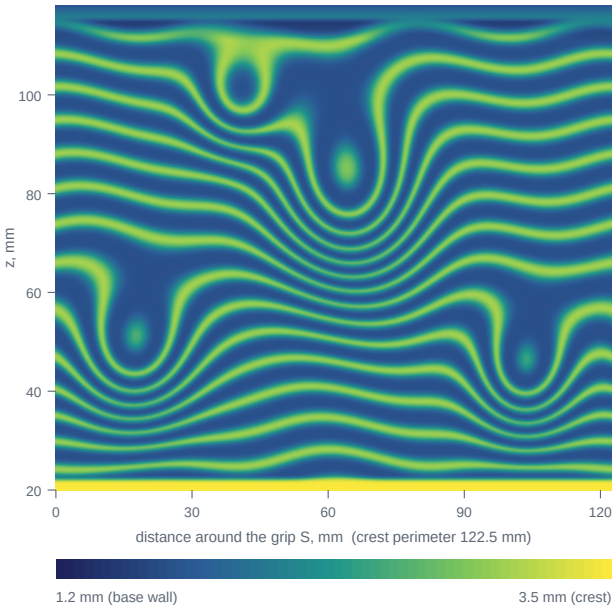


Four views, 90° apart · rendered from the STL, colours are preview only





UNROLLED SURFACE · HEIGHT ABOVE BASE



DESIGN PARAMETERS

Knots	**4**
Ridge spacing	**about 2.5 – 7 mm**
Ridge crest	**about 3.1 mm above base**
Valley floor	**1.66 mm above base**
Relief	**about 1.4 mm**
Textured band	**z 21.5 – 116.0 mm**
Colourway	**3 wood tones**

PRINT & VERIFICATION

Result: slot & hole / print-ready	**PASS / PASS**
Bodies · genus	**1 · 1 (screw hole)**
Paddle slot	**30.2 × 16.2 mm**
Envelope over reference	**+0.000 mm**
Material under 0.4 mm wide	**0.000 %**
Overhangs / bridges	**none**

09 ERGO-CONTOUR

Palm swell, four finger flutes and a thumb rest for hand conformance.

88.4 g

PLA · 100 % infill · solid

✓ SOLID · SLOT + SCREW HOLE



3/4 view · rendered from the STL, colours are preview only

AT A GLANCE

Overall height	**132.82 mm**
Max footprint (foot)	**42.63 × 36.61 mm**
Paddle slot	**30.2 × 16.2 mm, 127.8 deep**
Textured band	**z 21.5 – 116.0 mm (94.5 tall)**
Outer size in band (max)	**36.9 × 30.9 mm**
Peak texture height	**2.77 mm above base**
Screw hole (butt)	**Ø 3.5 mm, through z 0 – 5**
Volume / surface area	**71.33 cm³ / 293 cm²**

WEIGHT · 100 % INFILL

PLA	1.24 g/cm³		**88.4 g**
PETG	1.27 g/cm³		**90.6 g**
TPU	1.21 g/cm³		**86.3 g**
ABS	1.04 g/cm³		**74.2 g**

0 — 120 g

COLOURWAY

#424248

3MF print palette

PRINT PROFILE

Infill / layer height	**100 % / 0.20 mm**
Layers	**665**
Orientation	**upright, butt end down**
Supports	**none required**
Filament, 1.75 mm	**≈ 29.7 m (100 % flow)**

DESIGN INTENT

A smooth grip shaped for the hand rather than textured: an hourglass swell, a raised palm pad on the back, four finger flutes and a thumb dish on the front. There are no sharp edges anywhere on the surface.

The smoothest surface in the set; it prints with the least visible layer texture.

KEY FEATURES

- Four finger flutes at z = 44, 58, 72, 86 mm
- Thumb dish at z about 97 mm, palm pad at z about 62 mm
- Hourglass swell 1.73 – 2.33 mm, peaking at about 2.8 mm on the palm pad

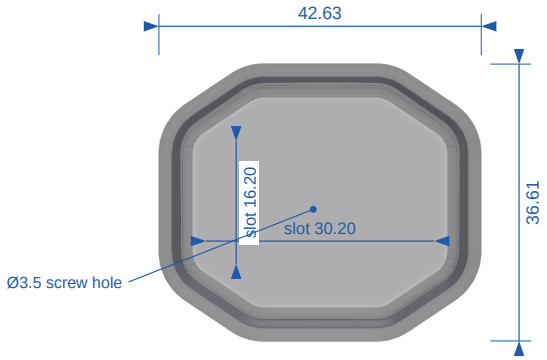
VERIFIED ON THE STL

Paddle slot	**30.2 × 16.2 mm (30 × 16 + 0.2)**
Screw hole	**Ø 3.5 mm through the butt**
Outer envelope vs reference	**+0.000 mm at most**
Bodies / enclosed voids	**1 / none**
Material under 0.4 mm wide	**0.000 % of volume**
Overhangs / bridges	**none**

FILES

stl/09_ergo_contour.stl	2.1 MB · 42,162 triangles
3mf/09_ergo_contour.3mf	0.5 MB · colour

✓ SLOT & HOLE VERIFIED ✓ SOLID · ONE CLOSED BODY ✓ PRINT-READY · 0.20 MM LAYERS ✓ 3MF VALIDATED

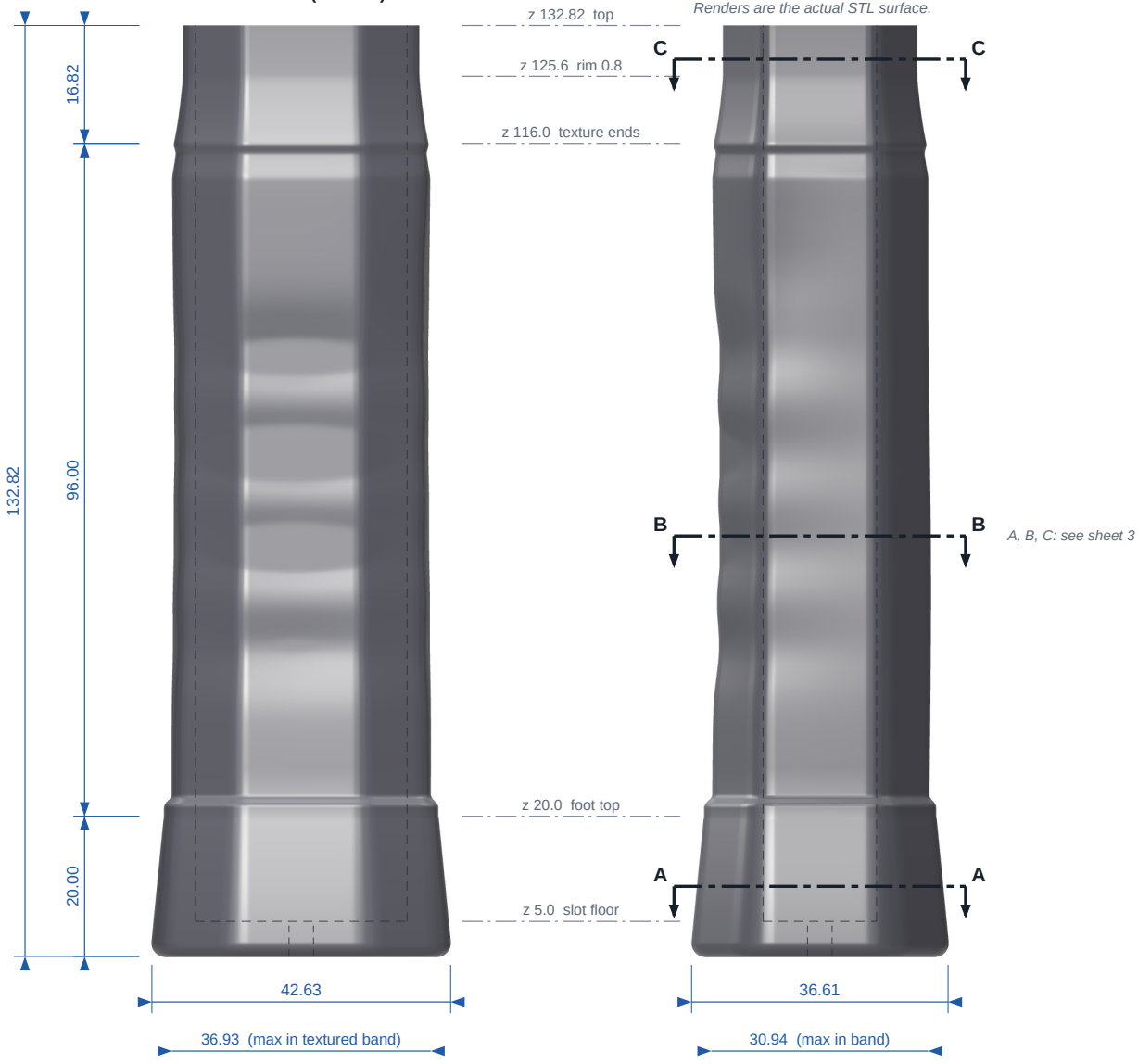


TOP VIEW (from +Z)

KEY DIMENSIONS

Overall height	**132.82**
Overall width × depth (foot)	**42.63 × 36.61**
Paddle slot width × depth	**30.20 × 16.20**
Slot depth (floor at z 5.0)	**127.82**
Outer W × D, textured band (max)	**36.93 × 30.94**
Foot top (z 20)	**39.15 × 33.13**
Butt face (z 0)	**40.56 × 34.54**
Peak texture height above base	**2.77**
Screw hole through butt (z 0 – 5)	**Ø 3.50**

All values in mm. Dashed lines: paddle slot and screw hole (hidden edges). Renders are the actual STL surface.



FRONT VIEW (from −Y)

RIGHT VIEW (from +X)

SCALE 1 : 1

0 10 20 30 40 50 mm

PART

ERGO-CONTOUR

Pickleball paddle grip sleeve, closed butt, open throat

SCALE	UNITS	PROJECTION
1 : 1	millimetres	
MATERIAL	**INFILL**	**LAYER HEIGHT**
PLA / PETG / TPU	100 %	0.20 mm
MASS (PLA)	**VOLUME**	**SURFACE AREA**
88.4 g	71.33 cm³	293 cm²
REVISION	**DATE**	**SHEET**
3	2026-10-17	2 of 4

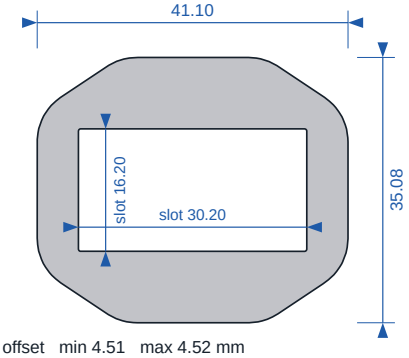
GENERAL NOTES

1. Views are true size: print this sheet at 100 %, do not fit to page.
2. Dimensions are measured from the STL, not nominal.
3. Solid body: only the 30.2 × 16.2 mm paddle slot and the Ø3.5 mm screw hole are open.
4. Outer envelope matches the reference PickleballGrip_1.3mf (rim held ≥ 0.8 mm).

Sections & profiles

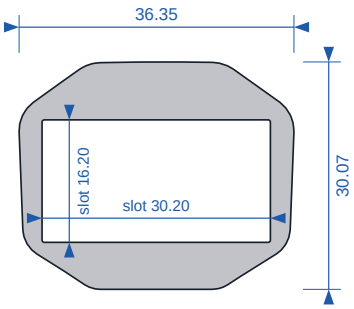
True-size cross-sections, how the outer surface and material change with height, and the dimension tables.

SECTION A–A z = 10 mm · foot



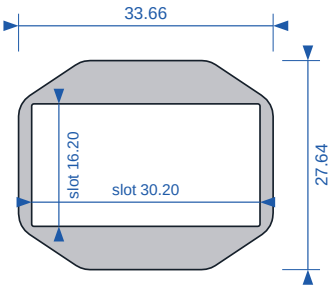
Skin offset min 4.51 max 4.52 mm
Area 780 mm² · outer perimeter 128.9 mm

SECTION B–B z = 60 mm · textured band



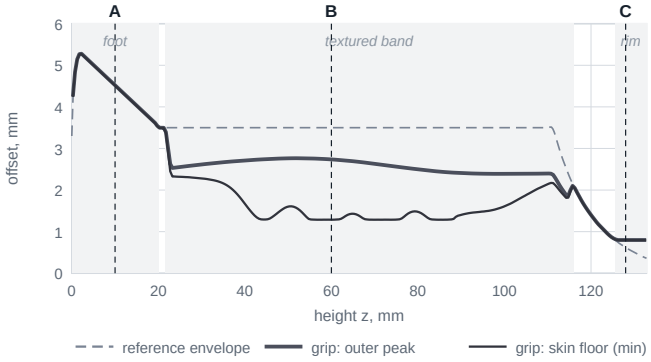
Skin offset min 1.29 max 2.74 mm
Area 469 mm² · outer perimeter 112.6 mm

SECTION C–C z = 128 mm · rim

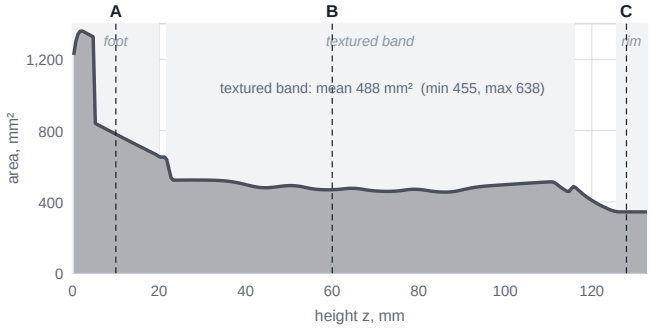


Skin offset min 0.79 max 0.80 mm
Area 344 mm² · outer perimeter 105.5 mm

OUTER OFFSET FROM THE REFERENCE OUTLINE VS HEIGHT



MATERIAL CROSS-SECTION AREA VS HEIGHT



OVERALL & SLOT

Overall length (Z)	**132.82 mm**
Overall width (X)	**42.63 mm**
Overall depth (Y)	**36.61 mm**
Slot width × depth	**30.20 × 16.20 mm**
Slot area / perimeter	**489.2 mm² / 92.3 mm**
Slot depth (floor at z 5.0)	**127.82 mm**
Screw hole (butt, z 0 – 5)	**Ø 3.50 mm**
Butt face (z 0)	**40.56 × 34.54 mm**
Outer W × D in band (max)	**36.93 × 30.94 mm**
Volume / surface area	**71.33 cm³ / 293.1 cm²**

ZONES BY HEIGHT

ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	**5.0**
Flared foot	0.0	20.0	**20.0**
Smooth collar	20.0	21.5	**1.5**
Textured band	21.5	116.0	**94.5**
Smooth neck (taper)	116.0	125.6	**9.6**
Rim (flat top, slot opening)	125.6	132.82	**7.3**

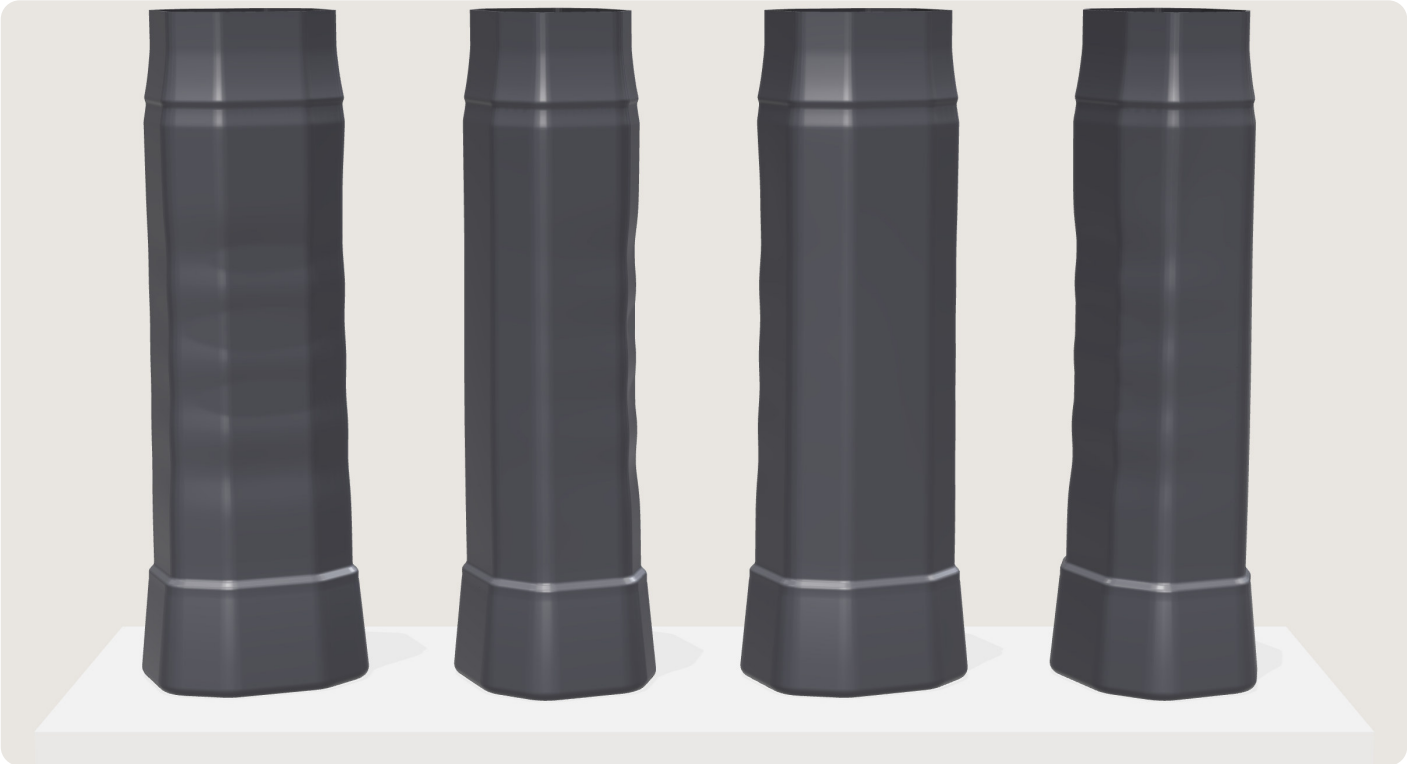
Relief floor above reference line (design / min)	**1.20 / 1.29 mm**
Min skin at the neck (design / measured)	**0.80 / 0.80 mm**
Peak texture height above base	**2.77 mm**

OUTER DIMENSIONS BY HEIGHT

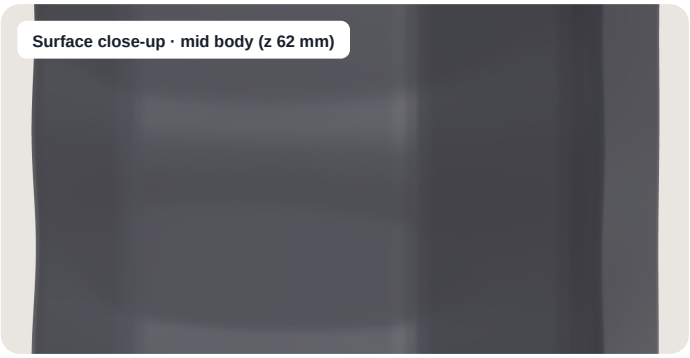
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM²	SKIN MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	1,084	5.00	5.01
10.0	41.10	35.08	128.9	780	4.51	4.52
20.0	39.11	33.09	122.6	655	3.52	3.52
30.0	36.93	30.94	115.6	523	2.28	2.62
45.0	36.60	30.16	113.3	480	1.30	2.74
60.0	36.35	30.07	112.6	469	1.29	2.74
75.0	36.18	29.90	112.2	461	1.29	2.56
90.0	36.27	30.01	112.5	468	1.40	2.41
105.0	36.66	30.68	114.6	505	1.88	2.39
116.0	36.21	30.19	113.5	484	2.07	2.08
125.0	33.77	27.75	105.8	350	0.85	0.85
132.0	33.66	27.64	105.5	344	0.79	0.80

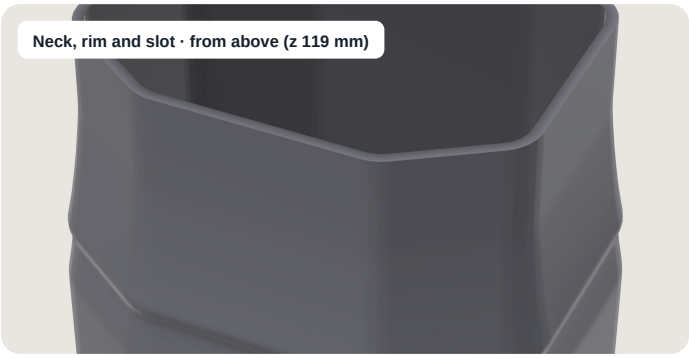
Views & surface

Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.

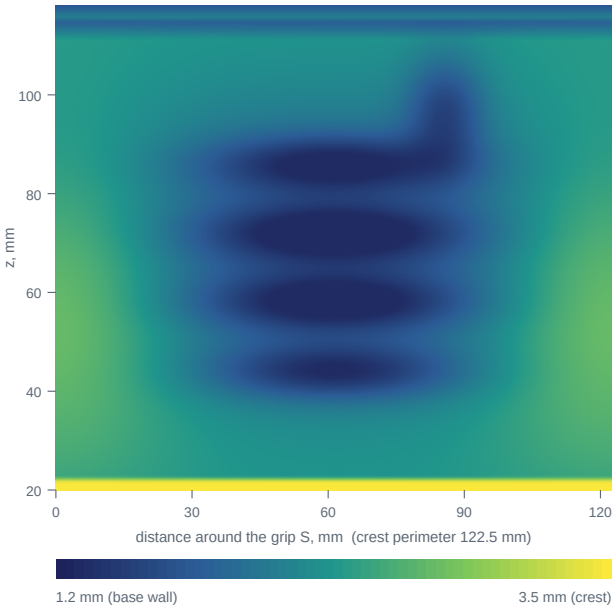


Four views, 90° apart · rendered from the STL, colours are preview only





UNROLLED SURFACE · HEIGHT ABOVE BASE



DESIGN PARAMETERS

Finger flutes	**4, at z 44 / 58 / 72 / 86**
Flute width / depth	**about 9 mm / 0.8 mm**
Thumb dish	**z 97 mm, depth 0.6 mm**
Palm pad	**back face, z 62 mm, +1.0 mm**
Hourglass swell	**1.73 – 2.33 mm above base**
Peak height	**about 2.8 mm above base**
Surface	**smooth, no sharp edges**

PRINT & VERIFICATION

Result: slot & hole / print-ready	**PASS / PASS**
Bodies · genus	**1 · 1 (screw hole)**
Paddle slot	**30.2 × 16.2 mm**
Envelope over reference	**+0.000 mm**
Material under 0.4 mm wide	**0.000 %**
Overhangs / bridges	**none**

10 HEXA-MOD

Modular hex pads on four height levels, colour-coded (red = tallest).

88.5 g

PLA · 100 % infill · solid

✓ SOLID · SLOT + SCREW HOLE



3/4 view · rendered from the STL, colours are preview only

AT A GLANCE

Overall height	**132.82 mm**
Max footprint (foot)	**42.63 × 36.61 mm**
Paddle slot	**30.2 × 16.2 mm, 127.8 deep**
Textured band	**z 21.5 – 116.0 mm (94.5 tall)**
Outer size in band (max)	**38.7 × 33.0 mm**
Peak texture height	**3.50 mm above base**
Screw hole (butt)	**Ø 3.5 mm, through z 0 – 5**
Volume / surface area	**71.36 cm³ / 315 cm²**

WEIGHT · 100 % INFILL

PLA	1.24 g/cm³	**88.5 g**
PETG	1.27 g/cm³	**90.6 g**
TPU	1.21 g/cm³	**86.3 g**
ABS	1.04 g/cm³	**74.2 g**

0 — 120 g

COLOURWAY

#1C1C1E	#464A4E	#74787E	#B0B4BA	#D63A28

3MF print palette

PRINT PROFILE

Infill / layer height	**100 % / 0.20 mm**
Layers	**665**
Orientation	**upright, butt end down**
Supports	**none required**
Filament, 1.75 mm	**≈ 29.7 m (100 % flow)**

DESIGN INTENT

A honeycomb of individual hex pads, each separated by a 1.7 mm gap and set at one of four heights. The height map is pseudo-random with a gentle swell along the handle; the tallest pads (red) reach the full 3.5 mm crest.

Pads are chamfered on the lower edge so they never overhang beyond 45 degrees.

KEY FEATURES

- 12 columns × 11 rows of pads, 10.2 mm across flats
- Four levels: 2.35 / 2.76 / 3.13 / 3.50 mm above the base
- Red pads mark the tallest level (about 6% of the band)

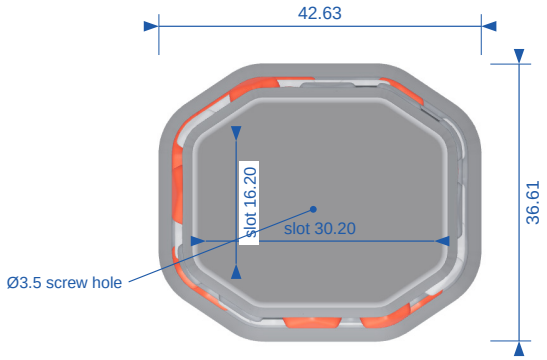
VERIFIED ON THE STL

Paddle slot	**30.2 × 16.2 mm (30 × 16 + 0.2)**
Screw hole	**Ø 3.5 mm through the butt**
Outer envelope vs reference	**+0.000 mm at most**
Bodies / enclosed voids	**1 / none**
Material under 0.4 mm wide	**0.000 % of volume**
Overhangs / bridges	**none**

FILES

stl/10_hexa_mod.stl	11.9 MB · 238,830 triangles
3mf/10_hexa_mod.3mf	2.6 MB · colour

✓ SLOT & HOLE VERIFIED ✓ SOLID · ONE CLOSED BODY ✓ PRINT-READY · 0.20 MM LAYERS ✓ 3MF VALIDATED

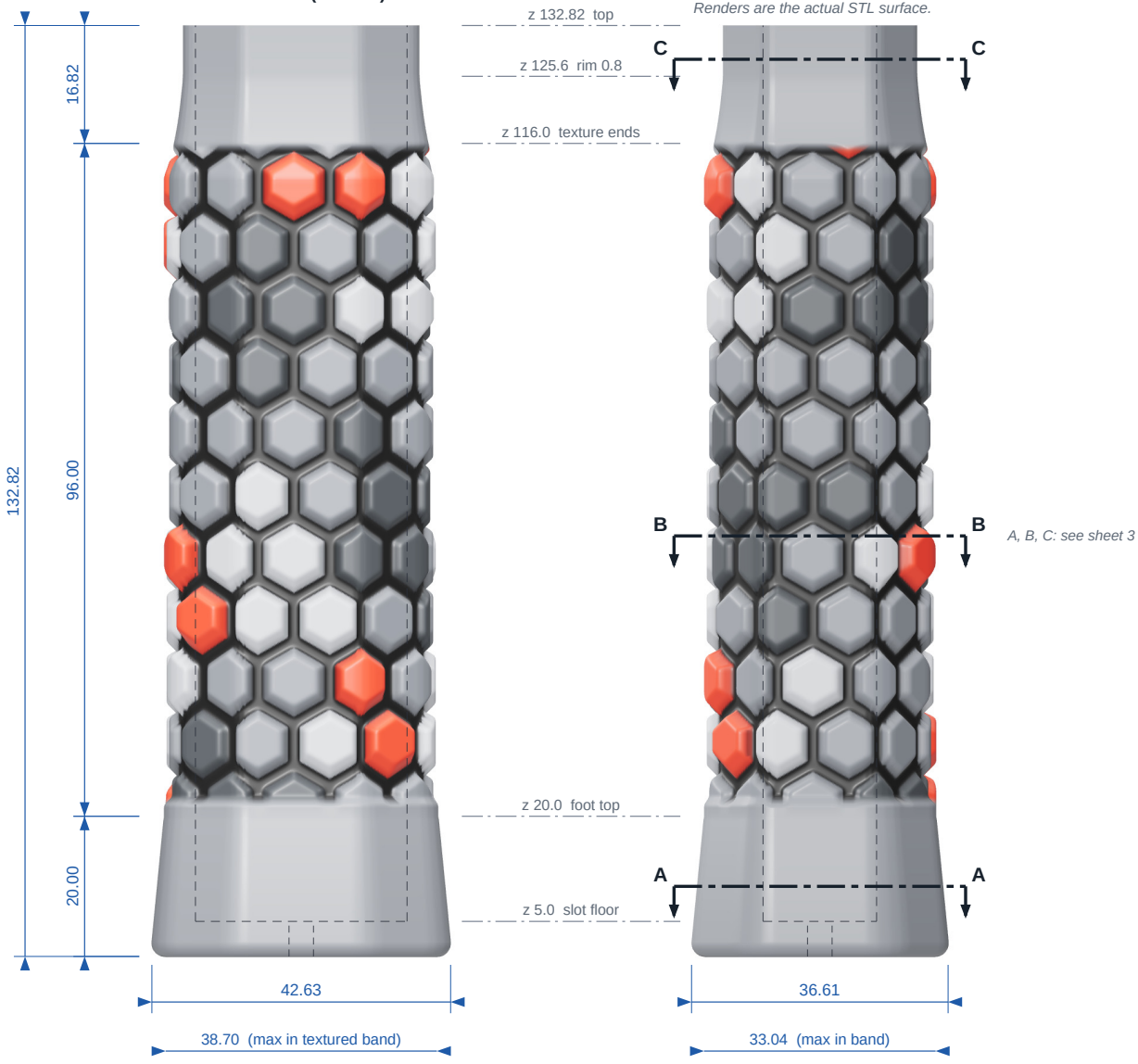


TOP VIEW (from +Z)

KEY DIMENSIONS

Overall height	**132.82**
Overall width × depth (foot)	**42.63 × 36.61**
Paddle slot width × depth	**30.20 × 16.20**
Slot depth (floor at z 5.0)	**127.82**
Outer W × D, textured band (max)	**38.70 × 33.04**
Foot top (z 20)	**39.16 × 33.13**
Butt face (z 0)	**40.56 × 34.54**
Peak texture height above base	**3.50**
Screw hole through butt (z 0 – 5)	**Ø 3.50**

All values in mm. Dashed lines: paddle slot and screw hole (hidden edges). Renders are the actual STL surface.



FRONT VIEW (from −Y)

RIGHT VIEW (from +X)

SCALE 1 : 1



PART

HEXA-MOD

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

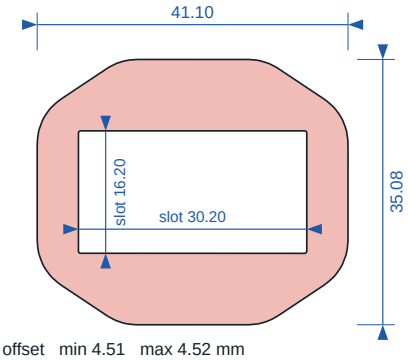
1. Views are true size: print this sheet at 100 %, do not fit to page.
2. Dimensions are measured from the STL, not nominal.
3. Solid body: only the 30.2 × 16.2 mm paddle slot and the Ø3.5 mm screw hole are open.
4. Outer envelope matches the reference PickleballGrip_1.3mf (rim held ≥ 0.8 mm).

SCALE **1 : 1**	UNITS **millimetres**	PROJECTION
MATERIAL **PLA / PETG / TPU**	INFILL **100 %**	LAYER HEIGHT **0.20 mm**
MASS (PLA) **88.5 g**	VOLUME **71.36 cm³**	SURFACE AREA **315 cm²**
REVISION **3**	DATE **2026-10-17**	SHEET **2 of 4**

Sections & profiles

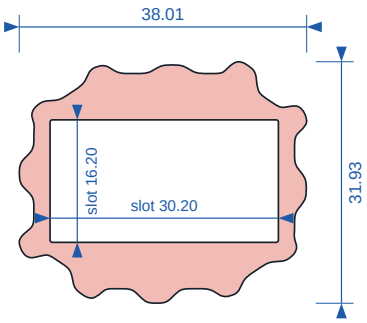
True-size cross-sections, how the outer surface and material change with height, and the dimension tables.

SECTION A–A z = 10 mm · foot



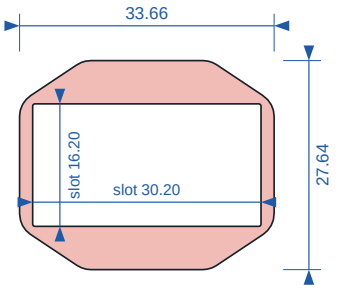
Skin offset min 4.51 max 4.52 mm
Area 780 mm² · outer perimeter 128.9 mm

SECTION B–B z = 60 mm · textured band



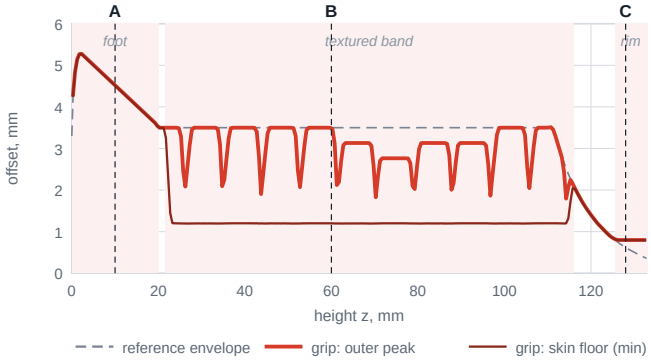
Skin offset min 1.20 max 3.48 mm
Area 465 mm² · outer perimeter 117.7 mm

SECTION C–C z = 128 mm · rim

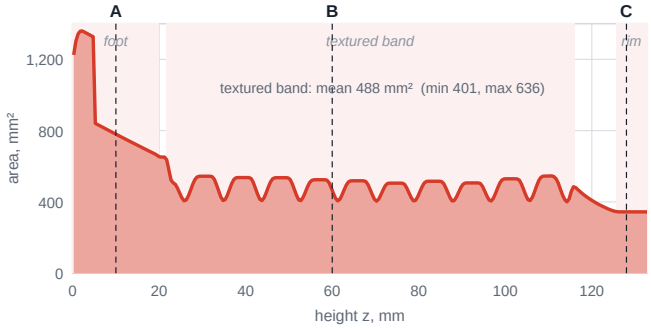


Skin offset min 0.79 max 0.80 mm
Area 344 mm² · outer perimeter 105.5 mm

OUTER OFFSET FROM THE REFERENCE OUTLINE VS HEIGHT



MATERIAL CROSS-SECTION AREA VS HEIGHT



OVERALL & SLOT

Overall length (Z)	**132.82 mm**
Overall width (X)	**42.63 mm**
Overall depth (Y)	**36.61 mm**
Slot width × depth	**30.20 × 16.20 mm**
Slot area / perimeter	**489.2 mm² / 92.3 mm**
Slot depth (floor at z 5.0)	**127.82 mm**
Screw hole (butt, z 0 – 5)	**Ø 3.50 mm**
Butt face (z 0)	**40.56 × 34.54 mm**
Outer W × D in band (max)	**38.70 × 33.04 mm**
Volume / surface area	**71.36 cm³ / 315.2 cm²**

ZONES BY HEIGHT

ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	**5.0**
Flared foot	0.0	20.0	**20.0**
Smooth collar	20.0	21.5	**1.5**
Textured band	21.5	116.0	**94.5**
Smooth neck (taper)	116.0	125.6	**9.6**
Rim (flat top, slot opening)	125.6	132.82	**7.3**

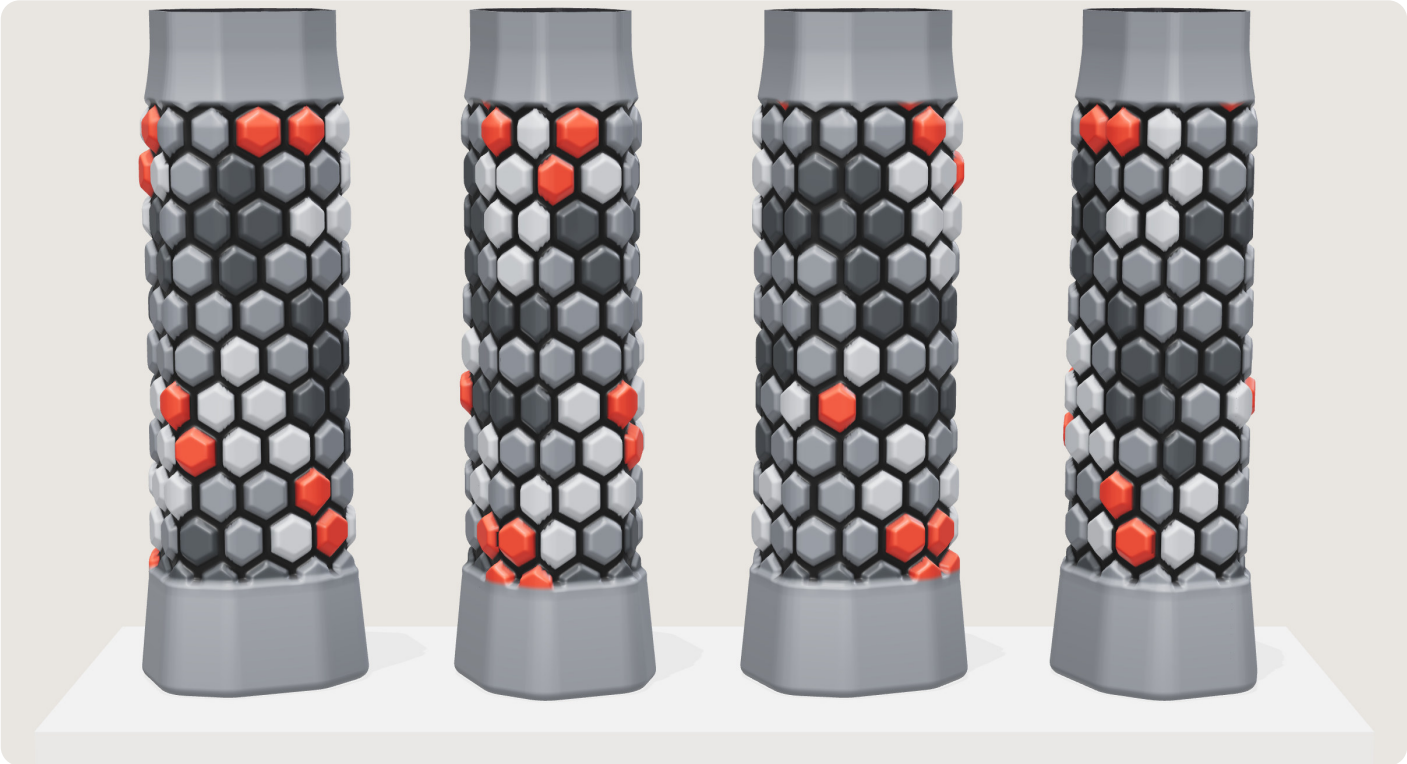
Relief floor above reference line (design / min)	**1.20 / 1.19 mm**
Min skin at the neck (design / measured)	**0.80 / 0.80 mm**
Peak texture height above base	**3.50 mm**

OUTER DIMENSIONS BY HEIGHT

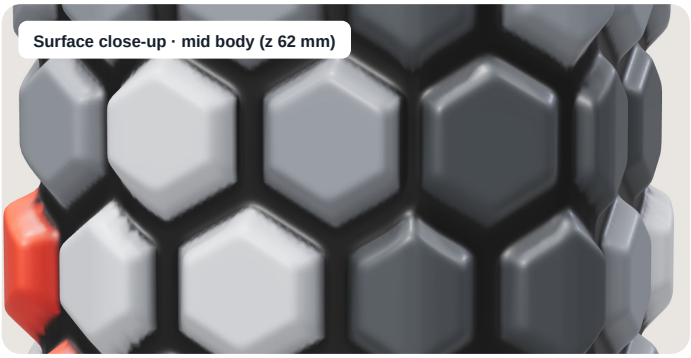
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM²	SKIN MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	1,084	5.00	5.01
10.0	41.10	35.08	128.9	780	4.51	4.52
20.0	39.11	33.09	122.6	655	3.52	3.52
30.0	38.33	32.67	118.8	544	1.20	3.50
45.0	37.80	32.03	115.1	469	1.20	3.13
60.0	38.07	31.91	117.6	465	1.20	3.48
75.0	37.59	31.49	117.0	506	1.20	2.76
90.0	37.15	32.20	117.7	491	1.20	3.13
105.0	37.70	31.09	112.9	425	1.20	2.94
116.0	36.26	30.24	113.4	483	2.04	2.10
125.0	33.77	27.75	105.8	350	0.85	0.85
132.0	33.66	27.64	105.5	344	0.79	0.80

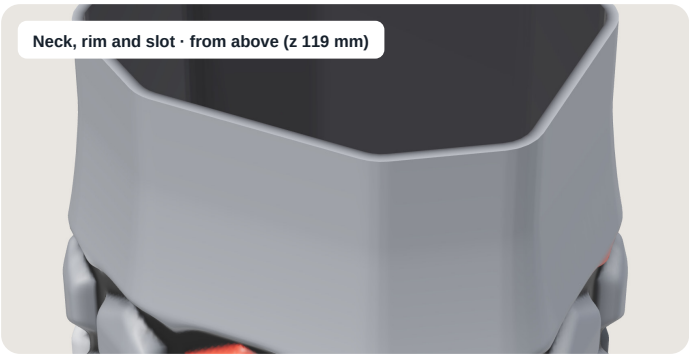
Views & surface

Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.

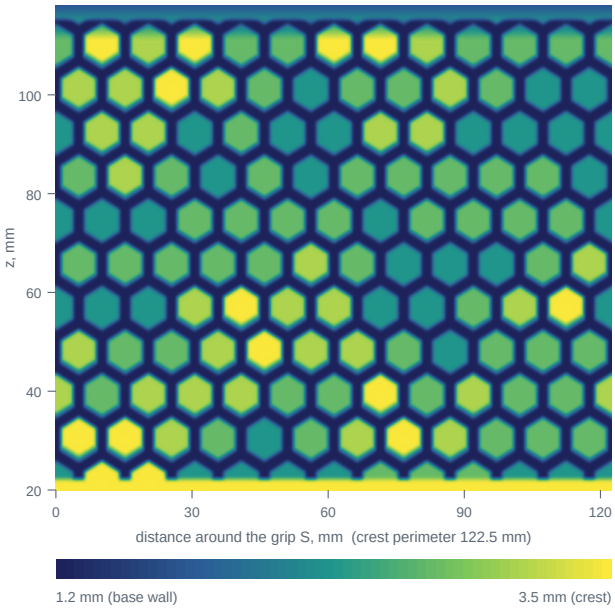


Four views, 90° apart · rendered from the STL, colours are preview only





UNROLLED SURFACE · HEIGHT ABOVE BASE



DESIGN PARAMETERS

Pad grid	**12 × 11 (pointy-top hex)**
Pad across flats	**10.21 mm**
Gap between pads	**1.7 mm**
Levels	**2.35 / 2.76 / 3.13 / 3.50 mm**
Gap floor	**1.20 mm (base wall)**
Pad edge	**chamfered, about 0.8 mm**
Level colours	**3 greys + red**

PRINT & VERIFICATION

Result: slot & hole / print-ready	**PASS / PASS**
Bodies · genus	**1 · 1 (screw hole)**
Paddle slot	**30.2 × 16.2 mm**
Envelope over reference	**+0.000 mm**
Material under 0.4 mm wide	**0.000 %**
Overhangs / bridges	**none**

Notes, method and file manifest

What the numbers mean, what was assumed, and what is in the package.

ABOUT THESE SHEETS

- All dimensions and volumes are measured directly from the STL files; design-intent values are only used for descriptions.
- Weight = volume × density at 100 % infill. Slicers land within a few percent, so budget ±3 %.
- “Above base” heights are measured outward from the reference bore line (the 32.06 × 26.04 mm octagon of PickleballGrip_1.3mf), which is now inside the solid body.
- Assumed: 0.4 mm nozzle, 1.75 mm filament, printed upright with the butt on the bed. Not sliced or printed yet.
- Renders and colours are a preview. The STLs are single-material; the 3MFs carry a small print palette per triangle.
- Sheet numbering: PG-01 to PG-10. Drawings are third-angle, true size at 100 % print scale.

DENSITIES USED (G/CM³)

PLA	1.24
PETG	1.27
TPU	1.21
ABS	1.04

FILES

stl/	ten single-material STLs
3mf/	ten colour 3MFs (materials extension)
specs/	ten sheets + this book
renders/	catalog renders and fitment overlay
scripts/	generator, verifiers, PDF builder

REVISION HISTORY

REV	CHANGE
1	Initial ten designs, STL + colour 3MF
2	Smooth finish, solid 1.2 mm walls, ≤ 50 g each, specification sheets
3	Solid body, 30 × 16 mm paddle slot, Ø3.5 mm screw hole; Voronoi windows removed

FILE MANIFEST

NO.	DESIGN	STL	SIZE	TRIANGLES	3MF	SIZE
PG-01	Vortex Grip	01_vortex_grip.stl	14.8 MB	295,950	01_vortex_grip.3mf	3.2 MB
PG-02	Cellular Mod	02_cellular_mod.stl	6.0 MB	119,164	02_cellular_mod.3mf	1.4 MB
PG-03	Tessel-Block	03_tessel_block.stl	15.0 MB	300,964	03_tessel_block.3mf	3.3 MB
PG-04	Logic-Grip	04_logic_grip.stl	14.4 MB	287,530	04_logic_grip.3mf	3.1 MB
PG-05	Neuro-Tread	05_neuro_tread.stl	18.6 MB	371,624	05_neuro_tread.3mf	4.1 MB
PG-06	Carbon Matrix	06_carbon_matrix.stl	15.4 MB	308,248	06_carbon_matrix.3mf	3.3 MB
PG-07	Voronoi-Core	07_voronoi_core.stl	7.9 MB	158,992	07_voronoi_core.3mf	1.8 MB
PG-08	Topo-Flow	08_topo_flow.stl	17.0 MB	339,714	08_topo_flow.3mf	3.8 MB
PG-09	Ergo-Contour	09_ergo_contour.stl	2.1 MB	42,162	09_ergo_contour.3mf	0.5 MB
PG-10	Hexa-Mod	10_hexa_mod.stl	11.9 MB	238,830	10_hexa_mod.3mf	2.6 MB

LIMITS OF THIS DOCUMENT

These sheets describe the geometry and its measured properties. They are not a print guarantee: material, printer, cooling and slicer settings all affect the result. Do a preview slice before committing filament. The slot is sized to a ruler measurement of 30 × 16 mm: check your paddle with calipers and adjust PADDLE_W / PADDLE_T in grip_core.py if it differs.